

PROFINET over wireless: Bridging the handover gap for seamless handover and redundancy

Implementation and Evaluation of Parallel Redundancy Protocol
over a Wi-Fi or Bluetooth Network for Industrial Automation

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Preface

Completing this master's thesis marks the culmination of my academic journey in the Electronics and ICT Engineering Technology programme at KU Leuven. The research presented in this document, titled "*PROFINET over wireless: Bridging the handover gap for seamless handover and redundancy*," bridges the gap between theoretical network engineering and industrial automation.

Working on a critical, hands-on challenge for ArcelorMittal Gent has been an incredibly rewarding and instructive experience. It allowed me to step outside the laboratory and apply my academic knowledge to a tangible industrial problem, specifically addressing the fail-safe standstills in the Steelshop. I hope that the empirical findings and deployment architectures proposed in this thesis will serve as a valuable foundation for the future of wireless operational technology.

I hope you enjoy reading this thesis.

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Summary

In industrial facilities like ArcelorMittal, automated machinery relies on wireless networks for operation and safety. Industrial protocols like PROFINET require strictly deterministic communication across both uplink and downlink. These systems are guarded by communication watchdogs. If signal delays exceed watchdog thresholds, the PROFINET connection drops and the machine reverts to a fail-safe state to prevent accidents. However, these abrupt stops cause expensive production delays, and the underlying loss of control creates potentially unsafe conditions. This research uses a moving crane as a case study to investigate and solve this issue.

Currently, the wireless networks in the ArcelorMittal Ghent Steelshop face a critical challenge. Large metal structures (e.g. other cranes) block wireless signals, creating "shadowing". As a mobile machine moves, it must switch to another network antenna. Typical access point settings hold the connection to a weak, distant antenna rather than switching early. This causes lower bit rates, which can cause packet loss and communication delays. When the machine finally switches Access Points, the delay and switching process often exceeds watchdog thresholds, triggering a fail-safe state.

This thesis aims to eliminate this switching delay, often called the "handover gap," by applying the Parallel Redundancy Protocol (PRP) to wireless links. PRP duplicates and sends every message simultaneously over two separate paths. If one path is blocked by another machine (such as a passing crane) or is switching antennas, the duplicate message arrives via the second path. At the receiving end, only the first uncorrupted message is forwarded, resulting in seamless switching.

Two wireless setups have been built as proof of concept. First, a Bluetooth setup used one-to-one connections to physically separate the paths. Because Bluetooth does not automatically switch antennas, a cascaded chain of devices seamlessly increased the coverage. Second, a Wi-Fi setup utilized Virtual Local Area Networks (VLANs) to virtually split the network into two separate channels. By physically staggering the antennas, the mobile device never switches antennas on both channels at the same time, ensuring a constant connection to both VLANs and their redundant paths.

Experimental evaluation demonstrated that both setups successfully utilize the duplicate paths to mask switching delays, achieving seamless handover and reducing maximum latency. However, analysis of the measurements revealed operational thresholds for each technology.

The Wi-Fi setup delivered superior performance, featuring a baseline latency of approximately 0.7 ms and high throughput, making it suitable for mixed Operational Technology (OT) and Information Technology (IT) traffic. In contrast, the Bluetooth setup is, unlike Wi-Fi, resistant to physical signal attenuation, but it exhibited physical layer bottlenecks: sub-1 Mbps throughput and latency spikes exceeding 30 ms, limiting its application to OT networks.

Ultimately, this research provides a validated framework for deploying wireless architectures capable of maintaining reliable PROFINET communication, minimizing compromises to safety requirements.

Keywords: PROFINET, Parallel Redundancy Protocol (PRP), Seamless Handover, Handover Gap, Deterministic Communication, Wi-Fi, Bluetooth

Samenvatting

In industriële omgevingen zoals ArcelorMittal zijn geautomatiseerde machines afhankelijk van draadloze netwerken voor hun werking en veiligheid. Industriële protocollen zoals PROFINET vereisen deterministische communicatie, zowel op de uplink als de downlink. Wanneer signaalvertragingen de vastgestelde drempelwaarden overschrijden, wordt de PROFINET-verbinding verbroken en keert de machine terug naar een fail-safe toestand om ongevallen te voorkomen. Deze stops leiden tot productieonderbrekingen en potentieel onveilige situaties. Dit onderzoek gebruikt een rijdende kraan als case study om dit probleem te onderzoeken en op te lossen.

De draadloze netwerken in de staalfabriek van ArcelorMittal Gent kampen momenteel met een uitdaging. Grote metalen constructies, zoals andere kranen, blokkeren draadloze signalen en veroorzaken "shadowing". Wanneer een mobiele machine zich verplaatst, moet ze overschakelen naar een andere antenne. Typische access point instellingen houden de verbinding vast aan een zwak, ver gelegen antenne in plaats van vroeg te wisselen. Dit resulteert in lagere bitsnelheden, wat packet loss en communicatievertragingen kan veroorzaken. Wanneer de machine uiteindelijk van access point wisselt, kan de vertraging en het wisselproces de watchdog-drempelwaarden overschrijden, wat een fail-safe toestand activeert.

Dit onderzoek heeft als doel deze wisselvertraging, ook wel de "handover gap" genoemd, te elimineren door het Parallel Redundancy Protocol (PRP) toe te passen op draadloze verbindingen. PRP dupliceert elk bericht en verstuurt het gelijktijdig over twee afzonderlijke paden. Als één pad geblokkeerd wordt door een passerende machine of bezig is met een antennewisseling, arriveert één van de kopiën via het tweede pad. Aan de ontvangstzijde wordt enkel het eerste onbeschadigde bericht doorgestuurd, wat zorgt voor naadloze handover.

Twee draadloze opstellingen zijn gebouwd als proof of concept. Ten eerste werd een Bluetooth-opstelling gebruikt met point-to-point verbindingen om de paden fysiek te scheiden. Omdat Bluetooth niet automatisch van antenne wisselt, werd een cascade van apparaten gebruikt om de dekking naadloos uit te breiden. Ten tweede maakte een Wi-Fi-opstelling gebruik van Virtual Local Area Networks (VLAN's) om het netwerk virtueel op te splitsen in twee afzonderlijke kanalen. Door de antennes fysiek te verspringen, wisselt het mobiele apparaat nooit gelijktijdig van antenne op beide kanalen, zodat het mobiele apparaat altijd verbonden is met beide VLAN's en hun redundante paden.

Experimentele evaluatie toonde aan dat beide opstellingen de dubbele paden succesvol be-

nutten om naadloze handover te realiseren en de maximale vertraging te verlagen. Analyse van de metingen bracht echter operationele drempelwaarden aan het licht. De Wi-Fi-opstelling leverde superieure prestaties met een baseline vertraging van ongeveer 0,7 ms en hoge throughput, wat ze wat ze geschikt maakt voor gemengd Operational Technology (OT) and Information Technology (IT) verkeer. De Bluetooth-opstelling daarentegen is, in tegenstelling tot Wi-Fi, bestand tegen fysieke signaaldempingen, maar vertoonde beperkingen: een throughput onder 1 Mbps en latency spikes van meer dan 30 ms, wat de toepassing beperkt tot OT netwerken.

Dit onderzoek biedt een gevalideerd framework voor de implementatie van draadloze architecturen die betrouwbare PROFINET-communicatie kunnen handhaven, met minimale concessies aan veiligheidsvereisten.

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List of Acronyms

AC Access Category.

ACK Acknowledgment.

AFH Adaptive Frequency Hopping.

AIFS Arbitration Inter-Frame Space.

AIFSN Arbitration Inter-Frame Space Number.

AP Access Point.

AR Application Relation.

BANY Bus Analyzer.

BER Bit Error Rate.

BNEP Bluetooth Network Encapsulation Protocol.

BPSK Binary Phase Shift Keying.

CA Collision Avoidance.

CAPEX Capital Expenditure.

CR Communication Relation.

CSMA Carrier-Sense Multiple Access.

DAN Dual Attached Node.

DCF Distributed Coordination Function.

Dev. Deviation.

DIFS DCF Interframe Space.

E2E End-to-End.

EDCA Enhanced Distributed Channel Access.

EDR Enhanced Data Rate.

EMI Electromagnetic Interference.

FE Fast Ethernet.

FEC Forward Error Correction.

FHSS Frequency Hopping Spread Spectrum.

FRER Frame Replication and Elimination for Reliability.

GI Guard Interval.

GTK Group Temporal Key.

HSR High-availability Seamless Redundancy.

IP Internet Protocol.

IQR Interquartile Range.

IRT Isochronous Real-Time.

ISI Inter Symbol Interference.

IT Information Technology.

L2CAP Logical Link Control and Adaptation Protocol.

LAN Local Area Network.

LoS Line of Sight.

LRE Link Redundancy Entity.

LSDU Link Service Data Unit.

MAC Media Access Control.

Mbps Megabits per second.

MCS Modulation and Coding Scheme.

MLO Multi-Link Operation.

MRM Media Redundancy Manager.

MRP Media Redundancy Protocol.

NAP Network Access Point.

NRT Non-Real-Time.

OFDM Orthogonal Frequency Division Multiplexing.

OFDMA Orthogonal Frequency Division Multiple Access.

OT Operational Technology.

P99 99th percentile.

PAN Personal Area Network.

PANU PAN User.

PCAP Packet Capture.

PDC Process Data Control.

PI PROFINET International.

PLC Programmable Logic Controller.

PN PROFINET.

pps Packets Per Second.

PROFINET Process Field Net.

PRP Parallel Redundancy Protocol.

PTK Pairwise Transient Key.

QAM Quadrature Amplitude Modulation.

QoS Quality of Service.

QPSK Quadrature Phase Shift Keying.

RCT Redundancy Control Trailer.

RF Radio Frequency.

RON Rest of Network.

RSSI Received Signal Strength Indicator.

RT Real-Time.

SAN Single Attached Node.

SAR Segmentation and Reassembly.

SIFS Short Interframe Space.

SNMP Simple Network Management Protocol.

SNR Signal-to-Noise Ratio.

Std. Standard.

TAP Test Access Port.

TCP Transmission Control Protocol.

TDMA Time Division Multiple Access.

TSN Time Sensitive Networking.

UDP User Datagram Protocol.

VLAN Virtual Local Area Network.

VRON VLAN Rest of Network.

WLAN Wireless Local Area Network.

WMM Wireless Multimedia Extensions.

WPS Wi-Fi Protected Setup.

Chapter 1

Introduction

1.1 Context and Motivation

The transition towards Industry 4.0 has fundamentally transformed modern manufacturing and heavy industry. A key pillar of this transformation is the integration of autonomous and semi-autonomous mobile machinery, which relies heavily on robust wireless communication networks. In environments such as the ArcelorMittal Ghent Steelshop, overhead cranes are essential for transporting heavy loads and liquid steel. To automate these critical assets, continuous, deterministic communication between the central Process Data Control (PDC) servers, the ground Programmable Logic Controllers (PLCs) and the mobile PLCs is an absolute necessity.

In industrial automation, communication is typically governed by Real-Time (RT) protocols such as Process Field Net (PROFINET (PN)). Unlike standard Information Technology (IT) networks where delayed packets merely result in slower loading times, PROFINET operates on deterministic cycle times monitored by communication watchdogs. These watchdogs act as a primary safety mechanism: if the communication delay exceeds a defined threshold, the system assumes a loss of control. To prevent accidents and equipment damage, the PLC immediately triggers a Fail-Safe state, bringing the machinery to a halt.

While this Fail-Safe mechanism is a necessity for operational safety, it ironically introduces new physical hazards to the workforce alongside production inefficiencies. A standstill of a massive overhead crane turns the stranded machinery into an unexpected obstacle. This creates immediate collision risks with other active cranes sharing the same runway, ground-based machinery, and personnel navigating the area below. Beyond these safety risks, in a continuous production environment like a steelshop, an unexpected crane standstill causes an immediate disruption to the production flow, significantly reducing the overall throughput of the facility. Furthermore, recovering a crane from a Fail-Safe standstill requires manual intervention, effectively negating the productivity benefits of the automated system. Therefore, the primary motivation for this research is to bridge the gap between the deterministic requirements of industrial protocols and the inherent volatility of wireless networks. The goal is

to ensure that automated mobile machinery can operate safely, predictably, and continuously, without these dangerous and costly interruptions.

1.2 Problem Statement

The existing single-path Wi-Fi infrastructure at the ArcelorMittal Gent steelshop suffers from critical reliability issues, preventing the safe automation of overhead cranes. Operating within a harsh, highly reflective industrial environment introduces three problems:

- **Dynamic Radio Frequency (RF) Shadowing:** The steelshop contains massive, mobile metallic structures. As adjacent cranes move along the runway, they act as transient physical barriers that block the Line-of-Sight (LoS) between a crane's wireless Client and the stationary Access Points (APs). This dynamic shadowing causes multipath fading and sudden collapses in signal strength.
- **Handover Latency and the "Sticky Client" Phenomenon:** As a mobile crane traverses the hall, its wireless Client must roam between different APs. To avoid handover disconnections, current clients exhibit "sticky" behavior, clinging to a distant AP until the signal drops to a critically low threshold. Operating on this minimal link margin leaves the connection vulnerable. Furthermore, when the handover finally triggers, the resulting disconnection period (the "handover gap") frequently exceeds the strict PROFINET watchdog timer.
- **Bitrate Fluctuations and MCS Fallback:** As the Signal-to-Noise Ratio (SNR) continuously fluctuates due to distance and shadowing, the wireless radios dynamically shift to lower Modulation and Coding Schemes (MCS). This drastically shrinks the available physical bandwidth. During these bitrate drops, IT traffic saturates the narrowed link, physically blocking time-critical PROFINET frames and triggering a crane Fail-Safe standstill.

These three factors create an unsustainable operational paradox. A single wireless path forces a critical trade-off: the system must either endure the latency of frequent handovers, or delay handover and operate on a weakened signal highly vulnerable to shadowing. Consequently, automated cranes are forced into Fail-Safe standstills multiple times a day. Relaxing the PROFINET watchdog timers to accommodate these delays is not an acceptable solution, as it fundamentally compromises the safety of the control loops. Therefore, a physical and logical architectural redesign of the wireless network is required to achieve high-availability automation.

1.3 Research Objectives

The primary goal of this master's thesis is to design, implement, and evaluate a redundant wireless architecture capable of guaranteeing continuous PROFINET connectivity for mobile machinery. The proposed solution must successfully mask the described problems to prevent Fail-Safe standstills.

To achieve this goal, the research is divided into five specific, measurable objectives:

1. **Field Audit and Baseline Analysis:** To quantify the limitations of the existing single-path infrastructure at the ArcelorMittal Steelshop. By correlating network diagnostic logs with PDC failure events, this objective aims to define the exact operational bottlenecks and establish the requirements for the new architecture.
2. **Architectural Design of Redundancy:** To design a multi-path wireless network architecture based on the principles of the Parallel Redundancy Protocol (PRP). The objective is to conceptualize a system where duplicated data frames are transmitted over logically and physically isolated paths, ensuring that a communication failure on one path is seamlessly masked by the other.
3. **Implementation of Wireless Testbeds:** To physically implement and configure the proposed redundant architectures in a controlled laboratory environment. This involves adapting two distinct wireless technologies, Wi-Fi and Bluetooth, to support seamless handover, compensating for their inherent limitations regarding handover and interference.
4. **Evaluation and Benchmarking:** To comparatively benchmark the implemented Wi-Fi and Bluetooth architectures against industrial PROFINET standards. This objective will evaluate key performance indicators (e.g. mean and maximum latency, jitter, ...) in different scenarios.
5. **Industrial Translation and Economic Analysis:** To translate laboratory findings into a practical physical deployment blueprint for ArcelorMittal Gent and giving a rough estimation of the Capital Expenditure (CAPEX) costs of the proposed redundant systems.

1.4 Thesis outline

This thesis is structured to guide the reader logically from problem identification to the technical implementation, validation, and practical deployment of the proposed solutions.

Chapter 2 provides the basics on the PROFINET protocol, PRP and Functional Safety to understand this thesis.

Chapter 3 explores the wireless fundamentals and the technologies used in this thesis.

Chapter 4 presents a data-driven field audit of the ArcelorMittal Steelshop, formally defining the limitations of the existing single-path network and proving the requirement for a solution.

Chapter 5 details the design and physical implementation of the proposed redundant Wi-Fi and Bluetooth architectures.

Chapter 6 evaluates the performance of the implemented solutions, benchmarking their performance under different circumstances and their success in masking handover gaps.

Chapter 7 translates the empirical findings into a blueprint for a full-scale rollout at the ArcelorMittal Gent, concluding with an estimation of the CAPEX.

Chapter 8 summarizes the final technical conclusions and provides suggestions for future research on this topic.

Chapter 2

PROFINET

2.1 Introduction to PROFINET

Process Field Net (PROFINET) is an Industrial Ethernet standard that bridges standard commercial IT networks and the deterministic requirements of Operational Technology (OT) control systems [1]. Standard commercial Ethernet (IEEE 802.3) operates on a best-effort basis, which introduces unpredictable delays. PROFINET overcomes this limitation by running proven industrial control logic over standard Ethernet infrastructure.

The protocol classifies network traffic into tiers based on real-time urgency. By using dedicated real-time communication channels, time-critical data bypasses standard Ethernet's non-deterministic TCP/IP stack. This architecture ensures high-speed, low-latency, and predictable communication for time-sensitive automation [1].

To support mobile crane applications over a volatile wireless link, it is necessary to analyze the core principles that govern PROFINET:

- **Guaranteed Determinism:** Industrial safety requires predictable delivery of time-critical data. PROFINET's cyclic exchange model guarantees that process data arrives within a predefined timeframe. This prevents latency and jitter from halting machinery [1].
- **IT/OT Coexistence:** PROFINET allows standard IT traffic and critical OT control data to share the same physical or wireless infrastructure. It achieves this by prioritizing real-time control traffic over background IT activity [1].
- **Fault Tolerance:** Because industrial networks do not tolerate single points of failure, PROFINET architectures integrate hitless redundancy mechanisms to ensure continuous operation, even if a cable or wireless link fails [1].

A typical PROFINET I/O system consists of an **IO-Controller** (usually a PLC) that manages the network, and connected **IO-Devices** (sensors, actuators, or remote I/O modules) that

interface with the physical process [2]. The controller and devices communicate via cyclic and acyclic Application Relations (ARs) over the network.

Structurally, an IO-Device consists of slots and subslots where physical field modules reside (see Figure 2.1). The modules located in these slots or subslots perform the actual I/O functions and define the specific amount of process data to transmit over the cyclic channels.

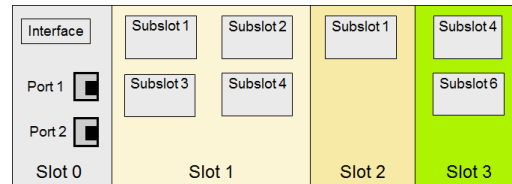


Figure 2.1: IO-Device with Slots and Subslots[2].

2.2 PROFINET Real-Time Communication Stack

PROFINET achieves real-time communication by bypassing the standard TCP/IP layers (OSI Layers 3 and 4) for cyclic data exchange. Instead, the protocol maps RT frames directly onto the Ethernet Data Link Layer (OSI Layer 2) using the unique EtherType 0x8892 and dedicated FrameIDs [3]. Bypassing the transport and network layers eliminates protocol overhead, reducing processing time and transmission latency at the endpoints [4]. Table 2.1 outlines the RT frame structure.

Table 2.1 RT Frame Structure [3][4].

Destination MAC	Source MAC	VLAN	EtherType	Frame ID	Application Data	Cycle Counter	Data Status	Transfer Status	FCS
6 bytes	6 bytes	4 bytes	2 bytes	2 bytes	40–1440 bytes	2 bytes	1 byte	1 byte	4 bytes

To coordinate this cyclic data exchange, PROFINET uses a timing hierarchy defined by three parameters:

1. **Send Clock:** The fundamental hardware heartbeat of the IO-Controller, defining the base transmission cycle (typically 1 ms or multiples thereof) [3].
2. **Reduction Ratio (2^n):** An integer multiplier applied to the Send Clock that determines the actual Update Time for a specific IO-Device [4]:

$$\text{Update Time} = \text{Send Clock} \times \text{Reduction Ratio} \quad (2.1)$$

For example, a Send Clock of 1 ms and a reduction ratio of 16 (2^4) yields a 16 ms Update Time. Figure 2.2 illustrates cycle timing examples for a 1 ms and a 4 ms cycle.

3. **Phase Offset:** A mechanism that staggers the transmissions of multiple devices to distribute bandwidth consumption evenly and prevent traffic bottlenecks at the switch

or wireless Access Point [4]. Figure 2.3 shows how phase offsets distribute the transmissions within a single cycle.



Figure 2.2: PROFINET Real-Time Communication Cycle Examples: clock cycle 1 ms (left) and 4 ms (right) [2].

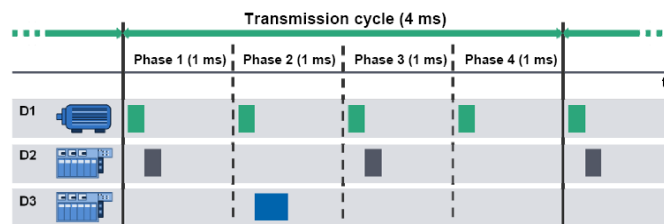


Figure 2.3: PROFINET transmission cycle and phase offsets [2].

To prioritize this critical OT data on a shared infrastructure, PROFINET RT employs IEEE 802.1Q VLAN tagging inserted into the Ethernet header [3]. This VLAN tag includes a 3-bit **Priority Code Point (PCP)** field, which PROFINET RT automatically sets to Priority 6 (Voice). Table ?? details the VLAN tag structure. This high-priority tag forces network switches and wireless Access Points to process and transmit the RT frames ahead of standard best-effort IT traffic.

Table 2.2 VLAN Tag Structure [3][4].

EtherType	PCP	DEI/CFI	VLAN-ID
2 Bytes	3-Bit	1-Bit	12-Bit

However, this tagging only guarantees soft Quality of Service (QoS). Because PROFINET RT does not reserve exclusive physical time slots on the medium, it cannot preempt an ongoing transmission. If a network node is already transmitting a large background packet (e.g., a 1500-byte UDP frame) when a Priority 6 RT frame arrives, the RT frame must wait in the buffer. This queuing delay introduces non-deterministic jitter into the cyclic exchange. This is a vulnerability that worsens over half-duplex, shared wireless channels.

2.3 System Reliability and Safety Limits: The Watchdog

Because industrial automation relies on strictly deterministic communication, the control loop must respond predictably to network anomalies. Over wireless channels, the transport

layer is highly volatile due to RF shadowing, interference, and handover delays.

To maintain safety during these disruptions, PROFINET implements a **Watchdog Timer** [3]. During startup, the IO-Controller and IO-Device agree on a maximum number of consecutive missed frames before assuming a link failure. The industry standard is three consecutive missed frames, which defines the Watchdog Time:

$$\text{Watchdog Time} = \text{Update Time} \times 3 \quad (2.2)$$

Both the PLC and the endpoint device continuously monitor this timer. If the receiver does not receive a valid, error-free RT frame before the watchdog expires, it immediately triggers a **Fail-Safe state**. For heavy machinery like an overhead crane, this halts all physical movement abruptly to prevent accidents, load drops, or mechanical damage.

Configuring the Watchdog Time over a wireless link requires a critical trade-off. Increasing the watchdog value allows the connection to tolerate temporary latency spikes (like standard Wi-Fi handovers), but it delays the emergency halt during a real physical accident. Consequently, the underlying wireless network must deliver cyclic data reliably within tight watchdog constraints to avoid nuisance tripping and costly production stoppages.

2.4 Parallel Redundancy Protocol

Standard redundancy protocols (such as MRP) require 50 ms to 200 ms to reconfigure a network path after a link failure. This recovery time exceeds typical PROFINET watchdog limits, tripping the safety loop. To achieve hitless, 0 ms recovery time, industrial networks employ the **Parallel Redundancy Protocol** standardized under **IEC 62439-3** [5].

2.4.1 PRP Architecture and Components

PRP achieves seamless redundancy by utilizing two physically separate, parallel networks: **LAN A** and **LAN B** [5]. These two networks operate independently and share no physical connections. Nodes in a PRP network are classified into three types based on their native support:

- **Dual Attached Node (DAN):** A device equipped with two independent network interfaces connected to LAN A and LAN B, managing frame duplication and deduplication internally.
- **Single Attached Node (SAN):** A standard device with a single interface that cannot connect to both networks natively.
- **Redundancy Box (RedBox):** A specialized hardware bridge that acts as a redundant proxy for SANs, mapping standard Ethernet traffic onto both PRP LANs transparently.

Figure 2.4 illustrates the PRP architecture connecting an IO-Controller and an IO-Device via independent parallel networks.

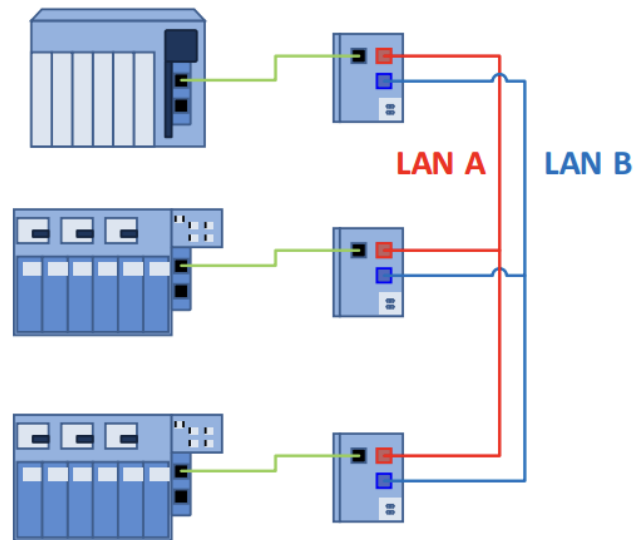


Figure 2.4: PRP parallel network architecture showing LAN A and LAN B physical separation to eliminate single points of failure [6].

2.4.2 Frame Duplication and Deduplication Mechanics

PRP achieves seamless, hitless redundancy using duplicate-and-discard logic at Layer 2:

1. **Duplication:** When a DAN (or RedBox) prepares to send a frame, it duplicates the packet into two identical copies.
2. **Identification:** The sender appends a 6-byte **Redundancy Control Trailer (RCT)** to the end of each frame, immediately before the Ethernet Frame Check Sequence (FCS) [6]. Figure 2.5 details the structure of the PRP-encapsulated frame. The RCT contains:
 - A 16-bit **Sequence Number** to identify duplicates.
 - A 4-bit **LAN Identifier** (value 1010 for LAN A, 1011 for LAN B) to specify the source interface.
 - A 12-bit **LSDU Size** indicating the size of the payload.
 - A 16-bit **PRP Suffix** (value 0x88FB) indicating a PRP frame.
3. **Simultaneous Transmission:** The sender transmits the LAN A frame and LAN B frame simultaneously. In wireless environments, these paths use different frequency bands or spatially separated Access Points to maximize transmission diversity.

4. **Reception and Deduplication:** The receiving node (or RedBox) accepts the first valid copy that arrives and processes it immediately. By checking the Sequence Number in the RCT, the receiver identifies the duplicate copy when it arrives via the alternative LAN and discards it silently.

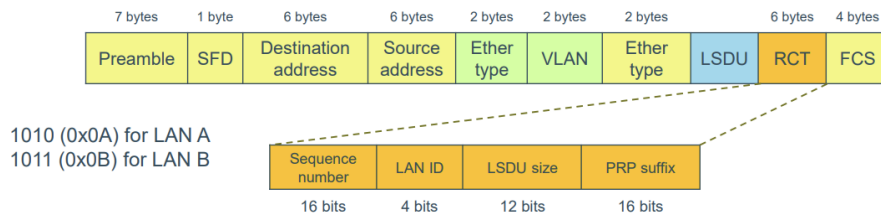


Figure 2.5: PRP-encapsulated Ethernet frame structure highlighting the Redundancy Control Trailer [6].

If a physical cable breaks or a network switch fails in LAN A, the duplicate frame still arrives uninterrupted via LAN B. The receiver accepts it with zero packet loss and 0 ms recovery delay, maintaining continuous communication.

Chapter 3

Wireless Technologies

3.1 Wireless Fundamentals

Transitioning from wired to wireless PROFINET replaces a controlled, point-to-point physical environment with an unpredictable, shared medium [7]. To explain why a wireless link behaves differently than a wired link, it is necessary to establish the physical and protocol-level differences.

3.1.1 The Industrial Wireless Challenge

Industrial wireless communication differs from standard consumer applications. While consumer Wi-Fi optimizes for total throughput (e.g., during file transfers) [7], industrial networks prioritize determinism, bounded latency, and high availability [3].

3.1.1.1 Shared Medium and Half-Duplex Nature

A wireless network replaces full-duplex switched Ethernet with a half-duplex shared medium [7], as Figure 3.1 illustrates. Unlike modern Ethernet switches that enable simultaneous, bi-directional communication across dedicated ports, a wireless transceiver can only transmit or receive at any single moment on a given channel. Consequently, wireless transceivers must constantly switch between receive and transmit modes.

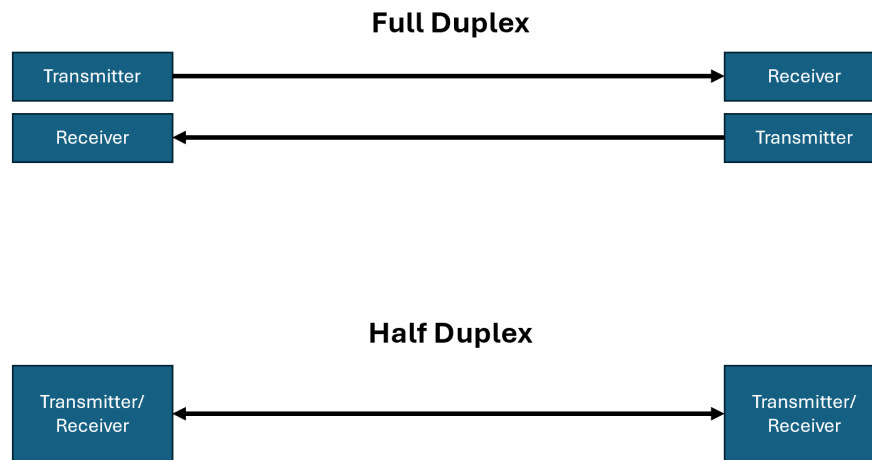


Figure 3.1: Comparison of Full-Duplex and Half-Duplex communication channels.

Furthermore, because only one device can occupy the medium within a wireless cell at any time, wireless communication treats airtime as a strict zero-sum resource. Every packet, from a 40-byte control frame to a 1500-byte data payload, consumes a finite amount of airtime to transmit over a frequency [7].

This physical limitation directly affects protocols like PROFINET, which rely on the continuous, cyclic exchange of small frames [3]. In a wireless environment, this characteristic is highly airtime-inefficient because every individual frame incurs fixed physical-layer overhead, such as preambles and headers, regardless of its payload size [7].

This inefficiency makes resource contention a major vulnerability. If non-critical IT traffic consumes the available airtime, the remaining capacity for the time-sensitive OT control loop diminishes. This contention raises latency and, in extreme cases, causes airtime starvation, leading to severe packet loss [7].

3.1.1.2 Determinism vs. Stochastic Behavior

While PROFINET RT requires strict cyclic data delivery within an Update Time boundary, wireless transmission is inherently stochastic [3]. During propagation, radio signals face four key physical phenomena:

- **Path Loss:** As a wireless signal propagates, its energy diminishes over distance and through physical obstacles. While this attenuation is gradual, its effect on digital links is abrupt. The link remains stable until the signal strength drops below the receiver's decoding threshold. [7].
- **Interference and Collisions:** Signals from other radios or Electromagnetic Interference (EMI) corrupt data packets, forcing the hardware to wait for a clear channel or execute

a backoff algorithm [7].

- **Multipath Fading:** In places like industrial facilities with large metal structures, signals reflect and reach the receiver via multiple paths. This multipath propagation causes destructive interference, resulting in severe signal fading and volatile reception quality [7].
- **Retransmissions and Jitter:** If a packet is corrupted over the air, the wireless hardware (Layer 2) automatically attempts to retransmit it. While this recovery improves the Packet Delivery Ratio, each attempt consumes additional airtime and introduces variable delays, directly increasing jitter [7].

3.1.1.3 The Handover Challenge

Mobile wireless PROFINET devices face a challenge when switching between different Access Points. This handover process, introduces a temporary disconnection gap where the Client cannot deliver frames [8]. If this gap exceeds the configured PROFINET watchdog time, the controller declares a communication loss and trips the safety loop, halting the machinery.

3.2 Wi-Fi

Standard Wi-Fi (IEEE 802.11a/b/g/h) provides a transparent Layer-2 bridge for Ethernet frames, making it compatible with the PROFINET protocol stack [3, 9]. However, because the 802.11 standard optimizes for high-throughput file transfers rather than real-time determinism, its medium access and modulation mechanisms introduce latency and jitter.

3.2.1 The 5 GHz Frequency Band

Wi-Fi operates across several frequency bands, but the 5 GHz band offers three key advantages for industrial deployments:

- **Spectrum Availability:** It provides a larger number of 20,MHz or 40,MHz non-overlapping channels (e.g., Channels 36, 40, 44, 48) than the crowded 2.4 GHz spectrum, which is limited to three non-overlapping channels (1, 6, and 11) [9].
- **Interference Mitigation:** Operating at 5 GHz avoids congestion from 2.4 GHz devices, such as Bluetooth nodes, Zigbee sensors, and microwave ovens [9].
- **High Attenuation:** Higher propagation losses limit signal spillage between adjacent cells, allowing dense Access Point deployment without mutual co-channel interference [7].

3.2.2 Medium Access Control: CSMA/CA

The Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA) mechanism governs medium access in standard Wi-Fi, introducing stochastic behavior. Wi-Fi devices share a single channel and must contend for access using a "Listen-Before-Talk" sequence [7]:

1. **Carrier Sensing:** The radio monitors the channel to verify it is idle.
2. **Inter-Frame Spacing:** The device waits for a mandatory period (e.g., Short Interframe Space (SIFS) or DCF Interframe Space (DIFS)) based on frame priority.
3. **Random Backoff:** If the medium is busy, the device enters a backoff state, waiting for a random number of time slots before re-sensing the carrier.

This shared access is non-deterministic. If a best-effort background packet occupies the channel, time-critical PROFINET frames must wait, introducing latency and jitter. CSMA/CA is also vulnerable to the **hidden node problem**, where two stations out of range of each other transmit simultaneously to the same Access Point (see Figure 3.2). These unmanaged collisions can cause packet loss and increase jitter [7].

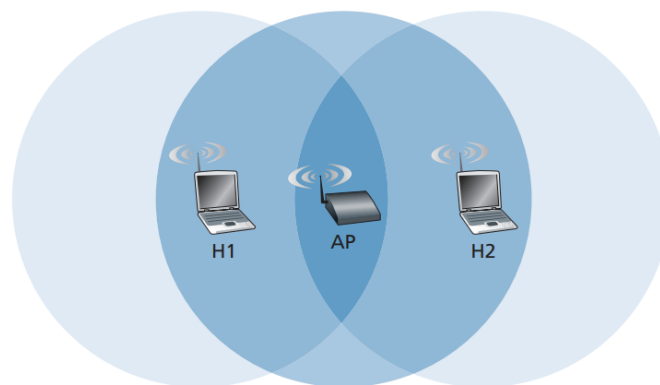


Figure 3.2: Hidden node example: H1 is hidden from H2, and vice versa [7].

3.2.3 Physical Layer Robustness: Orthogonal Frequency-Division Multiplexing (OFDM)

In any wireless transmission, RF signals reflect off physical obstacles in the environment. This creates a phenomenon known as multipath propagation, where multiple delayed copies of the same transmitted signal arrive at the receiver at slightly different times. This delay spread can cause Inter-Symbol Interference (ISI), a condition where consecutive data symbols physically overlap and corrupt the received signal [7, 9].

To mitigate ISI, Wi-Fi standards utilize Orthogonal Frequency-Division Multiplexing (OFDM) at the physical layer. Instead of transmitting data sequentially over a single high-speed carrier, OFDM divides the transmission channel (e.g., 20 MHz or 40 MHz) into dozens of narrower,

mathematically orthogonal subcarriers [7]. The data stream is multiplexed and transmitted in parallel across these subcarriers.

Because each parallel subcarrier operates at a significantly lower symbol rate than a single-carrier system would require, the physical duration of each symbol is mathematically lengthened. This longer symbol duration, combined with a protective Guard Interval (GI) inserted between transmissions, allows the delayed multipath reflections to dissipate before the receiving hardware samples the next symbol [7, 9]. Consequently, OFDM provides the fundamental physical-layer robustness required to maintain data integrity in highly reflective environments.

3.2.4 Signal Quality and Modulation and Coding Scheme

The determines the transmission rate and physical robustness of the wireless link by combining three parameters [9]:

- **Modulation Type:** The method of encoding digital bits onto the analog radio carrier. It ranges from robust, low-throughput schemes (Binary Phase Shift Keying (BPSK) and Quadrature Phase Shift Keying (QPSK)) to fragile, high-throughput schemes (16- and 64-State Quadrature Amplitude Modulation (QAM)).
- **Coding Rate:** The ratio of payload bits to total transmitted bits, representing the density of Forward Error Correction (FEC). A coding rate of 1/2 maximizes noise resilience but halves throughput, whereas 5/6 maximizes throughput but offers minimal bit-error protection.
- **Guard Interval and Bandwidth:** The channel width (typically 20 MHz or 40 MHz) and the brief inter-symbol pause (Guard Interval) that prevents inter-symbol interference from multipath reflections [9].

To maintain link stability, wireless transceivers employ dynamic rate adaptation algorithms that adjust the MCS based on the SNR and Received Signal Strength Indicator (RSSI) [9]. When a mobile Client moves away from the AP or behind physical obstructions, the SNR drops, raising the Bit Error Rate (BER). The transceiver counters this degradation by dynamically downgrading the MCS level (see Table 3.1).

Table 3.1 Typical IEEE 802.11a/g/n MCS Levels and Rate Adaptation Transitions [10].

MCS Index	Modulation	Coding Rate	Bitrate	Robustness	Required SNR
7	64-QAM	5/6	54 Mbps	Lowest	High (≥ 28 dB)
6	64-QAM	3/4	48 Mbps	Low	~ 24 dB
5	64-QAM	2/3	36 Mbps	Medium-Low	~ 20 dB
4	16-QAM	3/4	24 Mbps	Medium	~ 16 dB
3	16-QAM	1/2	18 Mbps	Medium-High	~ 12 dB
2	QPSK	3/4	12 Mbps	High-Medium	~ 10 dB
1	QPSK	1/2	9 Mbps	High	~ 7 dB
0	BPSK	1/2	6 Mbps	Highest	Low (≤ 5 dB)

Although downgrading the MCS stabilizes the link, it reduces the bitrate. This increases the airtime required to transmit each packet, as expressed by:

$$T_{\text{tx}} \approx \frac{L_{\text{packet}}}{R_{\text{bit}}} + T_{\text{overhead}} \quad (3.1)$$

Downgrading from 54 Mbps (MCS 7) to 6 Mbps (MCS 0) increases the transmission duration by a factor of nine. On a shared, half-duplex medium, this slow transmission consumes excessive airtime and delays other queued frames [9]. Consequently, a drop in signal quality does not immediately manifest as packet loss, but rather as a increase in latency and jitter.

3.2.5 Quality of Service: WMM and AP Bridging

The Access Point acts as a transparent Layer-2 bridge under IEEE 802.1D, mapping wired Media Access Control (MAC) addresses to active wireless Client associations [9]. When receiving a wired frame, the AP strips the 802.3 header, inspects its Layer-2 priority, and encapsulates the payload into an 802.11 frame [9].

To support real-time traffic, standard Wi-Fi employs Wi-Fi Multi-Media (WMM), implementing Enhanced Distributed Channel Access (EDCA) to prioritize traffic into four Access Categories (ACs) [10]. The AP maps PROFINET RT frames (tagged with wired PCP 6) directly to the highest-priority Voice category (AC_VO) [3].

WMM prioritizes traffic by assigning different MAC timing parameters to each queue. Compared to low-priority categories, high-priority queues use a smaller Arbitration Inter-Frame Space Number (AIFSN) and a smaller Contention Window (CW_{min} and CW_{max}), as shown in Table 3.2.

Table 3.2 IEEE 802.11 WMM Access Categories and Default EDCA Parameters [10].

Access Category (AC)	WMM AC	PCP Mapping	AIFSN	CWmin	CWmax
Voice	AC_VO	6, 7	2	3	7
Video	AC_VI	4, 5	2	7	15
Best Effort	AC_BE	0, 3	3	15	1023
Background	AC_BK	1, 2	7	15	1023

This timing configuration guarantees that a device transmitting a PROFINET frame in the AC_VO queue accesses the medium faster and with shorter backoff delays than best-effort traffic. However, WMM is non-preemptive. A high-priority frame cannot interrupt an on-going transmission; it must wait in the AP buffer until another frame finishes transmitting, introducing transient jitter [10].

3.2.6 Wi-Fi Startup and Client Association Sequence

Before transmitting PROFINET cyclic data, the wireless Client (operating as a workgroup bridge) must establish a Layer-2 connection with the AP through a four-phase startup sequence [9]:

1. **Scanning:** The Client locates APs using passive scanning (listening for periodic 100 ms Beacons) or active scanning.
2. **Authentication:** The Client exchanges Layer-2 authentication frames with the selected AP.
3. **Association:** The Client sends its supported data rates in an Association Request. The AP registers the Client, responds with an Association Response, and opens the Layer-2 bridge.
4. **Security Handshake:** If enabled, the devices execute a WPA2 4-way handshake to generate encryption keys (Pairwise Transient Key (PTK) and Group Temporal Key (GTK)).

Once this Layer-2 bridge is active, the IO-Controller can discover the IO-Device, assign addressing, and establish the cyclic AR.

3.2.7 Wi-Fi Handover

In mobile applications, a wireless Client must roam between adjacent APs to maintain coverage. Standard Wi-Fi employs a "Break-before-Make" model, which interrupts real-time control loops during the handover process [8]:

1. **Roaming Trigger:** The Client initiates a handover when the RSSI of its current AP drops below a threshold (typically -70 dBm to -75 dBm) [7].
2. **Scanning Jitter:** To locate candidates, the transceiver tunes away from its current channel to scan other frequencies. During this scanning period (often ≥ 100 ms), the Client cannot transmit or receive data, buffering and delaying all frames [9, 10].
3. **Handover Gap:** The Client disassociates from the old AP and executes the authentication, association, and WPA2 handshake on the new AP. This total communication gap ranges from 50 ms (with IEEE 802.11r Fast Transition) to over 300 ms (standard WPA2-PSK) [8–10].

If the scanning jitter and handover gap exceed the PROFINET watchdog threshold, the controller declares a communication loss. This trips the safety loop, triggering a Fail-Safe state. Eliminating or masking this handover gap represents the primary engineering challenge for wireless industrial Ethernet.

3.3 Industrial Bluetooth (IEEE 802.15.1)

Industrial Bluetooth (IEEE 802.15.1) offers a reliable, scheduled link and frequency agility. This thesis evaluates Bluetooth 2.1 + EDR (Enhanced Data Rate) with a 3 Mbps capacity [11] operating under the Personal Area Networking (PAN) profile.

3.3.1 Medium Access: Time Division Multiple Access (TDMA)

Bluetooth utilizes a TDMA architecture, dividing the wireless medium into discrete, sequential time slots [12]. Figure 3.3 illustrates a 4-slot sequence used to divide the medium. Under the PAN profile, the Network Access Point (NAP) acts as the master and schedules transmissions for all connected Client devices, known as PAN Users (PANUs) [13]:

- **Point-to-Point (1 NAP, 1 PANU):** The NAP and Client alternate transmissions in dedicated $625\mu\text{s}$ slots. This topology maximizes bandwidth and eliminates polling-induced jitter [12, 13].
- **Point-to-Multipoint (1 NAP, Multiple PANUs):** The NAP distributes time slots across all active PANUs. Each Client must remain silent until explicitly polled, which increases latency and jitter as the Client count rises [12, 13].

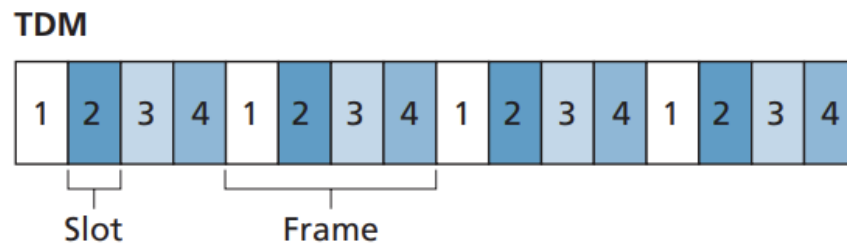


Figure 3.3: Visual representation of the TDMA slot and frame structure, showing a 4-slot sequence used to divide the shared medium [7].

This scheduled polling mechanism eliminates the hidden node problem and random backoff delays, providing superior jitter performance compared to contention-based Wi-Fi.

3.3.2 Interference Mitigation: Adaptive Frequency Hopping (AFH) and Frequency Hopping Spread Spectrum (FHSS)

To maintain link stability in the shared 2.4 GHz band, Bluetooth employs AFH, building upon FHSS [12]. FHSS rapidly switches the carrier frequency across 79 discrete 1 MHz channels up to 1600 times per second. This rapid hopping ensures that narrow-band interference only disrupts a single slot rather than breaking the connection.

AFH introduces dynamic spectrum awareness to FHSS by monitoring channel quality and blacklisting heavily interfered channels, such as those used by nearby Wi-Fi networks. The hopping sequence dynamically avoids these frequencies, utilizing only clean channels (see the spectrum map in Figure 3.4). This active evasion ensures that retransmissions occur on a clean frequency, minimizing packet loss and watchdog timeouts [12].

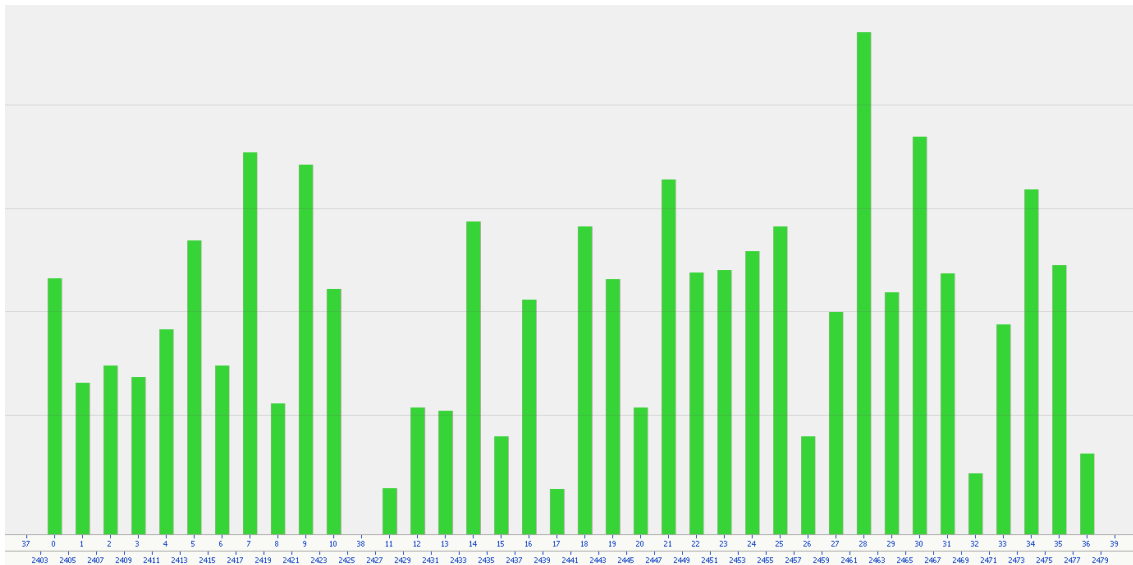


Figure 3.4: Example of a Bluetooth AFH spectrum map. The gaps in the channel distribution represent interfered frequencies that have been blacklisted to ensure reliable transmission [12].

3.3.3 Ethernet Bridging: BNEP and L2CAP Encapsulation

To achieve compatibility with Ethernet-based protocols like PROFINET, the PAN profile utilizes the Bluetooth Network Encapsulation Protocol (BNEP) [14]. BNEP performs Layer-2 translation, encapsulating standard Ethernet frames into Bluetooth packets. This encapsulation allows the NAP to act as a transparent Layer-2 bridge. The Ethernet stack treats the wireless link as a standard point-to-point Ethernet connection, requiring no protocol modifications.

Underlying this encapsulation process is the Logical Link Control and Adaptation Protocol (L2CAP). While BNEP handles the Ethernet-to-Bluetooth payload translation, L2CAP is responsible for adapting these potentially large payloads for the Bluetooth baseband layer [12]. Standard Ethernet frames can be up to 1500 bytes (or more with certain tags), which significantly exceeds the maximum payload capacity of underlying Bluetooth baseband packets. L2CAP bridges this gap by performing Segmentation and Reassembly (SAR) operations. It fragments the large, BNEP-encapsulated Ethernet frames into smaller packets for transmission over the air, and reassembles them at the receiving end. Furthermore, L2CAP provides protocol multiplexing, allowing the BNEP traffic to share the Bluetooth link seamlessly with other Bluetooth services.

Figure 3.5 and Figure 3.6 illustrate the BNEP protocol stack, including its position above L2CAP, and the frame encapsulation respectively.

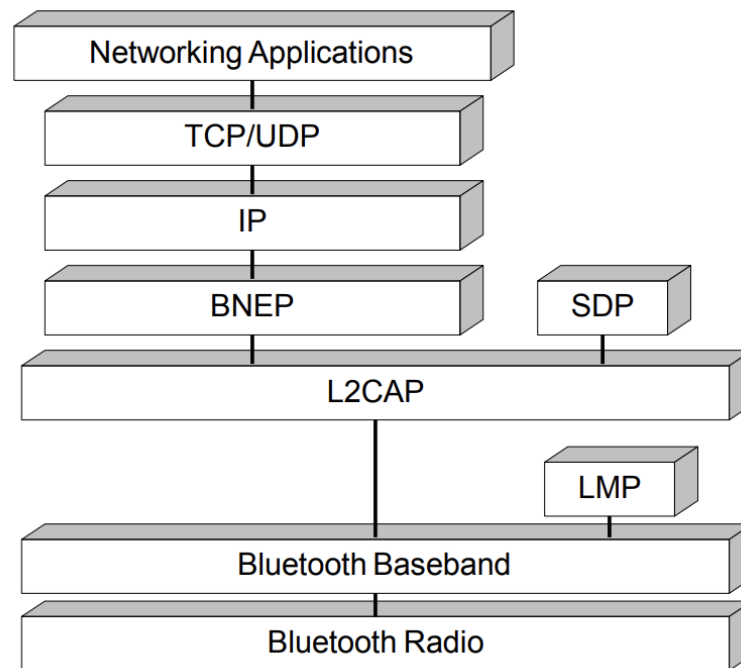


Figure 3.5: Bluetooth Protocol Stack including the BNEP layer [14].

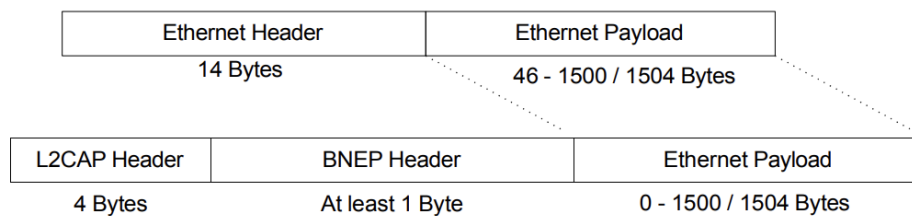


Figure 3.6: Encapsulation of Ethernet frames into BNEP [14].

3.3.4 Mobility Constraints

Unlike Wi-Fi, Bluetooth operating under the PAN profile lacks native fast-roaming mechanisms. If a PANU loses connection to its NAP, re-establishing a connection on a new NAP can take several seconds [13]. Because this delay far exceeds the PROFINET watchdog threshold, standard Bluetooth remains restricted to stationary "cable replacement" links or mobile units operating within the coverage radius of a single NAP.

Chapter 4

Field Audit: Performance Analysis of the Existing Infrastructure

To establish the solution's requirements, the reliability of ArcelorMittal Ghent's existing wireless infrastructure was quantified. Due to the frequent Watchdog Timeouts, leading to Fail-Safe states, this existing single-path network is currently deemed unsuitable for PROFINET RT data transmission to support automated crane operations. This chapter presents a data-driven audit correlating wireless network performance with industrial process timeouts to identify the network's current shortcomings.

4.1 Data Acquisition and Methodology

The ArcelorMittal Ghent automation department provided the audit data from two independent sources: wireless infrastructure management logs and PDC systems.

4.1.1 Operational Context and Subject Crane

The audit focuses on a single crane operating between two other cranes on the same runway. This configuration impacts the wireless analysis because:

- **Dynamic Obstructions:** The adjacent cranes are massive, mobile metallic structures that cause significant transient shadowing.
- **Restricted Line-of-Sight (LoS):** These cranes frequently block direct signal paths to the stationary Access Points (APs).

Analyzing network performance under these "sandwiching" conditions defines the solution's requirements.

4.1.2 Wireless Infrastructure Logs (Client Snapshots)

The Client modules' internal logging mechanism generates network data by polling at one-minute intervals. Each "snapshot" contains:

- **Timestamp:** The exact moment of polling.
- **Associated Access Point:** The unique identifier of the active AP.
- **Bitrate:** The actual link speed in Mbps.
- **Signal Strength (RSSI):** The Received Signal Strength Indication in dBm.

4.1.3 Process Data Control and Communication Logs

The PDC servers log the communication heartbeat between the control room and the crane. Captured at sub-second intervals, this dataset contains:

- **Watchdog and Communication Status:** Boolean flags indicating the health of the connection.
- **Positional Data (X, Y and Z Coordinates):** The physical position of the crane and its load.

4.1.4 Packet-Level Diagnostics

Packet-level diagnostics were conducted using an Allegro Network Multimeter and a PROFINET Inspector connected via a Network TAP just before the primary PLC.

The TAP captured all local crane traffic, making it impractical to isolate the specific packets dropped across the wireless link. However, the captures confirmed a critical network behavior: dynamic shadowing rarely causes a complete 802.11 disconnection. Instead, the metallic obstructions degrade the SNR, forcing the radio to downshift to a lower MCS.

This throttling introduces sufficient jitter to violate strict PROFINET RT Watchdog timers, ultimately triggering the Fail-Safe states.

4.2 Long-term Reliability and Spatial Distribution (Annual Data)

Annual data was analyzed to determine network stability and whether static coverage gaps or dynamic environmental factors caused Watchdog Timeouts

4.2.1 Annual Watchdog Timeout Frequency and Operational Impact

The dataset revealed 1,394 communication Watchdog Timeouts over 12 months (averaging 4 timeouts daily). This forces the crane into a Fail-Safe state multiple times a day, halting production and requiring constant manual intervention. Consequently, the current network cannot support automated crane operation.

4.2.2 Spatial Mapping of Fail-safe States

Correlating Fail-Safe states with XY-coordinates visualizes the Fail-Safe state distribution, determining whether Fail-Safe triggers cluster locally or spread across the entire working area.

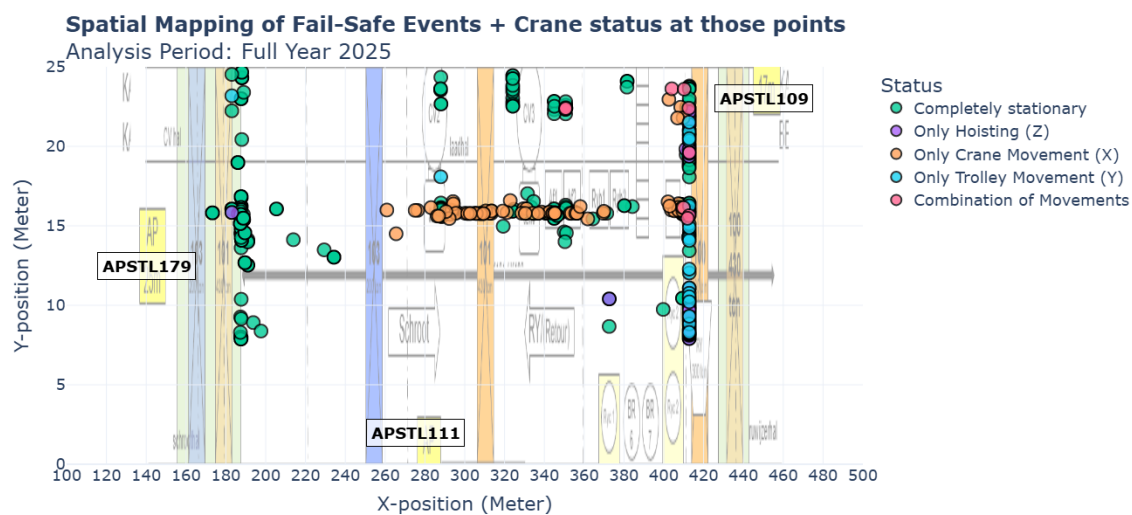


Figure 4.1: Annual spatial distribution of Fail-Safe states.

4.2.3 Conclusion: Static Obstructions vs. Dynamic Shadowing

Figure 4.1 shows Watchdog Timeouts occurring across the crane's entire trajectory. While some cluster around permanent physical obstructions like convector units (creating static "dead zones"), many occur outside these zones.

Automated operations require continuous connections; even a brief transit through a static dead zone can trigger a Fail-Safe state. However, the widespread timeouts indicate a more systemic issue: Dynamic Shadowing. The two adjacent metallic cranes constantly block the Line-of-Sight (LoS) to Access Points, creating moving shadows. This was also confirmed by the packet-level diagnostics and justifies the redundant architecture proposed in Chapter 5: while better AP placement might fix a static cluster, only a secondary, physically isolated path can mask unpredictable shadowing from mobile cranes.

4.3 Daily Case Study: Spatial, Link Dynamics and Handover Pathologies

To identify specific technical Fail-Safe triggers, data from a single operational day (24/03/2026) was analyzed. The PDC recorded three Watchdog Timeouts triggering Fail-Safe states. Correlating these events with diagnostic snapshots reveals two primary bottlenecks: bitrate collapse from environmental shadowing and inefficient handover behavior.

4.3.1 Spatial Distribution of Daily Watchdog Timeouts

Three Watchdog Timeouts are plotted on the crane's physical coordinates in Figure 4.2. These points serve as a specific case study for the Fail-Safe states occurring during a standard operational shift.

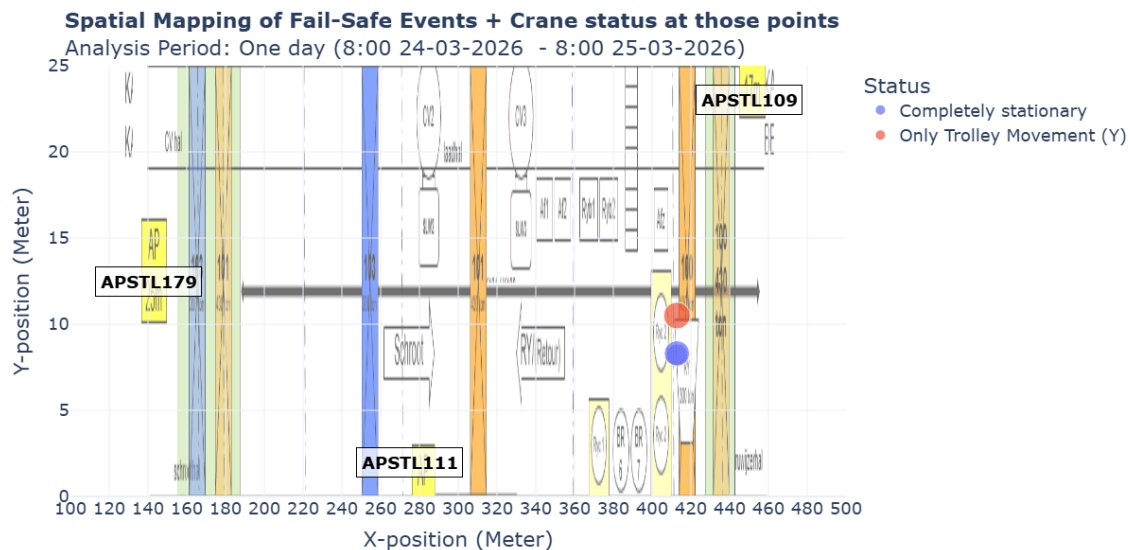


Figure 4.2: Spatial distribution of Fail-Safe states on 24/03/2026.

Mapping these three Watchdog Timeouts reveals a critical geometric weakness:

- **End-of-Track Shadowing:** All three Watchdog Timeouts occurred near the runway's end. Here, an adjacent stationary crane acts as a metallic shield between the subject crane and the active AP.
- **Line-of-Sight Blockage:** This creates a permanent "shadow pocket." The first Watchdog timeout (20:30) and the subsequent pair (07:00) all occurred within this zone.

Because parking a crane at the track's end predictably severs another crane's signal, a single wireless link is insufficient. A redundant setup with an alternative LoS is the only way to "see around" the obstructing crane.

4.3.2 Physical Layer Analysis: RSSI vs. Bitrate Discrepancy

A decoupled RSSI and Bitrate indicates a degraded link. While an RSSI of -65 dBm to -72 dBm theoretically supports stable communication, the daily logs show the bitrate frequently collapses to 24 Mbit/s or lower.

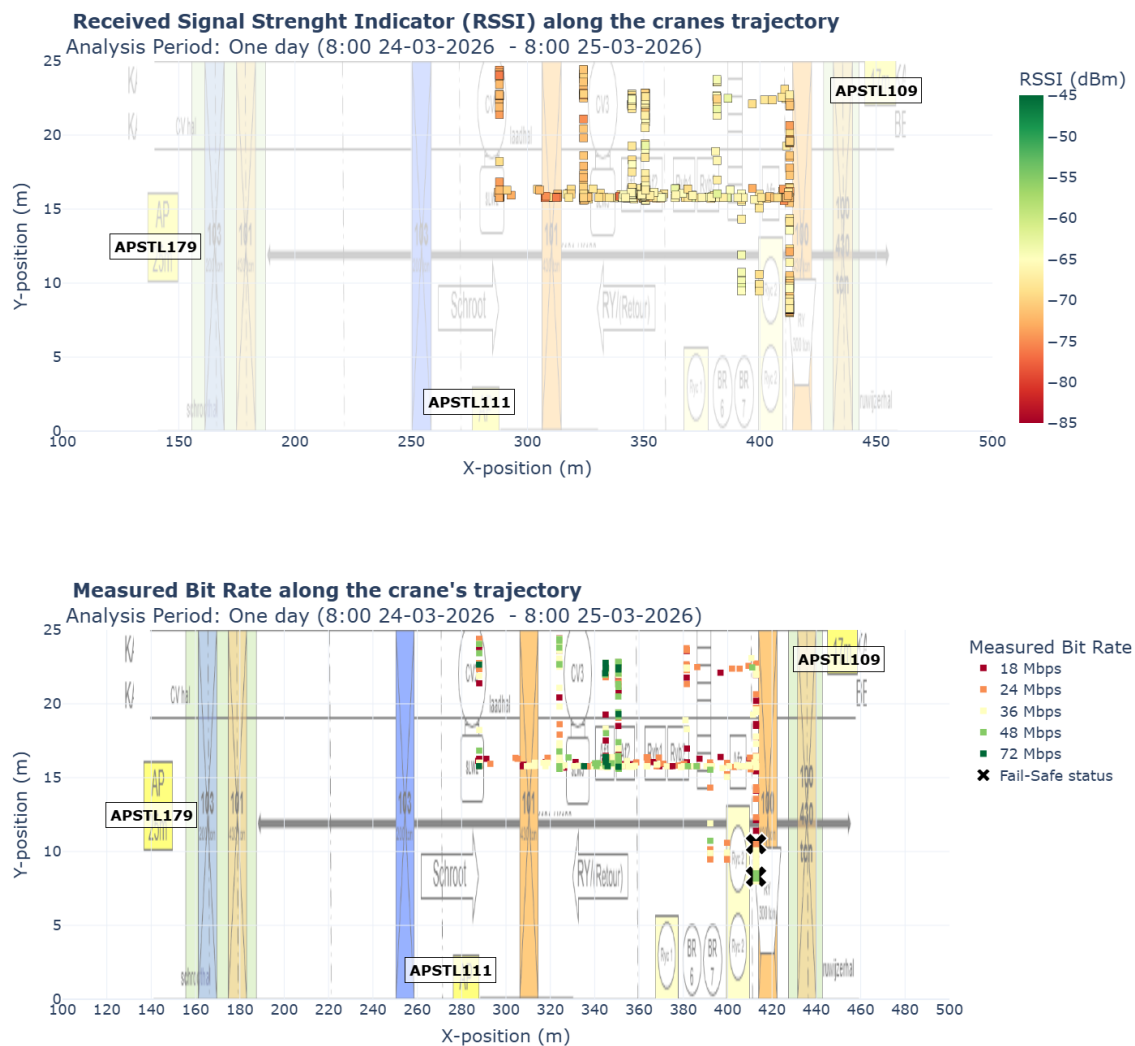


Figure 4.3: Correlation between Received Signal Strength (left) and Bitrate (right) during the audit day.

Figure 4.3 shows stable signal strength alongside fluctuating bitrates on certain moments. Varying noise levels (e.g. due to EMI, co-channel interference, multipath scattering, shadowing, etc.) and signal distortion force the Client to downshift its MCS, leading to lower bitrates. When this lower bitrate cannot handle PROFINET RT traffic within Watchdog timing constraints, a Fail-Safe state is triggered.

4.3.3 Handover Analysis: Roaming Thresholds and the "Sticky Client"

Figure 4.4 reveals a "Sticky Client" phenomenon: the wireless module remains associated with APSTL109 for disproportionately long periods, despite moving closer to other APs.

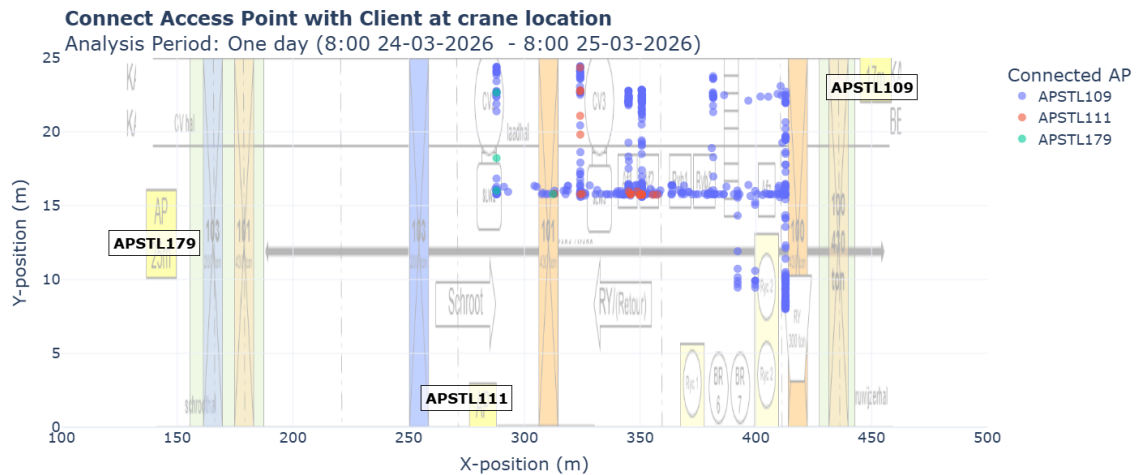


Figure 4.4: Access Point association relative to crane location. The data shows a persistent connection to APSTL109, indicating a reluctance to proactively switch to closer APs.

The handover events of that day are shown in Table 4.1.

Table 4.1 Log of Handover Events on 24/03/2026.

Time	Target AP	Previous AP	RSSI (Target AP)	X-Position (m)
09:39:15	APSTL111	APSTL109	-72 dBm	350.73
10:20:22	APSTL109	APSTL111	-68 dBm	412.85
15:41:18	APSTL111	APSTL109	-74 dBm	350.72
15:43:18	APSTL109	APSTL111	-70 dBm	324.11
15:51:19	APSTL109	APSTL111	-63 dBm	385.95
17:07:20	APSTL179	APSTL111	-69 dBm	287.96
17:11:20	APSTL109	APSTL179	-67 dBm	382.05
20:55:21	APSTL111	APSTL109	-69 dBm	348.82
21:00:33	APSTL109	APSTL111	-66 dBm	382.67

Figure 4.4 and Table 4.1 illustrates a deliberate configuration strategy. To avoid "ping-ponging" between APs and introducing communication delays, the system delays handovers until the signal reaches a critically low threshold.

While this strategy limits handovers, it forces the Client to be "Sticky" and to operate at the cell fringe. As seen in Table 4.1, when the handover occurs, the new connection often

starts with an already degraded signal (here, ranging from -66 dBm down to -74 dBm).

When the Client is clinging to a distant AP, a shadowing event can be enough to instantly crash the bitrate and trigger a Watchdog timeout.

This trade-off between handover delay and shadowing vulnerability proves a single-path network cannot support automation.

A dual-Client setup eliminates this conflict: one Client maintains a stable link while the secondary Client performs a seamless, masked handover in the background.

4.3.4 The necessary improvements for a solution

Both analyses confirm the single-path network cannot support automated crane operation. The proposed redundant architecture must integrate three critical improvements:

1. **Spatial Path Redundancy:** Single links are vulnerable to static dead zones and dynamic shadowing. A redundant, physically isolated path ensures an alternative LoS remains available when one path is obstructed.
2. **Handover Masking:** Single clients compromise between handover delays and vulnerable cell fringes. Two independent clients allow one to maintain a strong link while the other roams, masking handover latency.
3. **Bitrate Stabilization:** Fluctuating bitrates, caused by varying SNR, force the radio into lower MCS. Running two simultaneous, spatially diverse links allows the PRP modules to leverage the connection with the best SNR.

Chapter 5

Architectures & Implementation

To address the fluctuating bitrate, shadowing and handover reliability issues at ArcelorMittal Ghent, multiple laboratory testbeds are developed. These setups evaluate the suitability of different wireless technologies and network architectures for carrying PROFINET traffic under different conditions.

1. **Exploratory Setup: Redundant Point-to-Multipoint Bluetooth.** This configuration proved non-viable due to MAC flapping and the inability to implement VLANs.
2. **Setup 1: Redundant Point-to-Point Bluetooth Architecture.** This setup utilizes PRP over two independent Bluetooth point-to-point links to physically isolate the duplicated traffic.
3. **Setup 2: Redundant Wi-Fi Architecture.** This setup utilizes PRP to duplicate traffic over two separate Wi-Fi channels, which are VLAN-isolated.

5.1 Hardware of the Architectures

The experimental implementations utilize a selection of industrial automation and networking hardware, divided into three categories: components common to both setups, hardware unique to the redundant Wi-Fi solution, and hardware unique to the redundant Bluetooth solution.

5.1.1 Common Components

Both experimental setups use, where possible, the same base infrastructure to ensure that performance differences are not caused by different hardware components.

- **IO-Controller:** A **Siemens SIMATIC S7-1516-3 PN/DP** (CPU 1516) acts as the central logic unit, managing the cyclic exchange of PROFINET frames.

- **Distributed I/O (Remote Stations):** One or more IO-islands are used. The following IO modules are available and used in the lab:
 - **Siemens SIMATIC ET 200SP**
 - **Siemens SIMATIC ET 200S**
- **PRP Redundancy Boxes (RedBoxes):** Two "Phoenix Contact FL RED 2003E PRP" modules provide the necessary PRP logic to duplicate and discard frames according to the IEC 62439-3 standard.

5.1.2 Wi-Fi-Specific Hardware

The hardware utilized in the redundant Wi-Fi setup, detailed in Section 5.4, is:

- **Access Points:** Two **Siemens SCALANCE W788 2RR** units were used as wireless access points to create two separate cells.
- **Client Modules:** Three **Siemens SCALANCE W747 1RR** units were used as wireless bridges on the "Mobile" side of the link.
- **Managed Switch:** A **Siemens SCALANCE XC208** managed switch provides the wired infrastructure, supporting the VLAN tagging and trunking required for path isolation.

5.1.3 Bluetooth-Specific Hardware

The Bluetooth Point-to-Point setup and the Point-to-Multipoint setup make use of industrial Bluetooth 2.1 + EDR links. The hardware utilized for these setups, detailed in Section 5.3 and Section 5.2, is:

- **Network Access Points:** Two **Phoenix Contact FL EPA 2** modules, acting as APs, manage the Bluetooth connections.
- **PAN User Clients:** Two **Phoenix Contact FL BT EPA 2** modules, acting as the clients, bridge Ethernet traffic over the Bluetooth link via the BNEP.
- **Switch (Used in Exploratory Setup):** A **Siemens SCALANCE XC208**, used in the point-to-multipoint setup as an unmanaged bridge.

5.1.4 Equipment Summary

Table 5.1 provides a concise summary of the used manufacturers and models.

Table 5.1 Summary of industrial components used in the experimental setups.

Component Type	Model / Manufacturer	Application
PLC	Siemens SIMATIC S7-1516-3 PN/DP	Common
IO Devices	Siemens SIMATIC ET 200SP / ET 200S	Common
Redundancy Box	Phoenix Contact FL RED 2003E PRP	Common
Switch	Siemens SCALANCE XC208	Wi-Fi and BT Setup
Wi-Fi Access Point	Siemens SCALANCE W788 2RR	Wi-Fi Setup
Wi-Fi Client	Siemens SCALANCE W747 1RR	Wi-Fi Setup
Bluetooth NAP	Phoenix Contact FL EPA 2	BT Setup
Bluetooth PANU	Phoenix Contact FL BT EPA 2	BT Setup

5.2 Exploratory Setup: Point-to-Multipoint Bluetooth

Prior to the isolated redundant architectures, a Point-to-Multipoint Bluetooth configuration was evaluated. The rationale for this topology was to leverage the capabilities of the Bluetooth modules, which allow a single NAP to maintain connections with up to seven PANUs. This approach would simplify increasing the range by eliminating the need for cascaded RedBoxes or VLANs.

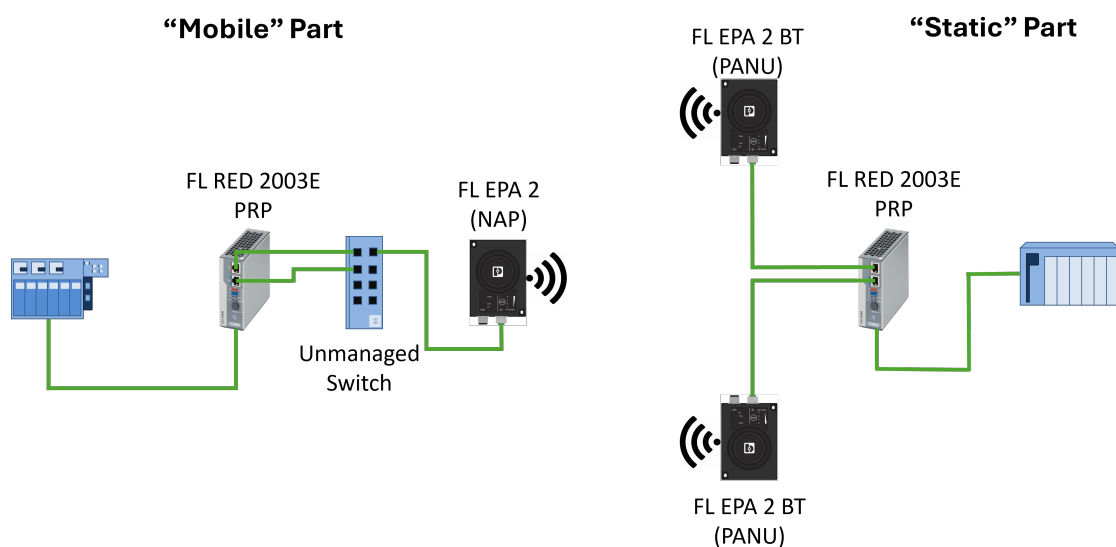


Figure 5.1: Topology of the exploratory Point-to-Multipoint Bluetooth setup. Both redundancy lines are funneled through an unmanaged switch and transmitted via a single NAP.

5.2.1 Physical Topology and Unmanaged Bridging

As illustrated in Figure 5.1, this setup abandons physical path isolation and uses a shared transmission medium.

- **"Mobile" Part:** The IO-Devices are connected to the a RedBox. Both the LAN A and LAN B redundancy ports of this RedBox are wired directly into the unmanaged switch. The switch merges these physical connections and routes them to a single NAP.
- **"Static" Part:** Along a trajectory, multiple PANUs (two in this baseline laboratory test) are connected to the central NAP. These PANUs are wired into the respective LAN A and LAN B ports of a Redbox, which in turn connects to the PLC.

5.2.2 Intended Data Flow and PRP Discard Logic

The redundancy paths are physically combined before transmission and separated upon reception. Because there is no logical isolation, the system relies entirely on the PRP's native error-handling at the receiving end.

The intended data flow and PRP discard logic of this setup is as follows:

1. The sending RedBox duplicates the PROFINET frame and transmits the copies simultaneously out of its LAN A and LAN B ports.
2. The unmanaged switch receives both frames, merges the traffic, and forwards it to the single NAP.
3. The NAP broadcasts the mixed traffic stream to all connected PANUs.
4. The PANUs receive the mixed frames and forward them to their physically connected ports on the receiving RedBox. Because the traffic is at the switch, the RedBox inevitably receives Line A frames on its LAN B port, and Line B frames on its LAN A port.
5. The receiving RedBox applies its PRP logic: it accepts the first uncorrupted frame it receives, actively discards the other duplicate frame, and discards any frames bearing the incorrect line identifier.
6. The single surviving frame is then successfully forwarded to the end device, closing the control loop.

While this setup succeeds in delivering the cyclic data, combining the PRP lines through an unmanaged switch fundamentally compromises the isolation of the network. Section 6.3.2 presents a detailed technical analysis of the resulting network behavior.

5.3 Setup 1: Redundant Point-to-Point Bluetooth Architecture

This setup evaluates the use of the PRP over Bluetooth 2.1 + EDR links operating in the 2.4 GHz band. This architecture achieves physical isolation by using point-to-point connections. This Bluetooth setup is only intended for the PROFINET control loop. No standard IT or management traffic is routed over these redundant links.

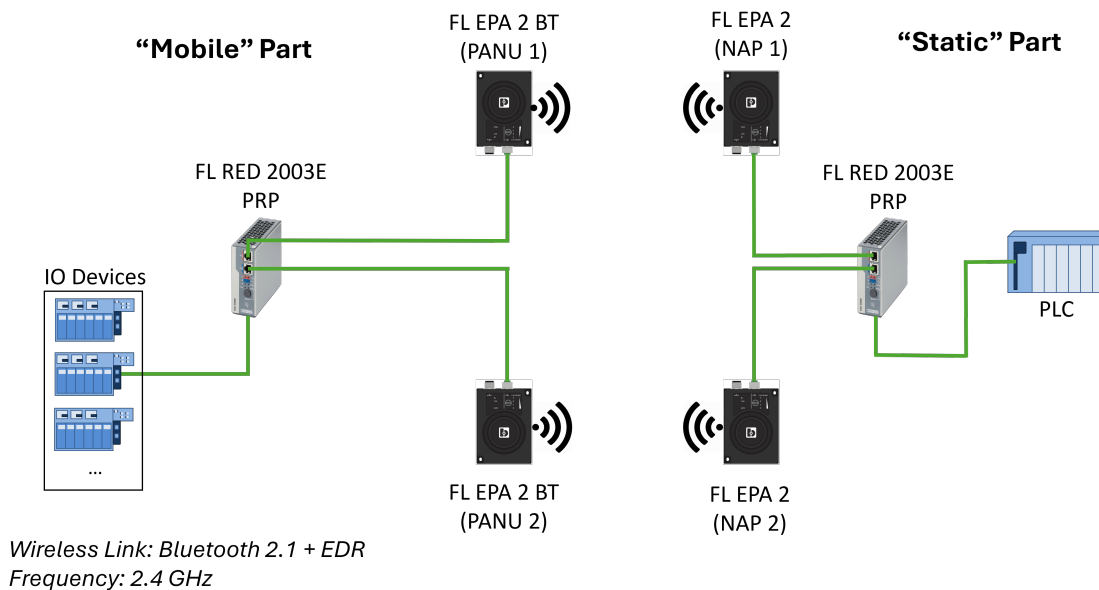


Figure 5.2: Topology of the Point-to-Point Bluetooth setup. Absolute path isolation is achieved without VLANs by physically separating the Line A and Line B networks between the "Static" and "Mobile" parts.

5.3.1 Physical Isolation and Device Configuration

Figure 5.2 illustrates the physical isolation strategy used in this setup. Because the hardware uses point-to-point links, there is no need for any VLAN configurations, which is not possible with Bluetooth.

This isolation is achieved as follows:

- **Line A (Network A):** The LAN A port of the RedBox on the "Static" side is connected to the first NAP.
- **Line B (Network B):** The LAN B port is connected to the second NAP.

The two NAPs establish two independent Bluetooth links, one for each PRP line. On the "Mobile" side, a PANU Client is deployed for each PRP line that connects to the NAP on the same PRP line. Each PANU is locked to its respective NAP by configuring it to connect exclusively to the device name of that target NAP.

To minimize transmission jitter and prioritize the cyclic real-time data exchange, the "Optimize for PROFINET" function of Phoenix Contact was enabled in the lab setup.

5.3.2 Scalability and Range Extension (Cascaded PRP)

Extending this setup's range requires a different approach than Wi-Fi. Bluetooth modules do not support handover between multiple NAPs.

To provide continuous coverage along a "Mobile" device's track, the architecture must be scaled by adding multiple Bluetooth zones. As shown in Figure 5.3, integrating these extra zones without creating network loops requires a Cascaded PRP topology.

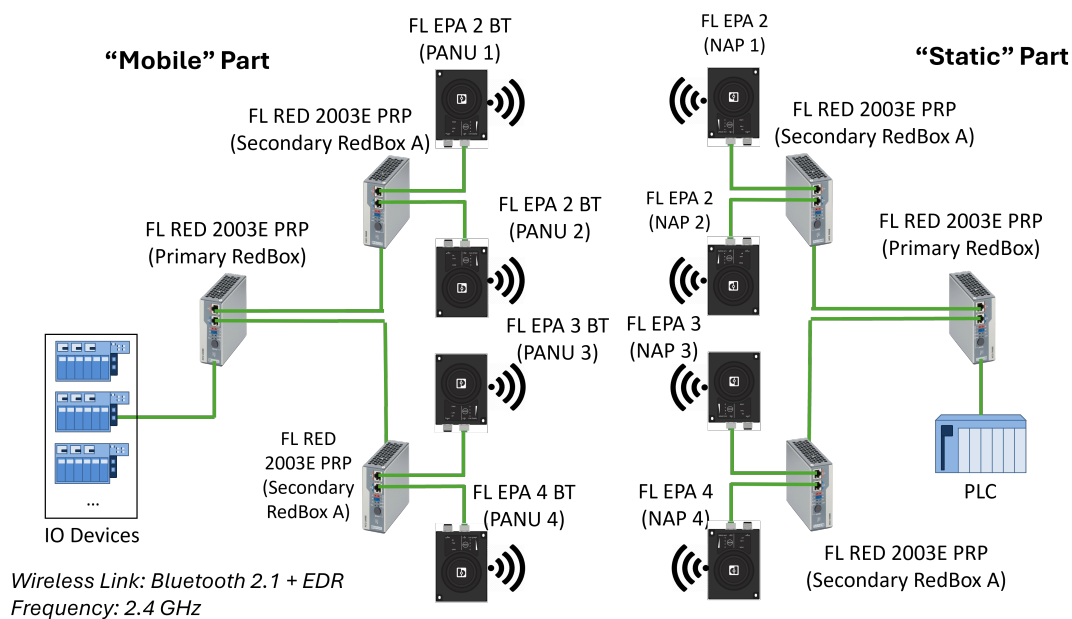


Figure 5.3: Extended range topology utilizing Cascaded PRP. Multiple RedBoxes are nested to combine multiple point-to-point Bluetooth zones along the track.

The primary RedBox on the "Static" side duplicates the cyclic data into Line A and Line B. However, instead of going to the wireless modules, these lines feed into a secondary tier of RedBoxes:

- **Secondary RedBox A ("Static" side):** Takes Line A and duplicates it, feeding **NAP 1** and **NAP 2**.
- **Secondary RedBox B ("Static" side):** Takes Line B and duplicates it, feeding **NAP 3** and **NAP 4**.

On the "Mobile" side, four corresponding PANUs receive the signals from the various zones along the track. These PANUs feed into secondary mobile RedBoxes, which discard the

duplicates within their specific zones, before passing the surviving frames to the primary RedBox on the "Mobile" side.

As the "Mobile" device moves down the track, it leaves the coverage zone of the first NAP and enters the next one. These overlapping coverage zones ensure that at least one of the links is active at any given moment, allowing the cascaded RedBoxes to seamlessly manage the handover.

5.3.3 Bidirectional Communication

This Bluetooth architecture operates fully bidirectionally. When the remote IO-Devices send cyclic response data back to the PLC, the logic operates in reverse:

1. The (primary) RedBox on the "Mobile" side duplicates the outgoing frame to the secondary mobile RedBoxes or PANUs.
2. In case of cascaded PRP, the secondary RedBoxes duplicate the frames again, transmitting them across all active links.
3. The NAPs on the "Static" side receive the BNEP payloads, unpack them into standard Ethernet frames, and pass them to the static RedBox(es).
4. In case of cascaded PRP, the secondary static RedBoxes discard the duplicates and forward the frames to the primary static RedBox.
5. The primary static RedBox discards the final duplicate, forwarding a single frame to close the PROFINET cycle.

5.4 Setup 2: Redundant Wi-Fi Architecture

This setup evaluates the use of PRP over a standard 802.11 5 GHz wireless infrastructure. Because Wi-Fi is a shared, contention-based medium, transmitting duplicate PRP frames over the same physical airspace would result in MAC flapping and network loops.

To ensure the two redundant paths remain isolated, this setup relies on logical isolation using IEEE 802.1Q Virtual Local Area Networks (VLANs).

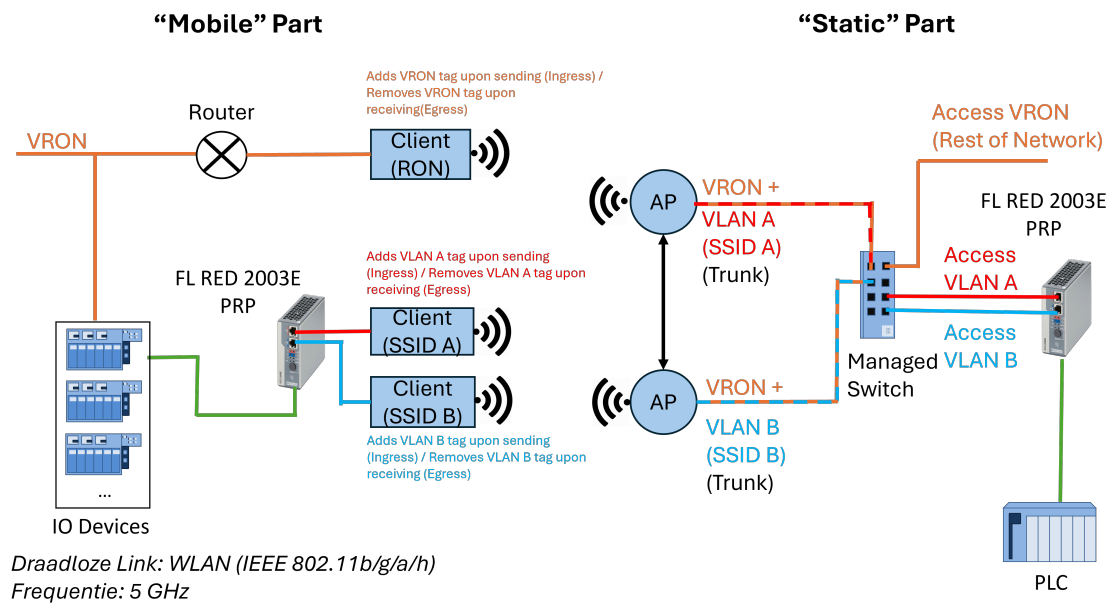


Figure 5.4: Logical and physical topology of the Wi-Fi setup with two APs used in the lab.

5.4.1 Logical Isolation and VLAN Configuration

Figure 5.4 shows that the isolation is handled by a managed switch. The switch separates the traffic into distinct VLANs:

- **VLAN A (Network A):** Dedicated VLAN for the first redundant PRP path (Line A), (VLAN 10 in lab implementation).
- **VLAN B (Network B):** Dedicated VLAN for the second redundant PRP path (Line B), (VLAN 20 in lab implementation).
- **VRON (VLAN Rest of Network):** A dedicated VLAN for IT or management traffic, isolated from the control loop (VLAN 30 in lab implementation).

When the RedBox on the “Static” side duplicates a frame, it sends one copy out of its LAN A and one out of its LAN B port. The managed switch is configured to apply a VLAN A tag to frames entering from LAN A, and a VLAN B tag to frames entering from LAN B. Once tagged by the switch, the frames are routed via trunk ports to the Access Points. The trunk ports are configured to combine the traffic of the VRON VLAN with the traffic of the redundant line VLANs (VLAN A or VLAN B). This ensures that a single physical cable can carry both traffic types.

5.4.2 Wireless Routing and Virtual Access Points (VAPs)

To maintain the isolation over the air, the physical APs are configured using VAPs.

The VAP allows a single physical hardware unit to logically partition its wireless interface into multiple virtualized networks. Within each physical AP, separate VAPs are instantiated and configured to broadcast isolated Service Set Identifiers (SSIDs):

- **VAP 1 (SSID A):** Logically mapped to VLAN A, handling the first PRP redundancy path (Line A).
- **VAP 2 (SSID B):** Logically mapped to VLAN B, handling the second PRP redundancy path (Line B).
- **VAP 3 (VRON):** Logically mapped to the VRON network for non-critical communication.

The architecture is designed to scale across larger environments as shown in Figure 5.5. To provide continuous coverage along an physical track, multiple APs can be deployed. However, when extending, spatial and frequency planning becomes critical. A Client must be prevented from forming simultaneous active connections with multiple APs broadcasting the same SSID. If this occurs, it creates a bridging network loop, leading to broadcast storms. To guarantee a single-path association, the following strategies can be implemented during deployment:

- **Static Channel Planning:** Overlapping APs must be locked to non-overlapping 5 GHz frequency channels. This prevents co-channel interference and ensures that the Client actively shifts its radio frequency to roam, breaking the connection before establishing the new one.
- **Cell Sizing:** The transmission power of each AP can be lowered to reduce its radius or APs can be placed in such a way to create bounded cells. This ensures the Client drops the previous connection just as it enters the range of an AP broadcasting the same SSID.

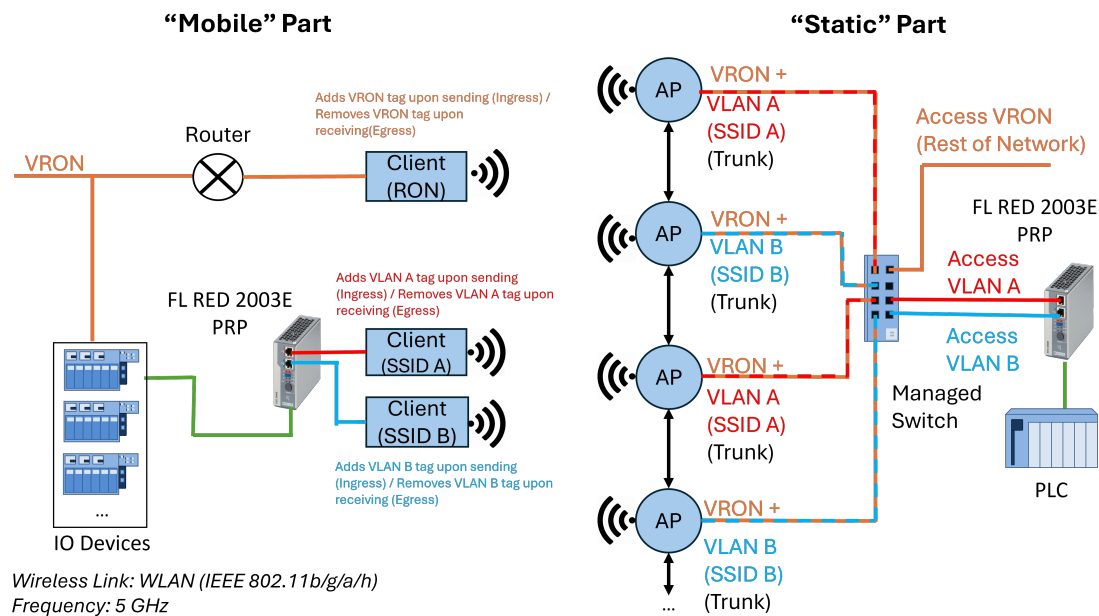


Figure 5.5: Logical and physical topology of the Wi-Fi setup with more than 2 APs to increase the working range of the "Mobile" part.

On the "Mobile" side, Client modules are deployed:

- **Client (SSID A)** associates only with the VAP broadcasting SSID A, ensuring it only receives VLAN A traffic.
- **Client (SSID B)** associates only with the VAP broadcasting SSID B, ensuring it only receives VLAN B traffic.
- **Client (RON)** connects to the VRON VAP for non-critical communications (e.g., camera feeds or maintenance access).

The clients independently roam between the APs. Because the infrastructure APs for Line A and Line B are physically offset, their coverage boundaries do not perfectly align. Consequently, a handover event for each Client occurs at different physical locations and times.

PRP masks these temporary disconnections, creating a seamless handover. While one Client scans, authenticates, and associates with a new AP, the PROFINET cycle remains uninterrupted because a duplicate frame arrives via the other path. When frames are received by the clients, they remove the VLAN tags before forwarding the untagged frames to the respective ports of the RedBox. The RedBox accepts the first arriving frame and discards the duplicate.

This communication process is fully bidirectional, as follows:

1. The RedBox at "Mobile" side duplicates the outgoing frame onto its LAN A and LAN B ports.

2. The respective wireless clients add the appropriate VLAN tags (VLAN A or VLAN B) upon ingress and transmit them to the APs.
3. The managed switch receives the frames via the trunk ports and strips the VLAN tags before routing them out of the designated access ports.
4. The untagged frames arrive at the RedBox at "Static" side , which discards the duplicate and forwards the surviving frame, closing the control loop.

5.5 Measurement Methodology and Diagnostic Infrastructure

To evaluate the performance and reliability of the architectures, a high-fidelity measurement infrastructure was required. Timing variations in the microsecond range can determine the stability of the control loop. Standard network interfaces and software-based capturing methods are insufficient due to the non-deterministic nature of operating system interrupt handling. This section details the rationale behind the selection and placement of measurement hardware and the subsequent analytical workflow.

5.5.1 Diagnostic Hardware and Placement Rationale

The measurement setup utilized a combination of specialized analyzers, passive components, and laboratory instruments, each selected for a specific role in the data acquisition and validation process.

- **Hilscher netAnalyzer (End-to-End (E2E) Latency Calculation):** The netAnalyzer was deployed at the boundaries of the network (between the PLC and the "Static" RedBox, and between the IO-Device and the "Mobile" RedBox).

Rationale: This device served as the primary source for quantitative results. Its hardware-generated 10 ns resolution timestamps provided the precision required for calculating the E2E latency. By capturing frames before they entered the redundancy infrastructure and after they exited, the total system delay was isolated.

- **Siemens Bus Analyzer (BANY) (PRP Verification):** The BANY was positioned to capture both independent lines of the PRP setup (Line A and Line B) simultaneously.

Rationale: Its primary role was to verify the internal logic of the Redboxes. By capturing the traffic on both redundant paths, it allowed for the verification of duplicate generation and the arrival order of frames. Furthermore, it was utilized for active packet insertion to test the network under load.

- **Digital Oscilloscope (Physical Validation):** A high-bandwidth oscilloscope was utilized to perform physical-layer measurements of the network signals.

Rationale: The oscilloscope served as a validation tool. By measuring the time delta between the transmitted frame and the received frame, the results obtained from the Python-based PCAP analysis could be empirically verified.

- **Siemens Ethernet TAPs (Functional Verification):** Standard hardware TAPs were inserted at specific troubleshooting points (e.g. VLAN trunking lines, connection between the NAP and the switch in the Point-to-Multipoint setup, ...).

Rationale: These TAPs were used exclusively for functional verification. Unlike the Hilscher or BANY data, these captures were not used for calculations.

- **Indu-Sol PROFINET Inspektor (Health and Event Monitoring):** The Inspektor was connected to the static infrastructure to provide a holistic view of the PROFINET environment.

Rationale: It was used to monitor the "Health" of the network (e.g. alarms, communication errors, netload, jitter, ...)

5.5.1.1 Visualization of Capture Points

Figures 5.6, 5.7, and 5.8 show the measurement points for the different setups. The blue TAP icons signify the points where traffic was branched off to the hardware analyzers.

Note on Visualization: To maintain visual clarity, not all diagnostic hardware is explicitly drawn in the figures. The rationale for this is as follows:

- **Siemens BANY:** The blue TAP icons in Figures 5.7 and 5.8 represent the BANYs used for PCAP capturing on the PRP lines. Although packet insertion was performed at the network boundary, the injection points are omitted from the schematics to keep them legible.
- **Indu-Sol PROFINET Inspektor:** This hardware was connected to a TAP within the "Static" segment. Because it monitors the global health of the network, visualizing it in the schematics would add unnecessary complexity.
- **Oscilloscope:** While used for verification, the oscilloscope is not depicted in the schemes as its application was localized to specific terminals and was used for verification rather than data collection.

5.5.2 Data Capture and Custom Analysis Workflow

The transition from raw network traffic to quantifiable performance metrics followed a three-step process: configuration, capturing, and analysis.

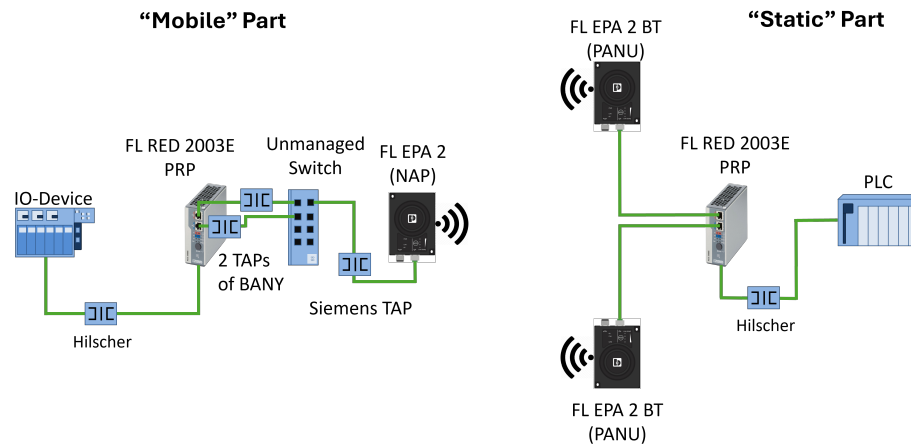


Figure 5.6: Measurement points for the exploratory Point-to-Multipoint Bluetooth setup.

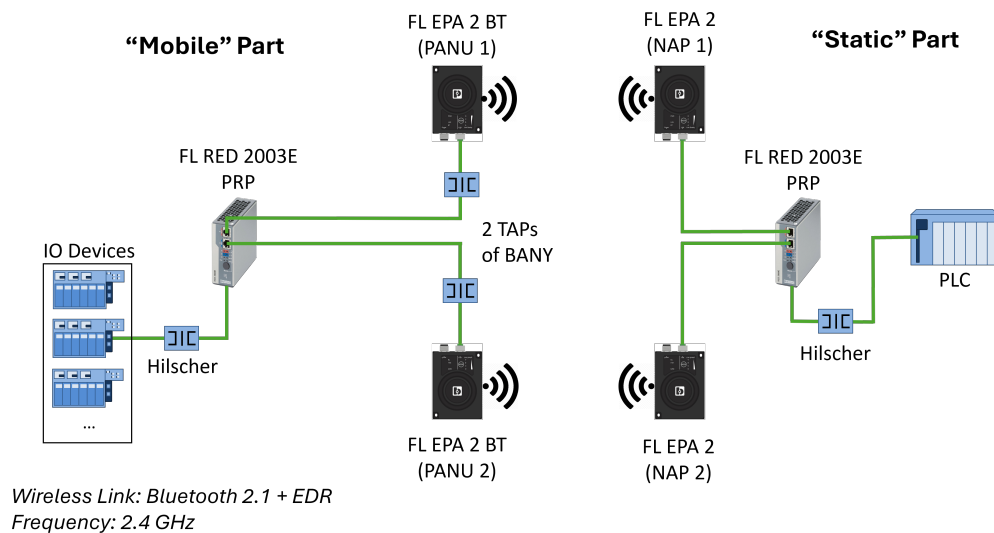


Figure 5.7: Measurement points for the Point-to-Point Bluetooth setup.

Stage 3: End-to-End Latency Calculation. The script correlates frames by matching the unique Cycle Counter values captured at the ingress and egress points. The E2E latency is defined as:

$$L_{E2E} = t_{\text{exit}} - t_{\text{entry}} \quad (5.1)$$

Stage 4: Aggregated Metric Computation and Visualization. From the filtered arrays, the script computes the full set of performance indicators required for the analysis and generates several types of visualizations to evaluate the system's behavior:

- **Latency Statistics:** Mean, minimum, maximum, standard deviation ((std. dev.) Jitter), and key percentiles (P50, P95, P99, P99.9) of the E2E latency distribution.
- **RFC 3550 Jitter:** Calculated as the mean of absolute consecutive latency differences, $\text{Jitter} = \overline{|L_{i+1} - L_i|}$, quantifying the cycle-to-cycle timing variation.
- **Packet Loss:** Derived from the ratio of received versus expected cycle identifiers over the capture span.
- **Network Load:** Aggregates the total per-second byte and packet counts across all captured traffic to characterize the offered netload.
- **Packet Rate:** The total number of packets per second (pps) observed in the capture, providing insight into the traffic intensity and potential congestion levels.
- **Generated Visualizations:** To structurally analyze these metrics, the script outputs multiple graphical representations:
 - **Histogram and KDE Plot:** To illustrate the probability density, baseline concentrations, and spread of the latency.
 - **CCDF (Complementary Cumulative Distribution Function):** To visualize the exact probability of latency spikes exceeding critical network thresholds.
 - **Box Plots:** To highlight the Interquartile Range (IQR) and expose variance expansion.
 - **Bar Charts:** To provide a direct, high-level comparison of the mean and absolute maximum latencies per scenario.

Chapter 6

Measurements and Results of the Proposed Architectures

This chapter presents the quantitative evaluation of the wireless architectures described in Chapter 5. Rather than re-explaining the underlying theory, each section directly reports the measured performance under controlled laboratory conditions and interprets the results in the context of the industrial requirements established in the field audit (Chapter 4).

The chapter is structured progressively: first, a wired baseline is established as the performance reference. Next, the wireless technologies are compared under ideal (baseline) conditions, followed by increasingly challenging scenarios: network load and signal attenuation. The chapter concludes with a comparative summary.

Note: This chapter presents the most critical findings. An extensive dataset of additional measurements and individual latency graphs across various cycle times is available in the supplemental GitHub repository. The repository includes a metadata table mapping every test scenario, physical parameter (e.g., RSSI, attenuation, UDP load), and statistical outcome (average/max latency, jitter, packet loss) to its raw PCAP capture file. The complete replication package is accessible at: <https://github.com/EliasNeels/PROFINET-over-wireless-Bridging-the-handover-gap-for-seamless-handover-and-redundancy>

6.1 Measurement Strategy and Test Scenarios

6.1.1 Measurement Equipment and Capture Points

The diagnostic hardware and its placement rationale were detailed in Section 5.5. In summary, the Hilscher netAnalyzer (10 ns resolution) served as the primary source for E2E latency calculations, positioned at the outermost network boundaries. The Siemens BANY was used for PRP verification and active packet injection for load testing. All raw traffic was exported as PCAP files with hardware timestamps.

6.1.2 Defined Test Scenarios

To systematically evaluate the architectures, four test scenarios were defined. Each scenario targets a specific requirement of the solution:

Table 6.1 Overview of the four test scenarios and their relation to the field audit findings.

ID	Scenario	Description	Cycle Times	Requirement Reference
P1	Baseline	PROFINET-only, no background traffic	16, 32, 64, 128 ms	General link viability
P2	Network Load	PROFINET + UDP background traffic	64 ms	IT/OT coexistence on shared medium
P3	Attenuation	RF attenuators on one or both lines	64 ms	Mitigating Dynamic shadowing (Sec. 4.2)
P4	Seamless Handover	Sequential cable pulls simulating handover	64 ms	Mitigating Handover gaps (Sec. 4.3)

6.1.3 Reported Metrics

For each test, the following Key Performance Indicators (KPIs) are reported:

- **End-to-end Latency:** mean, minimum, maximum, and key percentiles (e.g., P99) of the E2E delay through the complete wireless infrastructure.
- **Jitter (Standard Deviation):** The statistical standard deviation of the E2E latency, quantifying the overall cycle-to-cycle timing variation. This metric is reported as a percentage relative to the configured PROFINET update cycle (% cycle) to normalize timing stability across different test scenarios.
- **RFC 3550 Jitter:** A continuous measurement of network stability that tracks the variation in packet arrival times between two consecutive frames. This variation is smoothed over time using a statistical moving mean and expressed directly as a percentage of the nominal update cycle time (% cycle).
- **Packet Loss:** Percentage of PROFINET cycles where no valid frame was received.
- **Network Load (Netload):** The total volume of traffic traversing the network in % of the maximum wired throughput of 100 Mbps Fast Ethernet.
- **Total Packet Rate:** The total number of packets per second on the network.

6.1.4 Configuration Parameters

Unless stated otherwise, all tests used the following PROFINET configuration: a single IO-Controller (Siemens S7-1516) communicating with one IO-Device (ET 200SP) for baseline tests, or five IO-

Devices for load tests. The Watchdog multiplier was set to 1920 ms, representing the worst-case allowable threshold for the ArcelorMittal Gent infrastructure.

6.2 Wired Reference Baseline

Before evaluating the wireless architectures, a wired reference was established. The primary objective of this reference is to quantify the latency of a purely wired equivalent and determine whether propagation delay and the processing delays of the PRP hardware are mathematically relevant when compared to the delays inherent to wireless transport.

By defining this reference, it becomes possible to isolate the latency contribution of the wireless links from the fundamental transmission delays of the copper medium and the redundancy hardware.

To achieve this, three physical wired configurations were constructed and measured, as illustrated in Figure 6.1.

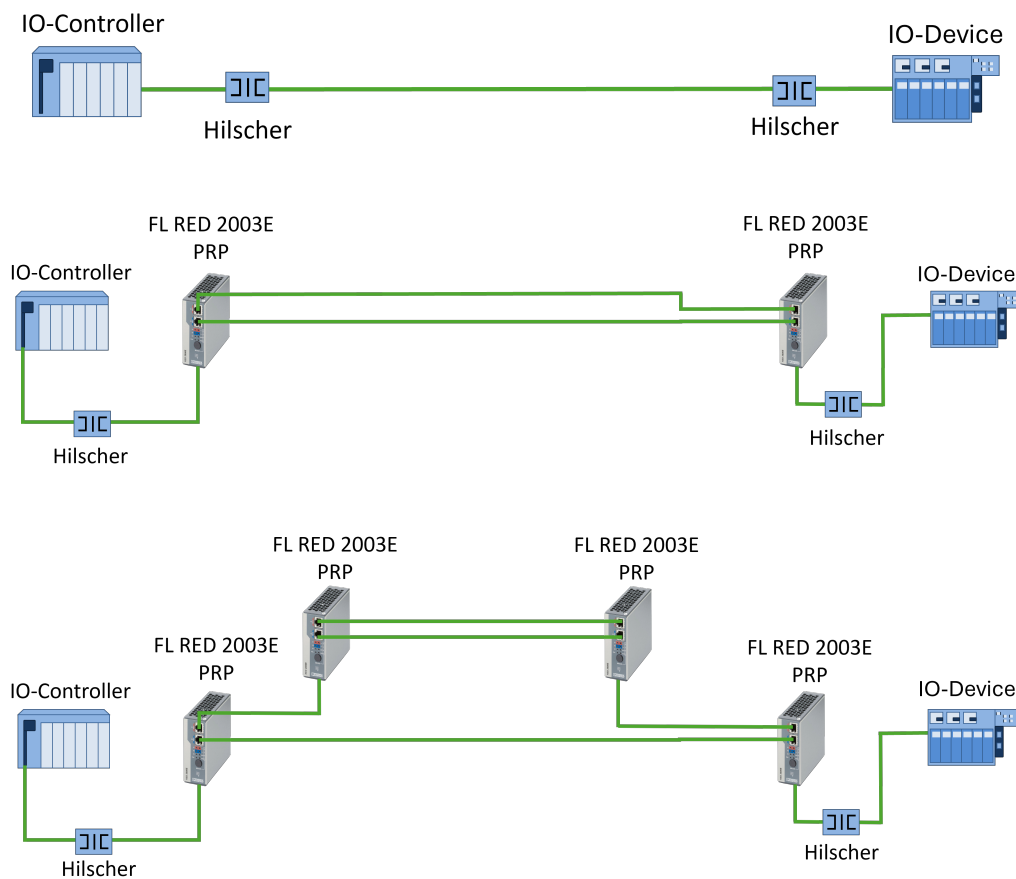


Figure 6.1: Physical laboratory setups for the wired baseline measurements. **Top:** Direct (No PRP) baseline measurement. **Middle:** Single PRP Pair configuration. **Bottom:** Cascaded PRP configuration utilizing two nested RedBox pairs.

Note on Latency Dispersion: *The microscopic 9.3 ns variation (0.1327 to 0.1420 μs) in the No*

Table 6.2 Wired reference measurements at a 64 ms cycle time. These values illustrate the linear scaling of hardware processing overhead in cascaded redundancy architectures.

Configuration	Net Load (% of 100 Mbps)	Total Rate (pps)	Mean (μ s)	Max (μ s)	P99 (μ s)	Std Jitter (% cycle)	RFC 3550 (% cycle)	PN Loss (%)
Direct (No PRP)	0.0208	31.2	0.1327	0.1420	0.1420	0.0000	0.0000	0.0000
Single PRP	0.0209	31.2	17.6166	17.7740	17.7340	0.0001	0.0001	0.0000
Cascaded PRP	0.0211	31.2	34.6625	34.9940	34.8840	0.0002	0.0002	0.0000

Wired Reference Measurements (64ms Cycle and 1 IO-Island)

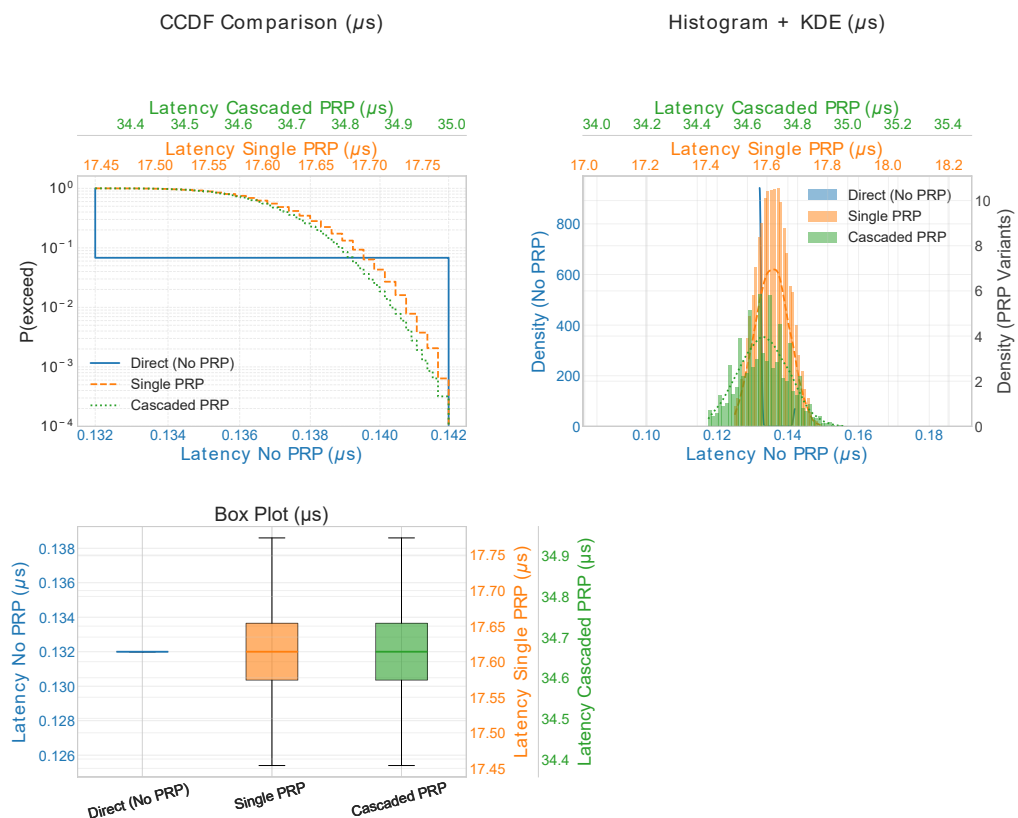


Figure 6.2: Wired Reference statistical analysis (μ s). **Top Left:** CCDF of latency thresholds. **Top Right:** Histogram and KDE of latency distributions. **Bottom Left:** Box plots of latency distributions. **Bottom Right:** Mean and maximum latency bar chart.

PRP baseline is not a processing delay, but rather PHY-layer clock quantization jitter (a 100 Mbps Ethernet clock cycle is ~ 8 ns) [7]. The wider dispersion observed in the PRP configurations is a direct consequence of the RedBoxes' active store-and-forward architecture. Buffering the frames, manipulating the RCT trailers, and re-serializing the payload introduces minor timing variations as the frames pass through each hardware tier.

6.2.1 Wired Performance and Hardware Overhead Analysis

The results presented in Table 6.2 and Figure 6.2 establish the performance floor of the experimental testbed by quantifying both the propagation delay and the processing overhead introduced by the Redboxes.

For the direct wired connection (No PRP), the system demonstrates a baseline, characterized by a mean latency of $0.1327 \mu\text{s}$ and a jitter representing an imperceptible 0.0000% of the update cycle. This distribution represents the propagation delay across the 25 m copper medium.

Introducing PRP predictably shifts this latency floor. One RedBox pair (Single PRP) elevates the mean latency to $17.6166 \mu\text{s}$. This roughly $17.5 \mu\text{s}$ penalty reflects the hardware's internal processing cycle. In the cascaded configuration, this processing delay scales linearly to a mean of $34.6625 \mu\text{s}$, as each subsequent RedBox executes the identical cycle. Evaluating the other KPIs as well reveals that using a RedBox and expanding the redundancy tiers does not degrade network stability; the maximum latency remains close to the respective mean latency, and the jitter stays exceptionally low, consuming a maximum of only 0.0002% of the cycle time.

The microsecond-scale hardware profile provides the necessary context for evaluating the industrial wireless architecture. In control systems, Maximum Latency and Jitter are the paramount metrics that govern Watchdog timeouts. Because the final system relies on wireless transport, which inherently introduces millisecond-scale delays and substantial variability, the physical propagation and PRP encapsulation delays of the wired components are mathematically negligible. Ultimately, this baseline proves that the redundancy hardware itself adds no significant timing uncertainty to the safety loop, allowing the subsequent evaluation to focus entirely on the dominant variable: the wireless transport layer.

6.3 Architectures Baseline Performance

The baseline scenario evaluates the fundamental latency characteristics of the Bluetooth and Wi-Fi architectures under ideal conditions: a single PROFINET IO-Device communicating with a PROFINET IO-Controller through the wireless link with no background traffic or intentional signal degradation.

6.3.1 Bluetooth Baseline

Table 6.3 and Figure 6.3 present the Bluetooth baseline results across all tested cycle times.

Bluetooth (BT) Baseline — Latency Comparison at different Cycle Times (1 IO-Island)

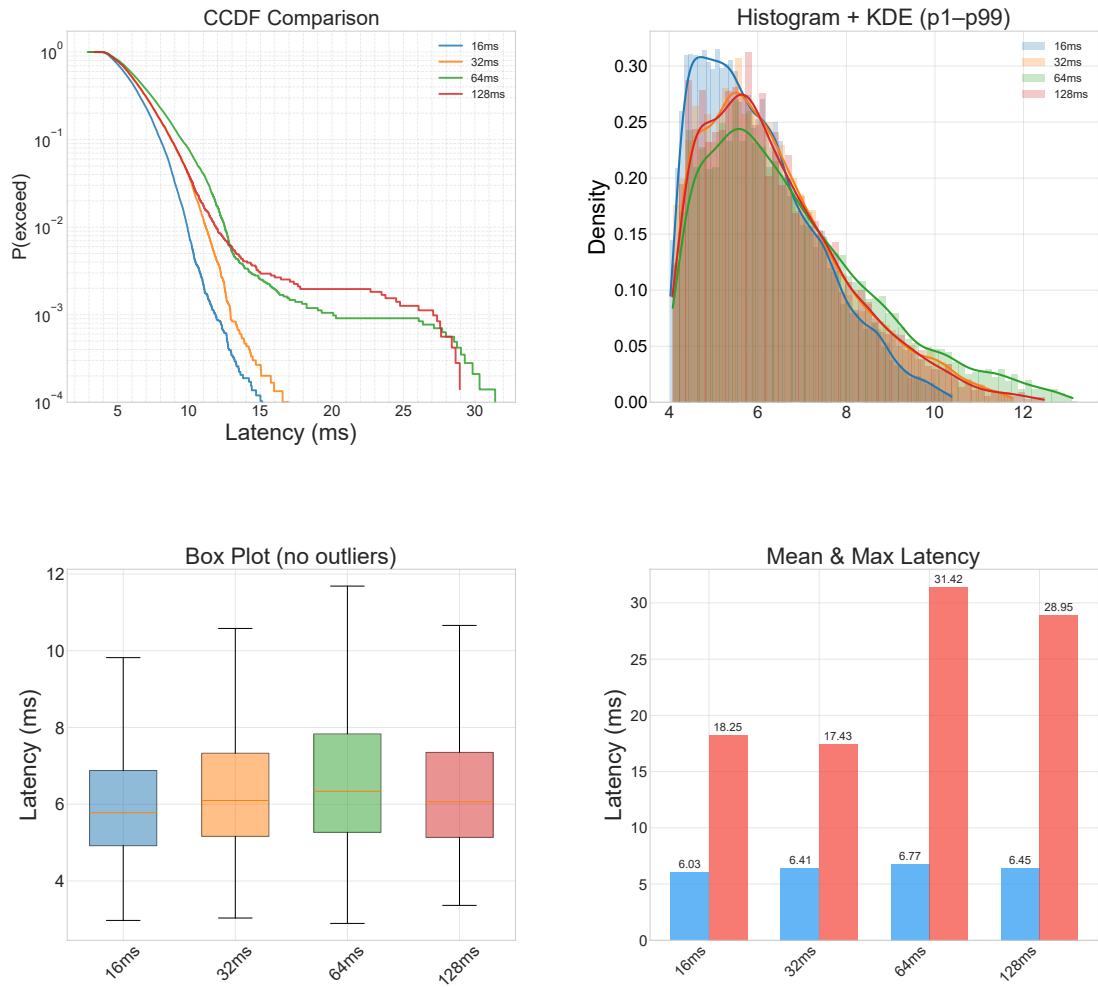


Figure 6.3: Bluetooth baseline statistical analysis. **Top Left:** CCDF of latency thresholds. **Top Right:** Histogram and KDE of latency distributions. **Bottom Left:** Box plots of latency distributions. **Bottom Right:** Mean and maximum latency bar chart.

Table 6.3 Bluetooth's baseline KPIs across different PROFINET cycle times.

Cycle Time	Net Load (% of 100 Mbps)	Total Rate (pps)	Mean (ms)	Max (ms)	P95 (ms)	P99 (ms)	Std Jitter (% cycle)	RFC 3550 (% cycle)	PN Loss (%)
16 ms	0.0738	125.0	6.0267	18.2481	8.6725	9.8794	8.729	9.221	0.0000
32 ms	0.0378	62.5	6.4099	17.4348	9.6967	11.1738	5.171	5.650	0.0000
64 ms	0.0198	31.3	6.7680	31.4247	10.6283	12.4916	3.218	3.405	0.0000
128 ms	0.0108	15.6	6.4487	28.9468	9.6950	11.8640	1.489	1.472	0.0000

6.3.1.1 Mean Latency and the TDMA Processing Floor

The packet rate increases eightfold across the four cycle times, rising from 15.6 pps at 128 ms to 125.0 pps at 16 ms. Despite this, the mean latency remains constant between 6.02 ms and 6.76 ms. The Bluetooth TDMA MAC layer causes this stable floor. The master's polling interval and BNEP encapsulation overhead impose a fixed 6 ms processing delay, completely independent of the PROFINET cycle time.

6.3.1.2 Distribution Spread and Jitter

The latency distributions (Figure 6.3) skew right, peaking uniformly between 4.5 ms and 6.5 ms. The core distributions, represented by the IQRs, remain nearly identical across all four cycle times. The primary difference lies entirely in the extreme outliers. The 64 ms and 128 ms configurations have higher maximum latencies (31.42 ms and 28.95 ms) than the 16 ms and 32 ms configurations (~18 ms).

Because the absolute timing variation remains stable across all tests, the relative jitter decreases at slower cycle times. Standard deviation jitter drops from 8.729% (16 ms) to 3.218% (64 ms), and RFC 3550 jitter declines from 9.221% to 3.405%. A fixed absolute variation simply consumes a much smaller percentage of a longer cycle time budget.

Without over-the-air RF protocol analyzers, the exact cause of the higher outlier spikes at 64 ms and 128 ms cannot be definitively proven. However, based on the Bluetooth specification, these spikes likely result from a combination of asynchronous polling alignment and FHSS retransmissions. If a packet arrives just after a polling window closes, or if localized noise forces the radio to buffer a corrupted frame and wait for the next clean frequency hop, it generates an absolute delay spike. At slower cycle times, this disrupts the sparse packet interval more severely [12].

6.3.1.3 CCDF Analysis of Maximum Latency

The Bluetooth link achieved 0.000% PROFINET packet loss across all configurations. Maximum latency peaked at 31.42 ms for the 64 ms cycle and 28.95 ms for the 128 ms cycle. The CCDF (Figure 6.3) illustrates the probability of exceeding these latency thresholds. The 64 ms and 128 ms

curves exhibit prominent tails compared to the steeper 16 ms and 32 ms curves, indicating a higher probability of latencies between 20 ms and 30 ms. While the measurements cannot isolate the exact RF mechanism, these spikes align perfectly with expected FHSS buffering and MAC-layer retransmission behaviors.

6.3.1.4 Impact on System Stability and Watchdog Timers

These baseline measurements dictate the minimum safe PROFINET Watchdog settings. At a 16 ms cycle time with a standard 3× Watchdog multiplier (48 ms limit), the 18.25 ms maximum latency consumes 38% of the safety budget. The system remains stable but leaves a limited margin for external interference. Conversely, a 64 ms cycle time with a 3× multiplier provides a 192 ms limit. The 31.42 ms maximum latency consumes only 16% of this budget, leaving a generous 160 ms safety margin. Slower cycle times provide greater system resilience. They offer a larger absolute time window to absorb unexpected latency spikes.

6.3.2 Point-to-Multipoint Bluetooth Topology

Empirical testing of the exploratory point-to-multipoint Bluetooth topology, revealed (Figure 5.1) physical and logical layer conflicts. These conflicts render the architecture unfeasible.

Table 6.4 Bluetooth KPI comparison: Point-to-Point vs. Point-to-Multipoint.

Architecture Time	Net Load (% of 100 Mbps)	Total Rate (pps)	Mean (ms)	Max (ms)	P95 (ms)	P99 (ms)	Std Jitter (% cycle)	RFC 3550 (% cycle)	PN Loss (%)
2AP 2CL (Standard Baseline)	0.0198	31.2	6.7680	31.4247	10.6283	12.4916	3.218	3.405	0.000
1AP 2CL (MAC Flapping Baseline)	0.0212	31.2	7.9212	91.3247	13.3866	33.8341	8.160	5.793	0.000

6.3.2.1 Mechanisms of Failure: MAC Flapping, Isolation Loss, and Diagnostic Degradation

PRP requires strict physical or logical isolation between LAN A and LAN B paths. Connecting both RedBox ports to a single unmanaged switch and routing traffic through a single Access Point breaks this isolation.

PCAP analysis between the unmanaged switch and the wireless links confirmed this broken path isolation. The captures revealed that frames appended with the LAN A RCT and frames with the LAN B RCT were arriving on the same physical line, merging the two independent parallel paths into a single collision domain.

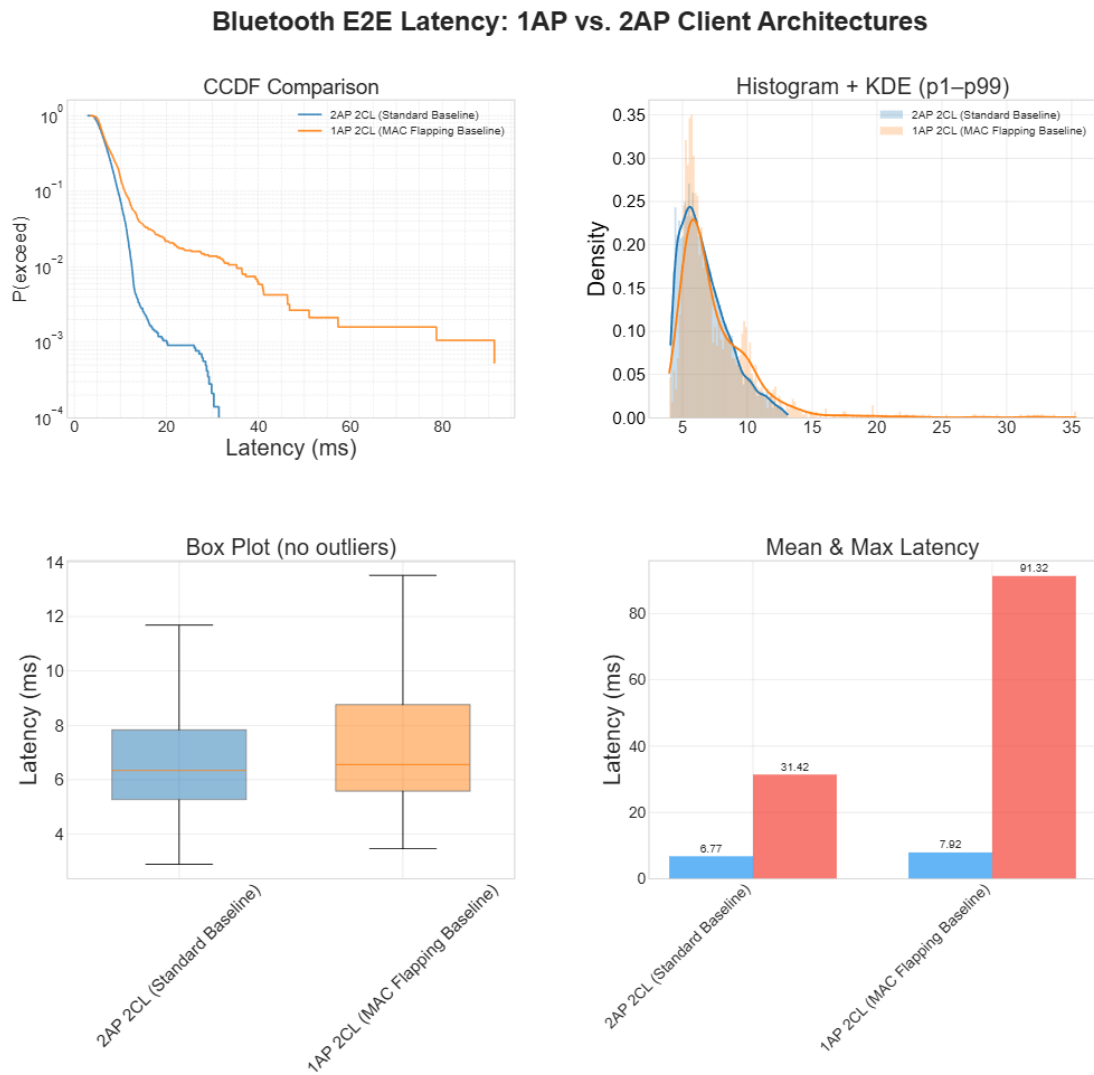


Figure 6.4: Bluetooth End-to-End latency comparison between the Point-to-Point architecture and the Point-to-Multipoint architecture. **Top Left:** CCDF of latency thresholds. **Top Right:** Histogram and KDE of latency distributions. **Bottom Left:** Box plots of latency distributions. **Bottom Right:** Mean and maximum latency bar chart.

This physical loop causes MAC flapping inside the unmanaged switch. The switch receives frames containing identical source MAC addresses alternating between different physical ports. Consequently, the switch continuously overwrites its address routing table and floods frames to all ports to ensure delivery, illuminating its hardware error LED.

The RedBoxes also detect this cross-talk. When a RedBox receives a Line A frame on its Line B port, it logs an isolation fault and illuminates its local error indicator. Although PRP deduplication successfully discards these misrouted frames and maintains 0.000% packet loss (Table 6.4), the persistent hardware error states make this architecture unsuitable for industrial deployment.

6.3.2.2 Latency and Jitter Degradation

The resulting switch flooding and table corruption degrade PROFINET timing determinism (Table 6.4 and Figure 6.4). Compared to the point-to-point architecture, the point-to-multipoint configuration increases the mean latency from 6.77 ms to 7.92 ms. The maximum latency nearly triples, rising from 31.42 ms to 91.32 ms.

Two primary mechanisms drive this latency inflation. First, routing both redundancy lines through a single NAP doubles the packet volume on that single radio interface. Because the Bluetooth master must serialize these frames sequentially within its TDMA schedule, the increased packet density creates queuing delays inside the buffer. Second, the switch's constant table overwrites and frame flooding add processing overhead.

The box plot and histogram show that while the core IQRs overlap, this setup develops an high-latency tail. The 95th percentile latency rises from 10.63 ms to 13.39 ms, while the 99th percentile nearly triples from 12.49 ms to 33.83 ms. This variance more than doubles the standard deviation jitter, driving it from 3.218% up to 8.160%. The CCDF comparison curve illustrates this probability shift, tracking a long staircase profile up to the 91.32 ms extreme. While the PRP layer prevents packet drops, this latency inflation introduces unacceptable timing risks for real-time industrial control loops.

6.3.3 Wi-Fi Baseline

Table 6.5 and Figure 6.5 presents the Wi-Fi baseline results across the same cycle times under the identical single IO-Device condition. The bitrate in these measurements was 54 Mbps.

Table 6.5 Wi-Fi's baseline KPIs across different PROFINET cycle times.

Cycle Time	Net Load (% of 100 Mbps)	Total Rate (pps)	Mean (ms)	Max (ms)	P95 (ms)	P99 (ms)	Std Jitter (% cycle)	RFC 3550 (% cycle)	PN Loss (%)
16 ms	0.0720	125.0	0.7750	35.6941	1.2848	4.8664	5.810	1.382	0.0000
32 ms	0.0370	62.5	0.6015	1.2297	0.6232	0.7361	0.087	0.046	0.0000
64 ms	0.0195	31.3	0.6057	1.2316	0.6257	0.7121	0.043	0.024	0.0000
128 ms	0.0107	15.6	0.7888	30.8645	1.3054	4.5549	0.790	0.174	0.0000

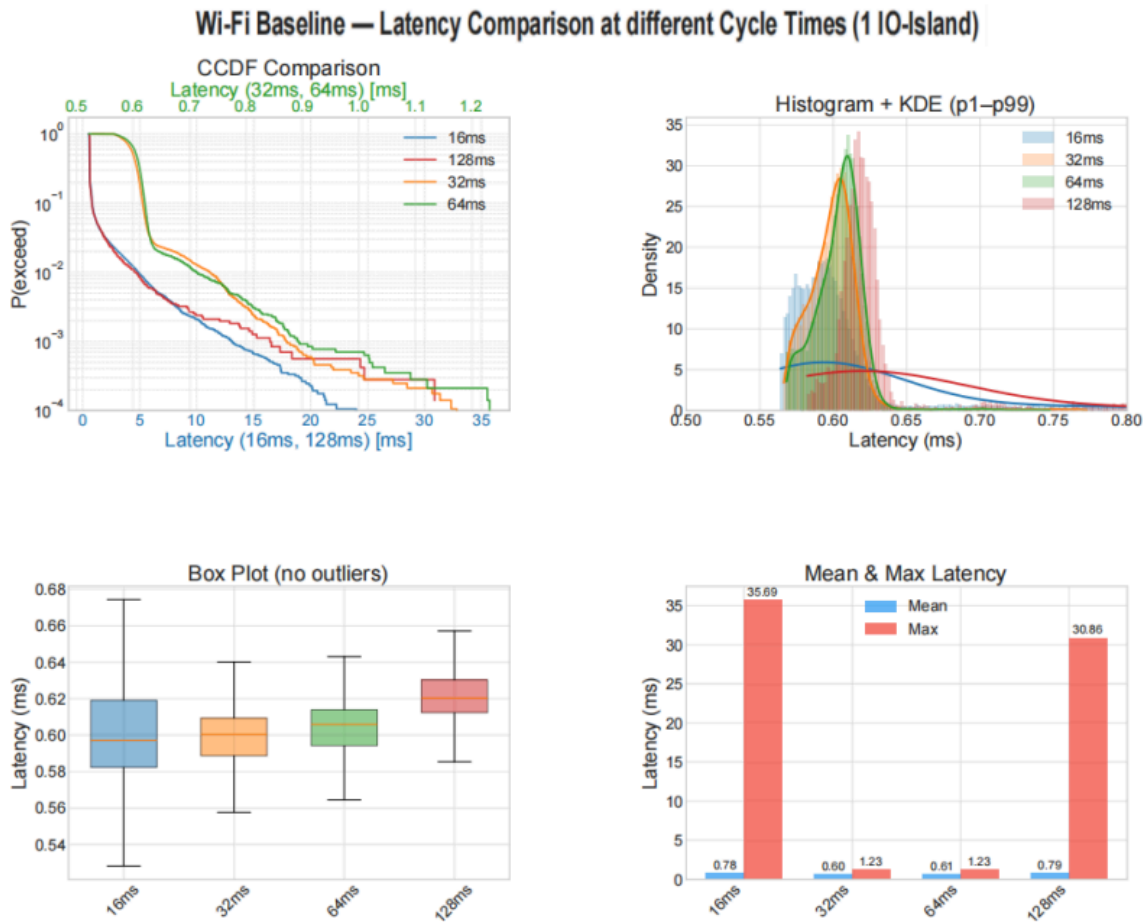


Figure 6.5: Wi-Fi Baseline statistical analysis. **Top Left:** CCDF of latency thresholds. **Top Right:** Histogram and KDE of latency distributions. **Bottom Left:** Box plots of latency distributions. **Bottom Right:** Mean and maximum latency bar chart.

6.3.3.1 Mean Latency and CSMA/CA Performance

The Wi-Fi link maintains a highly consistent core processing delay. Unlike Bluetooth's rigid TDMA schedule, Wi-Fi uses CSMA/CA. On a clean 5 GHz medium, the radio transmits frames instantly without artificial buffering, yielding a sub-millisecond mean latency (~ 0.6 ms) across all configurations. While rare outliers slightly inflate the mean latency at the 16 ms and 128 ms cycles (0.78 ms and 0.79 ms), the histograms confirm these are isolated disruption events, not fundamental degradations of the baseline radio performance.

6.3.3.2 Distribution Spread and Variance Analysis

The 32 ms and 64 ms configurations exhibit tightly clustered distributions with virtually no variance. Maximum latencies peak at 1.23 ms, and both jitter metrics (Standard Deviation and RFC 3550) remain near zero.

Conversely, the 16 ms and 128 ms datasets expose rare but severe maximum latency spikes of 35.69 ms and 30.86 ms, respectively. This introduces a significant divergence in the jitter metrics. For the 16 ms cycle, standard deviation jitter spikes to 5.810%, while RFC 3550 jitter stays low at 1.382%. Because standard deviation squares the variance from the mean, while RFC 3550 averages consecutive packet differences, this divergence mathematically proves that sudden, isolated latency spikes (not persistent medium degradation) cause the variance.

6.3.3.3 PRP Masking and the Origin of Latency Spikes

The exact origin of these extreme spikes remains unverified without over-the-air PCAP. However, the PRP architecture limits the possible RF causes. Because PRP transmits identical frames simultaneously across two independent Wi-Fi links, the system masks transient delays on one path. Therefore, for a > 30 ms latency spike to manifest, the delay must either result from a statistically improbable simultaneous disruption on both independent wireless channels, or originate outside the RF medium entirely.

The absence of spikes in the 32 ms and 64 ms datasets likely stems from the fixed 15-minute measurement window. The 16 ms configuration generates significantly more packets, mathematically increasing the statistical probability of capturing a rare hardware or environmental event causing these spikes. The single spike in the sparse 128 ms dataset confirms these occur by chance, independent of the PROFINET cycle time.

6.3.3.4 Impact on Safety Budgets and System Engineering

These isolated latency spikes dictate the safety boundary conditions for industrial system design. Under optimal conditions (32 ms and 64 ms cycles), the Wi-Fi link is exceptionally robust. Maximum latencies of ~ 1.23 ms consume a negligible fraction of the PROFINET Watchdog Timer, leaving nearly 100% of the safety budget available to absorb environmental disturbances.

However, engineers must factor these rare latency anomalies into their network planning. While a 35.69 ms spike at a 16 ms cycle time does not inherently trigger a Fail-Safe state, it drastically narrows the remaining time margin. These diminished margins highlight a key characteristic of industrial Wi-Fi: while it excels in average processing delay and network throughput, its safety margins temporarily fluctuate. To guarantee stability in mobile environments, engineers must actively tailor critical parameters, such as AP placement, roaming thresholds, and PROFINET Watchdog multipliers, to accommodate these transient delays.

6.3.4 Comparative Analysis: Bluetooth vs. Wi-Fi Baseline

Both architectures achieve 0.000% PROFINET packet loss under ideal conditions, but they exhibit fundamentally different latency profiles. Table 6.6 summarises and compares the two.

6.3.4.1 Low Latency and Baseline Stability

In an uncontended RF environment, Wi-Fi offers lower latency and jitter due to CSMA/CA. Conversely, Bluetooth's TDMA imposes a permanent ~ 6 ms latency floor and higher jitter, regardless of

Table 6.6 Comparative summary of Bluetooth and Wi-Fi baseline performance.

Characteristic	Bluetooth Architecture	Wi-Fi Architecture	Best Architecture
Mean Latency	~6.0 – 6.8 ms TDMA polling and BNEP overhead impose a fixed floor.	~0.6 – 0.8 ms Uncontended CSMA/CA provides instantaneous medium access.	Wi-Fi
Jitter	Higher (~1.5 – 8.7%) Fixed timing variance consumes a larger percentage of short cycles.	Near-Zero (<0.1%) Distribution remains extremely tight during standard operation.	Wi-Fi
Maximum Latency	~31.4 ms FHSS MAC-layer retries generate predictable latency tails.	~35.7 ms Infrequent isolated spikes.	Tie
Safety Margin	Moderate The 6 ms floor continuously consumes part of the Watchdog budget.	High but Volatile Retains ~99% of the budget, except during isolated anomaly events.	Wi-Fi

the network load.

6.3.4.2 Worst-Case Latency and Watchdog Constraints

Both systems have maximum latencies between 30 ms and 35 ms. Bluetooth has more predictable spikes due to its FHSS retransmission mechanism, creating a more right-skewed distribution. Wi-Fi experiences these latency spikes more sporadically. The wide gap between Wi-Fi's standard deviation and RFC 3550 jitter mathematically proves these 35 ms delays are infrequent rather than continuous medium degradation.

Because these > 30 ms worst-case spikes are inherent to both systems, neither technology provides a sufficient safety margin for low cycle times (e.g., 16 ms) paired with low Watchdog limits (e.g., 48 ms). To guarantee operational stability, both technologies require higher cycle times (e.g., 32 ms or 64 ms) or higher Watchdog limits.

6.4 Architectures under Load (P2)

In an industrial deployment, the operational intent dictates the network design. Wi-Fi typically functions as a shared infrastructure, handling both the OT and IT traffic. Conversely, Bluetooth architectures serve as dedicated, point-to-point cable replacements.

Subjecting both architectures to a UDP background load provides a stress test to empirically evaluate their maximum throughput limits and protocol overhead vulnerabilities.

The test expanded the baseline configuration to one IO-Controller and five IO-Devices. The system injected unidirectional UDP background traffic at the PLC, extracting it just before the IO-Device. The test utilized a fixed 64 ms cycle time and a 1920 ms Watchdog Timer. The UDP injection rate progressively increased until the system triggered a Watchdog Timeout. To comprehensively evaluate the hardware, the test executed two traffic profiles: large 1400-byte payloads (simulating data-heavy applications) and small 64-byte payloads (simulating high-frequency control traffic).

6.4.1 Bluetooth Under Network Load

Because Bluetooth serves as a dedicated PROFINET link, this stress test does not simulate a designed mixed-traffic environment. Rather, it exposes the limits of the Bluetooth 2.1 + EDR protocol and the FL EPA 2 modules when subjected to certain traffic conditions.

6.4.1.1 Large UDP Payload (1400 Bytes) - IT Traffic Simulation

Table 6.7 Bluetooth's KPIs under increasing UDP load with 1400-byte payloads (64 ms cycle, 5 IO-Devices).

5 IO + UDP Load	Net Load (% of 100 Mbps)	Total Rate (pps)	Mean (ms)	Max (ms)	P95 (ms)	P99 (ms)	Std Jitter (% cycle)	RFC 3550 (% cycle)	PN Loss (%)
5IO + No UDP	0.1061	156.3	6.1911	35.8249	9.5873	14.8899	3.590%	2.861%	0.000%
5 IO + 0.1 Mbps	0.2053	165.2	8.8689	49.1580	13.0052	22.5592	4.868%	4.400%	0.000%
5 IO + 0.2 Mbps	0.3042	174.1	9.3646	50.8493	14.5250	23.7805	5.660%	5.010%	0.000%
5 IO + 0.3 Mbps	0.3954	183.0	11.8838	167.7294	18.6152	33.3071	7.926%	5.783%	0.197%

The 1400-byte payload results (Table 6.7 and Figure 6.6) define the absolute physical throughput ceiling of the Bluetooth link.

The system remains relatively stable up to 0.2 Mbps of background load. At this threshold, the total packet rate reaches 174.1 pps, mean latency rises to 9.36 ms, and the link maintains 0.000% packet loss.

However, injecting 0.3 Mbps triggers total physical saturation. The CCDF (Figure 6.6, Top Left) displays a severe fat-tail distribution. Maximum latency spikes to 167.73 ms, exceeding the 64 ms cycle limit by more than 2.6×, and the system drops 0.197% of PROFINET packets.

Saturating a theoretical 3 Mbps Bluetooth link with a mere 0.3 Mbps payload exposes the severe protocol penalties inherent to this architecture. The failure stems from byte volume rather than packet quantity:

1. **Narrow Physical Channels:** Unlike Wi-Fi, which operates on wide 20 MHz or 40 MHz channels, Bluetooth utilizes narrow 1 MHz physical channels. This physical layer constraint limits the maximum throughput before any protocol overhead is even applied.

Bluetooth E2E Latency vs UDP Load (64 ms Cycle Time, 1400 Bytes UDP Packets)

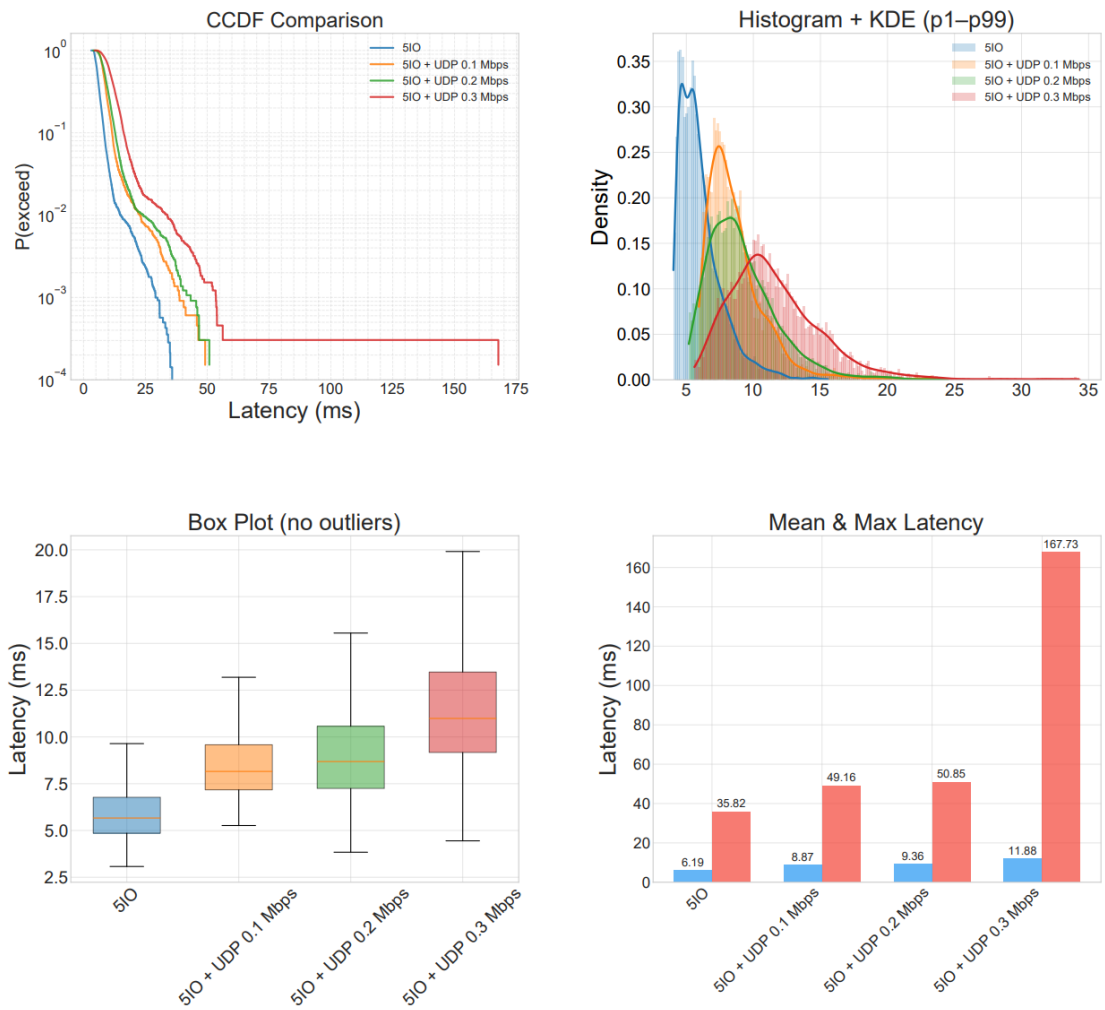


Figure 6.6: Bluetooth End-to-End latency distribution under an increasing 1400-byte UDP background load. **Top Left:** CCDF of latency thresholds. **Top Right:** Histogram and KDE of latency distributions. **Bottom Left:** Box plots of latency distributions. **Bottom Right:** Mean and maximum latency bar chart.

2. **PRP RedBox Overhead:** The RedBoxes append a RCT to every Ethernet frame, increasing the byte footprint before wireless transmission.
3. **Bluetooth Encapsulation:** The Bluetooth modules wrap the heavy Ethernet frames inside BNEP and L2CAP headers.
4. **TDMA Fragmentation:** Bluetooth's TDMA scheduler cannot transmit a 1400-byte frame in a single time slot. The hardware must fragment the large payload across multiple consecutive baseband slots. This fragmentation monopolizes the airtime, physically blocking time-critical PROFINET frames from accessing the medium.

6.4.1.2 Small UDP Payload (64 Bytes) - OT Traffic Simulation

Table 6.8 Bluetooth's KPIs under increasing UDP load with 64-byte payloads (64 ms cycle, 5 IO-Devices).

5 IO + (UDP Load)	Net Load (% of 100 Mbps)	Total Rate (pps)	Mean (ms)	Max (ms)	P95 (ms)	P99 (ms)	Std Jitter (% cycle)	RFC 3550 (% cycle)	PN Loss (%)
5 IO + No UDP	0.1061	156.3	6.1911	35.8249	9.5873	14.8899	3.590%	2.861%	0.000%
5 IO + 0.05 Mbps	0.1472	253.9	14.9987	35.5159	19.9588	25.0586	4.340%	4.596%	0.000%
5 IO + 0.10 Mbps	0.1882	351.6	17.1829	48.3672	21.9715	27.5852	4.756%	3.983%	0.000%

The 64-byte payload tests (Table 6.8 and Figure 6.7) prove that Bluetooth stability depends heavily on packet quantity, completely independent of total data volume.

Injecting just 0.10 Mbps of 64-byte packets barely moves the physical data load on the cable (0.188%), but it pushes the total packet rate to 351.6 pps. This high-frequency injection alters the latency profile. The density peak (Figure 6.7, Top Right) shifts abruptly from ~6 ms to ~15 ms. The mean latency nearly triples to 17.18 ms, the P99 latency reaches 27.58 ms and the max latency reaches 48.37 ms. This degradation highlights a critical vulnerability in the Bluetooth TDMA structure: it cannot handle high-frequency, asynchronous traffic. At 351.6 pps, the payload-to-overhead ratio collapses. Every single 64-byte UDP packet requires its own BNEP/L2CAP encapsulation, and its own dedicated TDMA slot.

Unlike the 1400-byte test where large frames blocked the medium through fragmentation, the 64-byte test exhausts the actual schedule. The hardware consumes all available TDMA polling slots transmitting these UDP packets. Consequently, the cyclic PROFINET frames back up in the hardware queues, missing multiple polling cycles before securing a transmission slot. This proves that unoptimized, high-frequency traffic, even in negligible data volumes, destroys Bluetooth determinism.

Bluetooth E2E Latency vs UDP Load (64 ms Cycle Time, 64 Bytes UDP Packets)

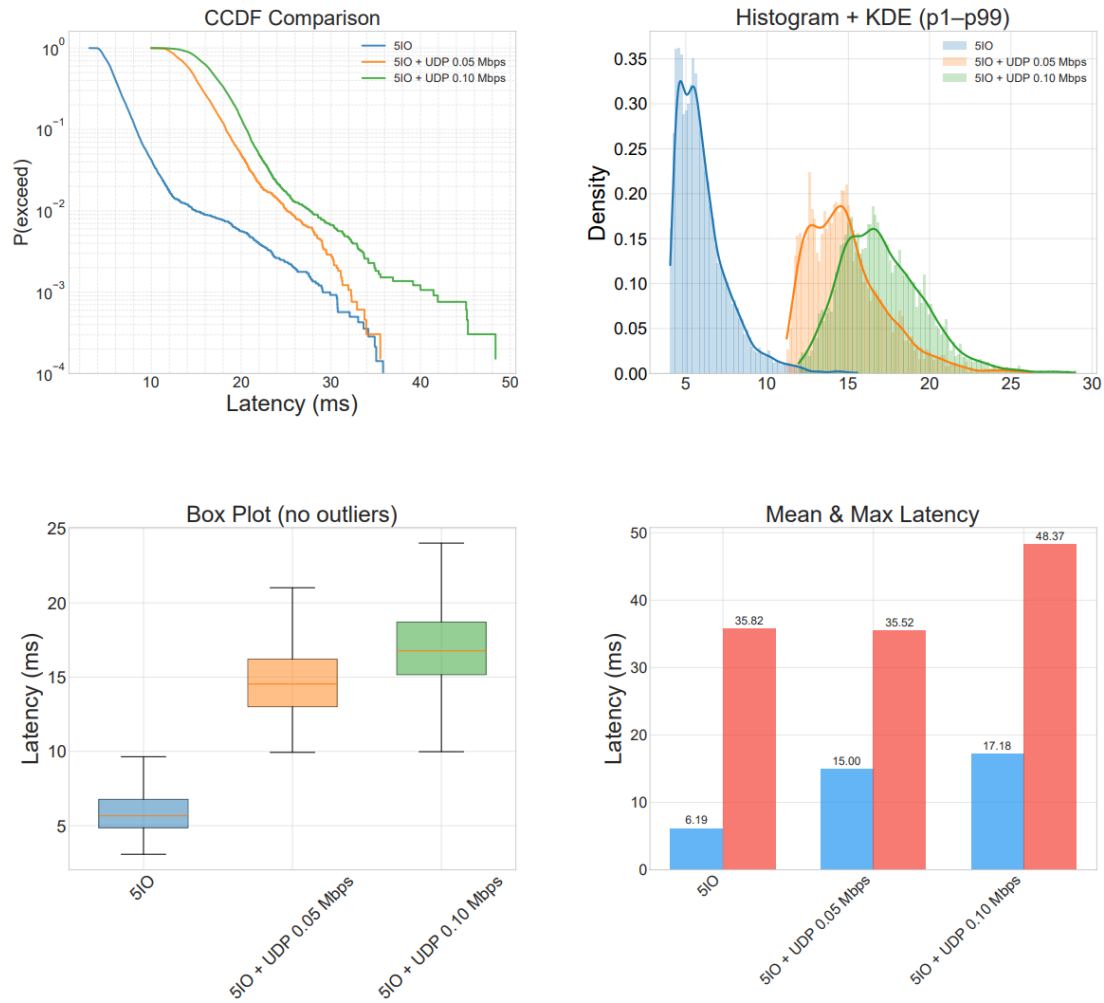


Figure 6.7: Bluetooth End-to-End latency distribution under an increasing 64-byte UDP background load. **Top Left:** CCDF of latency thresholds. **Top Right:** Histogram and KDE of latency distributions. **Bottom Left:** Box plots of latency distributions. **Bottom Right:** Mean and maximum latency bar chart.

6.4.1.3 Engineering Assessment for Bluetooth Capacity

These load tests establish strict architectural boundaries for Bluetooth deployments. Because both high data volume (TDMA fragmentation) and high packet frequency (schedule exhaustion) quickly saturate the link, Bluetooth serves strictly as a low-throughput solution. To guarantee PROFINET determinism, engineers must isolate the Bluetooth network from background IT traffic and limit the number of connected IO-Devices or utilize a slower cycle time (16ms update cycle, 5 IO-Devices gives watchdog timeouts).

6.4.2 Wi-Fi Under Network Load (Without QoS)

Unlike Bluetooth, Wi-Fi operates as a shared infrastructure designed to handle both the IO and IT traffic simultaneously because it can handle higher throughputs. These measurements were conducted under optimal RF conditions, with a RSSI between -30 and -40 dBm and a bitrate of 54 Mbps. The excellent signal strength guarantees that any observed latency degradation or packet loss is due to Medium Access Control (MAC) contention and hardware queuing, rather than RF fading or signal-induced retransmissions. These load tests evaluate the CSMA/CA throughput limits without IEEE 802.11e QoS (WMM) enabled.

6.4.2.1 Large UDP Payload (1400 Bytes) - IT Traffic Simulation

Table 6.9 Wi-Fi's KPIs under increasing UDP load with 1400-byte payloads, No QoS (64 ms cycle, 5 IO-Devices).

5 IO + (UDP Load)	Net Load (% of 100 Mbps)	Total Rate (pps)	Mean (ms)	Max (ms)	P95 (ms)	P99 (ms)	Std Jitter (% cycle)	RFC 3550 (% cycle)	PN Loss (%)
5 IO + No UDP	0.1034	156.3	0.6094	2.6781	0.7005	0.8187	0.098	0.039	0.000
5IO + 5 Mbps	5.1035	602.7	0.6459	2.8836	0.8814	0.9626	0.138	0.074	0.000
5IO + 10 Mbps	10.1036	1049.1	0.6040	2.6590	0.8029	0.9521	0.161	0.141	0.000
5IO + 20 Mbps	20.1034	1942.0	0.6820	4.4968	0.9880	1.5415	0.299	0.285	0.000
5IO + 40 Mbps	40.1035	3727.7	0.7864	18.2848	1.0443	1.4735	0.552	0.262	0.000
5IO + 60 Mbps	60.1035	5513.4	2.2568	10.7121	6.1041	6.6183	2.547	1.228	0.000
5IO + 65 Mbps	65.0913	5959.8	3.2234	9.0091	5.8919	6.3031	2.423	1.609	0.000
5IO + 67 Mbps	67.0670	6138.4	3.3611	64.5936	5.9390	6.3330	2.994	2.296	3.420

The Wi-Fi architecture handles large data volumes efficiently, sustaining a maximum of 65 Mbps of UDP traffic with absolute 0.000% packet loss.

However, the visual distributions in Figure 6.8 reveal that up to a netload of 40.1035% (40 Mbps), it handles the mixed traffic flawlessly. The Histogram and KDE plot show that the primary density peak

Wi-Fi E2E Latency vs UDP Load (64 ms Cycle Time, 1400 Bytes UDP Packets) NoQoS

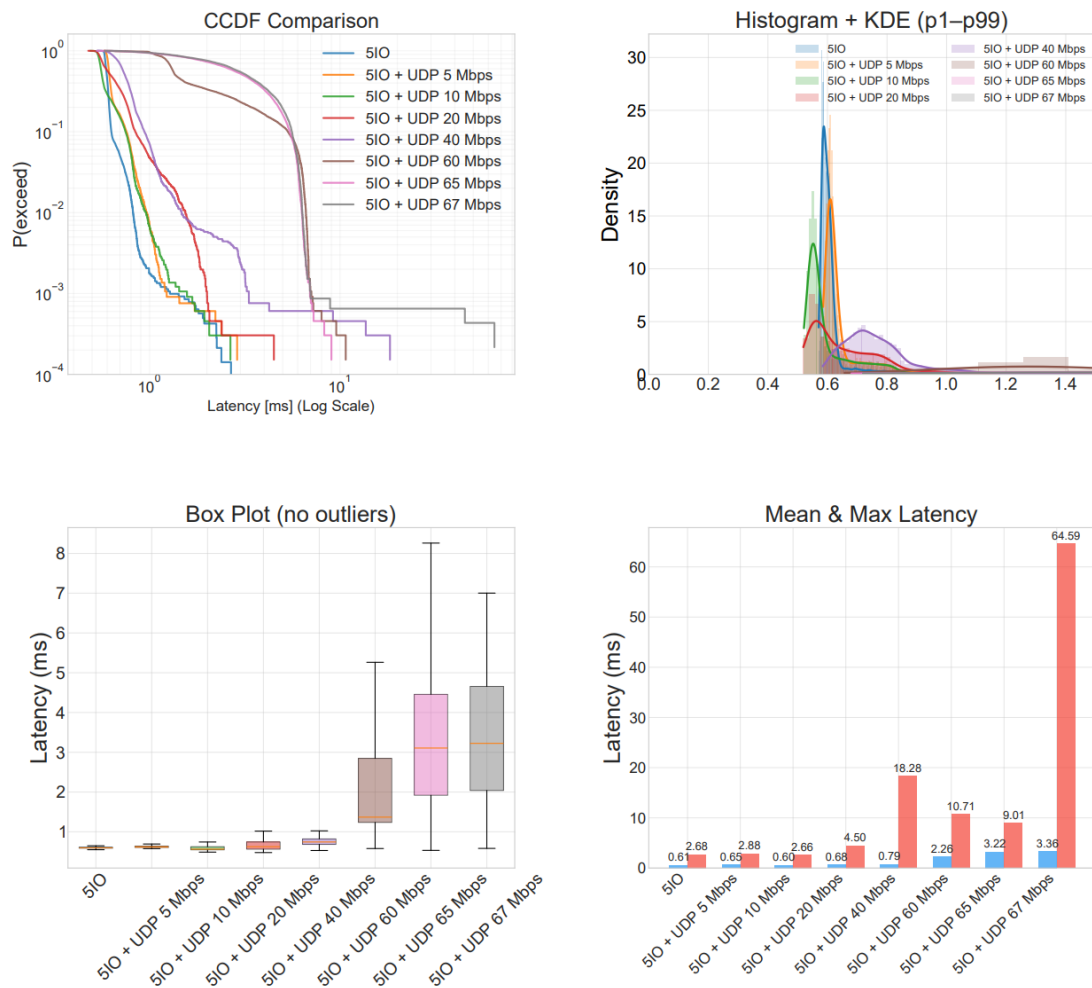


Figure 6.8: Wi-Fi (No QoS) End-to-End latency distribution under an increasing 1400-byte UDP background load. **Top Left:** CCDF of latency thresholds. **Top Right:** Histogram and KDE of latency distributions. **Bottom Left:** Box plots of latency distributions. **Bottom Right:** Mean and maximum latency bar chart.

remains below 1.0 ms. The Box Plot confirms this stability, showing a small IQR as in the baseline 5 IO-Devices measurement. Because of the optimal RSSI, which gives us a 54 Mbps bitrate, the wireless medium has enough idle airtime that PROFINET frames rarely have to wait behind long queues of IT traffic.

But the throughput can not exceed the 54 Mbps bitrate threshold. Above a netload of 54% (54 Mbps), the Histogram starts detaching from the sub-millisecond baseline, forming a wider density peak around 2.2 ms to 3.2 ms. The Box Plot shows an expansion of the IQR, and the P99 latency leaps from 1.04 ms to over 6.30 ms. At this load, the available airtime is nearly exhausted and the network transitions from a state of "immediate access" to a state of "persistent contention". PROFINET frame now sometimes need to wait in the hardware buffers while these large UDP frames serialize over the air interface. Despite this queuing delay, the hardware buffers are large enough, and the CSMA/CA algorithm is efficient enough, to transmit all PROFINET frames before the 1920 ms Watchdog Timer expires, maintaining 0.000% loss.

At a netload of 67% (67 Mbps), the CCDF plot illustrates the breaking point: the curve develops a "fat tail", indicating that packets are experiencing high delays compared to before. The maximum latency spikes to 64.59 ms, and the system suffers a 3.420% PROFINET packet loss. The volume of incoming UDP traffic saturates the wireless bridge hardware. The PROFINET frames become trapped in the buffers longer than the strict 1920 ms Watchdog limit, resulting in Fail-Safe timeouts.

6.4.2.2 Small UDP Payload (64 Bytes) - OT Traffic Simulation

Table 6.10 Wi-Fi's KPIs under increasing UDP load with 64-byte payloads, No QoS (64 ms cycle, 5 IO-Devices).

5 IO + (UDP Load)	Net Load (% of 100 Mbps)	Total Rate (pps)	Mean (ms)	Max (ms)	P95 (ms)	P99 (ms)	Std Jitter (% cycle)	RFC 3550 (% cycle)	PN Loss (%)
5 IO + No UDP	0.1034	156.3	0.6094	2.6781	0.7005	0.8187	0.098	0.039	0.000
5 IO + 2.5 Mbps	2.6035	5039.1	0.8110	13.9367	1.1262	1.4885	0.416	0.293	0.000
5 IO + 5.0 Mbps	5.0876	9921.9	2.8671	14.3394	5.2343	5.6683	2.205	2.641	0.000
5 IO + 7.5 Mbps	7.5511	14804.7	3.3708	177.351	55.4941	5.9103	8.701	3.964	7.736

The 64-byte payload tests (Table 6.10 and Figure 6.9) reveal a critical CSMA/CA vulnerability: packet quantity, not data volume, dictates airtime saturation. The Wi-Fi bridge comfortably sustains 65 Mbps using large 1400-byte packets, but small packets degrade performance at lower throughputs. As background load increases, the corresponding surge in packets per second rapidly deteriorates the wireless link:

- **At 2.5 Mbps (5039 pps):** Maximum latency jumps to 13.93 ms, indicating the increased queue build-up.

Wi-Fi E2E Latency vs UDP Load (64 ms Cycle Time, 64 Bytes UDP Packets) NoQoS

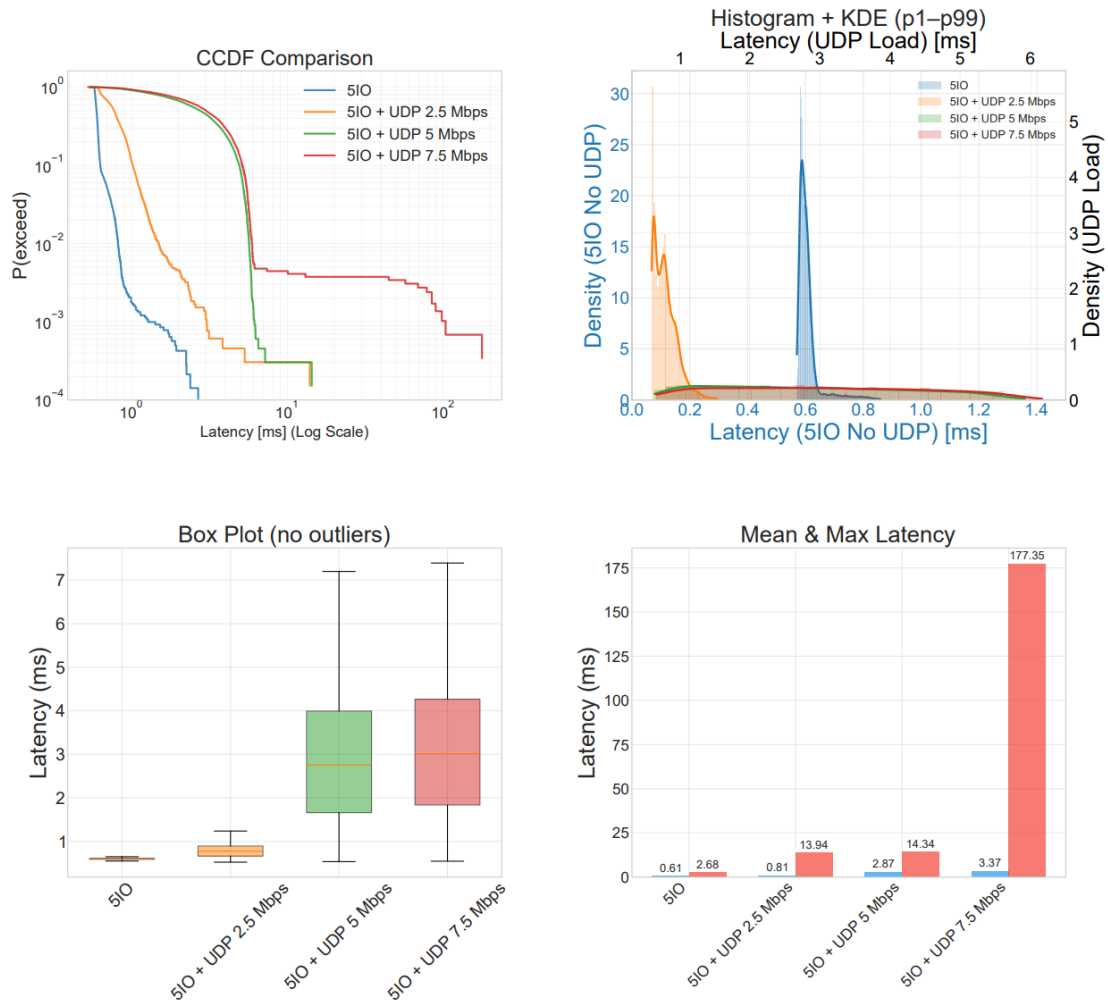


Figure 6.9: Wi-Fi (No QoS) End-to-End latency distribution under an increasing 64-byte UDP background load. **Top Left:** CCDF of latency thresholds. **Top Right:** Histogram and KDE of latency distributions. **Bottom Left:** Box plots of latency distributions. **Bottom Right:** Mean and maximum latency bar chart.

- **At 5.0 Mbps (9921 pps):** The primary latency density peak shifts from the sub-millisecond baseline to a dispersed concentration around 2.8 ms. Standard deviation jitter increases to 2.205%. Despite queuing pressure, the bridge hardware serializes packets fast enough to prevent 1920 ms watchdog timeouts and maintains 0.000% PROFINET packet loss.
- **At 7.5 Mbps (14804 pps):** The CCDF plot shows a severe flat tail extending beyond 10^2 ms. Maximum latency spikes to 177.35 ms, and the system drops 7.736% of PROFINET packets, triggering fail-safe states.

IEEE 802.11 MAC layer overhead causes this massive discrepancy between the 65 Mbps (large frame) and 7.5 Mbps (small frame) failure points. Every wireless transmission requires a fixed temporal investment regardless of payload size: a physical layer preamble, a DIFS, random backoff contention windows, and a mandatory MAC-level acknowledgment (ACK) separated by a SIFS [7].

To generate 5.0 Mbps of throughput, the radio processes roughly 450 pps using 1400-byte payloads, but nearly 10,000 pps using 64-byte payloads. At these extreme packet densities ($> 14,000$ pps at 7.5 Mbps), protocol headers and mandatory inter-frame silent periods consume almost a lot of available airtime. Actual payload data occupies only a tiny fraction of the channel.

Consequently, media contention overhead, rather than raw throughput, bottlenecks the Wi-Fi link. The radio simply runs out of available airtime to negotiate channel access for the massive volume of frames. Because the hardware cannot transmit frames onto the wireless medium as fast as they arrive from the wired network, the transmit queues quickly overflow. This buffer saturation is the direct cause of the discarded packets and the extreme latency spikes observed in the data.

6.4.3 Effect of QoS (WMM) on Wi-Fi Load Resilience

To evaluate if IEEE 802.11e QoS (WMM) mitigates the physical saturation points identified previously, the load tests were repeated with WMM enabled. PROFINET RT frames, tagged with VLAN PCP 6, are mapped to the high-priority "Voice" AC (AC_VO), granting them shorter Arbitration Inter-Frame Space (AIFS) and smaller contention windows compared to the best-effort UDP traffic. UDP has no special priority and is mapped to the "Best Effort" AC [10].

Table 6.11 and Table 6.12 present the performance metrics for the QoS-enabled architecture under large and small payload profiles, respectively.

Comparing these results to the No-QoS baseline shows that WMM prioritization fails to increase the physical load ceiling. The performance curves and breaking points remain identical similar to the No-QoS architecture.

6.4.3.1 Architectural Explanation of QoS Ineffectiveness

Virtual AP (VAP) separation, detailed in Section 5.4, causes this lack of QoS effectiveness.

WMM (EDCA) normally provides QoS by dividing a single, unified network queue into priority-based access categories. However, the VAP architecture bypasses this mechanism by creating isolated logical queues for each broadcasted SSID. Because PROFINET and IT traffic use separate VAPs, they never share the same logical queue.

Consequently, WMM prioritization becomes ineffective. Tagging a PROFINET frame with AC_VO priority only moves it to the front of the isolated PROFINET queue. It establishes no priority over the

Table 6.11 Wi-Fi's KPIs with QoS enabled under 1400-byte UDP load (64 ms cycle, 5 IO-Devices).

5 IO + (UDP Load)	Net Load (% of 100 Mbps)	Total Rate (pps)	Mean (ms)	Max (ms)	P95 (ms)	P99 (ms)	Std Jitter (% cycle)	RFC 3550 (% cycle)	PN Loss (%)
5 IO + No UDP	0.1035	156.2	0.5672	1.5492	0.6361	0.7644	0.067	0.022	0.000
5 IO + 5 Mbps	5.1035	602.7	0.5806	1.6316	0.7531	0.8501	0.111	0.081	0.000
5 IO + 10 Mbps	10.1035	1049.1	0.6688	2.6978	0.8666	1.0545	0.171	0.124	0.000
5 IO + 20 Mbps	20.1035	1942.0	0.7570	3.3994	1.0187	1.2651	0.247	0.232	0.000
5 IO + 40 Mbps	40.1034	3727.7	1.0174	14.8731	1.9872	2.6480	0.797	0.517	0.000
5 IO + 60 Mbps	60.1035	5513.4	1.9248	8.7942	3.9959	5.4281	1.588	1.528	0.000
5 IO + 65 Mbps	65.0922	5959.8	3.1970	15.9715	5.9440	6.3575	2.601	2.119	0.000
5 IO + 67 Mbps	67.0856	6138.4	3.3429	88.1641	5.9679	6.4758	3.832	2.469	6.543

Table 6.12 Wi-Fi's KPIs with QoS enabled under 64-byte UDP load (64 ms cycle, 5 IO-Devices).

5 IO + (UDP Load)	Net Load (% of 100 Mbps)	Total Rate (pps)	Mean (ms)	Max (ms)	P95 (ms)	P99 (ms)	Std Jitter (% cycle)	RFC 3550 (% cycle)	PN Loss (%)
5 IO + No UDP	0.1035	156.2	0.5672	1.5492	0.6361	0.7644	0.067	0.022	0.000
5IO + UDP 2.5 Mbps	2.6035	5039.1	0.8762	6.6135	1.3986	1.8358	0.441	0.319	0.000
5IO + UDP 5 Mbps	5.0872	9921.9	2.9592	6.1980	5.3008	5.6627	2.293	1.886	0.000

Wi-Fi E2E Latency vs UDP Load (64 ms Cycle Time, 1400 Bytes UDP Packets) QoS

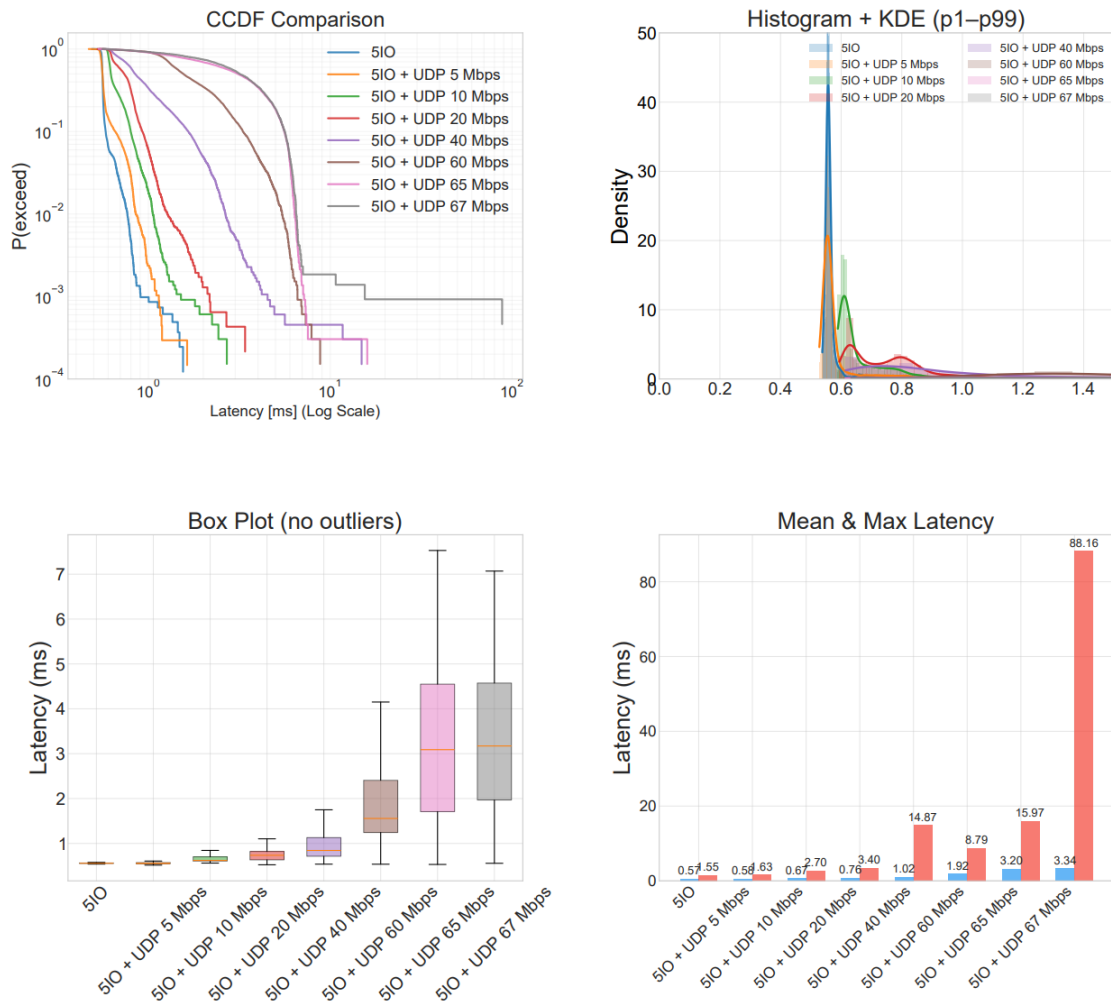


Figure 6.10: Wi-Fi (QoS Enabled) End-to-End latency distribution under an increasing 1400-byte UDP background load. **Top Left:** CCDF of latency thresholds. **Top Right:** Histogram and KDE of latency distributions. **Bottom Left:** Box plots of latency distributions. **Bottom Right:** Mean and maximum latency bar chart.

Wi-Fi E2E Latency vs UDP Load (64 ms Cycle Time, 64 Bytes UDP Packets) QoS

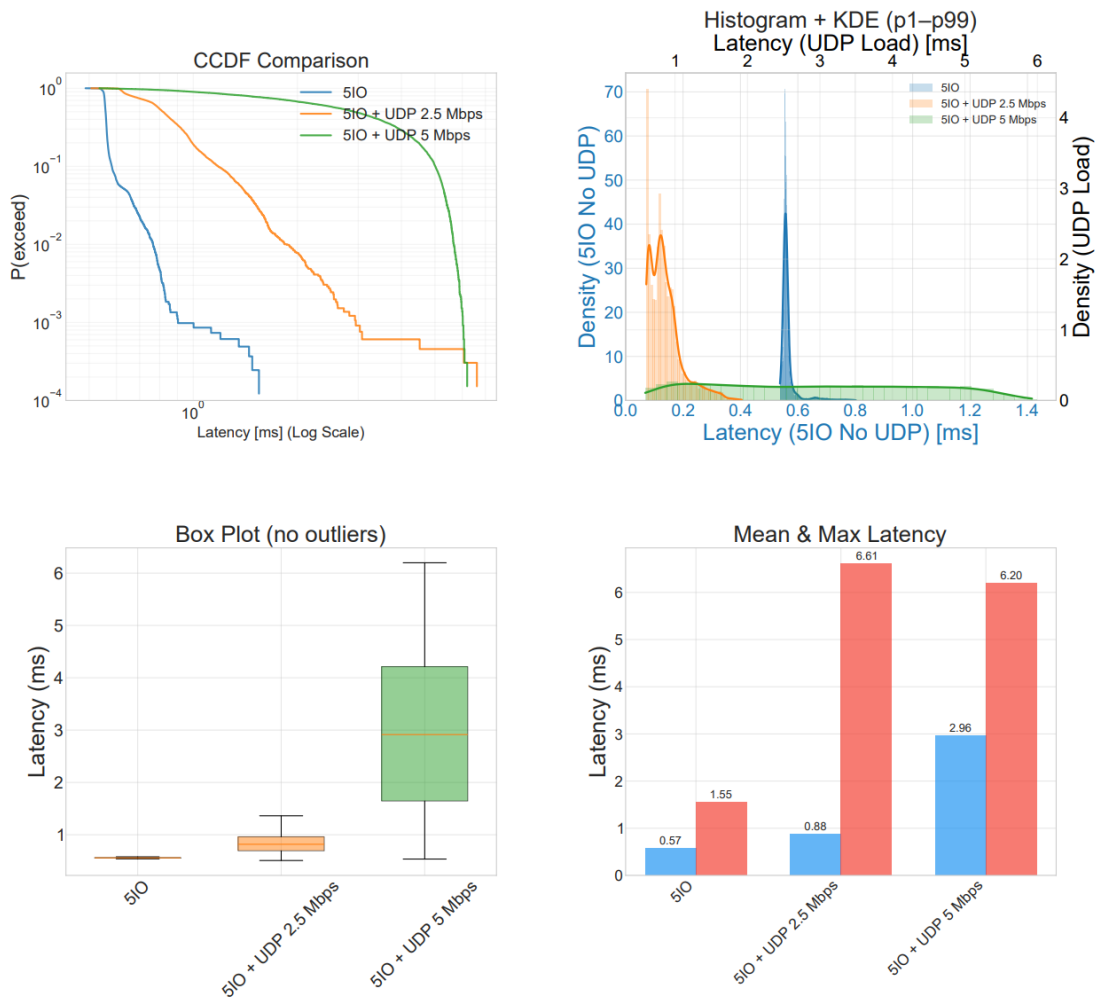


Figure 6.11: Wi-Fi (QoS Enabled) End-to-End latency distribution under an increasing 64-byte UDP background load. **Top Left:** CCDF of latency thresholds. **Top Right:** Histogram and KDE of latency distributions. **Bottom Left:** Box plots of latency distributions. **Bottom Right:** Mean and maximum latency bar chart.

UDP traffic residing in the separate IT queue. Ultimately, while VAP separation provides necessary logical isolation, it completely neutralizes WMM cross-traffic prioritization.

6.4.4 Comparative Analysis: Bluetooth vs. Wi-Fi Under Network Load

The network load measurements reveal different characteristics for both architectures. Table 6.13 summarizes their performance under stress.

Table 6.13 Comparative summary of Bluetooth and Wi-Fi under network load.

Characteristic	Bluetooth Architecture	Wi-Fi Architecture	Best Architecture
IT Traffic Capacity (Large UDP Packets)	Extremely Low Narrow 1 MHz physical channels, TDMA fragmentation and BNEP overhead limit the total throughput.	High CSMA/CA efficiently serializes large frames when the channel is secured, limited by negotiated bitrate.	Wi-Fi
OT Traffic Capacity (Small UDP Packets)	Extremely Low High packet frequency instantly exhausts the fixed TDMA polling slots.	Moderate High packet frequency exhausts physical airtime due to protocol overhead	Wi-Fi

6.4.4.1 Wi-Fi: The Shared Infrastructure Standard

The data proves Wi-Fi is the only viable architecture for a shared industrial network. CSMA/CA efficiently handles bulk data transfers, stably routing up to 54 Mbps of background IT traffic alongside the PROFINET traffic. However, high-frequency traffic (e.g. More PROFINET cyclic frames or small data packets) overwhelms Wi-Fi at a lower netload than with bigger less frequent payloads. Because Wi-Fi imposes protocol overhead on every frame, injecting large quantities of small payloads quickly exhausts the physical airtime well below its theoretical throughput limit.

6.4.4.2 Bluetooth: For Deterministic, Low Throughput, Dedicated Links

The load tests explicitly confirm Bluetooth 2.1 + EDR cannot support mixed-traffic environments. Narrow 1 MHz physical channels, multi-layer encapsulation (BNEP/L2CAP) and TDMA scheduling cause the link to choke under mere kilobits of UDP traffic. So it is a cheaper alternative for low throughput, deterministic, dedicated OT links.

6.4.4.3 Engineering Assessment for Network Load

If an industrial deployment requires converging OT and IT traffic over a single wireless medium, engineers must select the Wi-Fi architecture. To guarantee stability, network administrators must prevent the total throughput from exceeding the negotiated bitrate at all time and high-frequent traffic from causing airtime starvation. Conversely, if the design physically separates IT and OT traffic,

deploying the Bluetooth architecture as an isolated, dedicated OT link can be a more deterministic alternative.

6.5 Architectures under Attenuation (P3)

Dynamic shadowing causes Fail-Safe states in the ArcelorMittal Steelshop (Section 4.2). To replicate this, copper shielding and RF-blocking fabric isolated the attenuated Client signal. Adjusting this shielding degraded the RSSI, mimicking LoS obstructions and distance-based path loss. All measurements used a 64 ms cycle time, a 1920 ms Watchdog Timer, and a 5 IO-Device load.

Note: The lab environment lacked sufficient RF-blocking material to shield both redundant lines equally. To prevent the PRP deduplication mechanism from skewing the results by masking a degraded link with a healthier one, the test methodology intentionally disconnected Line A during extreme attenuation scenarios (marked as “OFF”). This isolated Line B, allowing accurate measurement of the absolute failure thresholds of a single link.

6.5.1 Bluetooth Performance Under Attenuation

While Bluetooth radio waves physically attenuate when obstructed, the protocol maintains a highly stable latency profile regardless of signal strength. Table 6.14 and Figure 6.12 detail the E2E latency distributions across seven attenuation states.

Table 6.14 Bluetooth’s KPIs under various attenuation scenarios. “OFF” indicates that the line was disconnected.

Line A /Line B RSSI (dBm)	Net Load (% of 100 Mbps)	Total Rate (pps)	Mean (ms)	Max (ms)	P95 (ms)	P99 (ms)	Std Jitter (% cycle)	RFC 3550 (% cycle)	PN Loss (%)
5 IO Baseline	0.1061	156.2	15.5906	49.1604	20.6739	29.2022	4.979	4.169	0.000
A:−40 / B:−75	0.1083	156.2	14.3431	29.8358	19.8258	24.6862	4.168	3.516	0.000
A:−75 / B:−40	0.1066	156.2	15.1707	40.2862	25.2771	31.5638	6.524	5.499	0.000
A:−65 / B:−65	0.1066	156.2	14.6643	32.8752	19.5059	22.9793	3.891	3.563	0.000
A:−75 / B:−75	0.1083	156.2	14.4214	30.9216	20.0766	23.0829	4.000	3.638	0.000
A:OFF / B:−85	0.1081	156.2	15.3048	36.9520	21.0239	26.0105	4.455	3.844	0.000
A:OFF / B:−92	0.1081	156.2	19.0799	60.7747	29.5992	37.4764	8.331	8.083	0.101

6.5.1.1 Latency Determinism and PRP Masking under Attenuation

The latency density curves (Figure 6.12, (Top right)) remain nearly identical from the baseline down to the ‘OFF / −85 dBm’ configuration. Mean latency stays bounded between 14.34 ms and 15.59

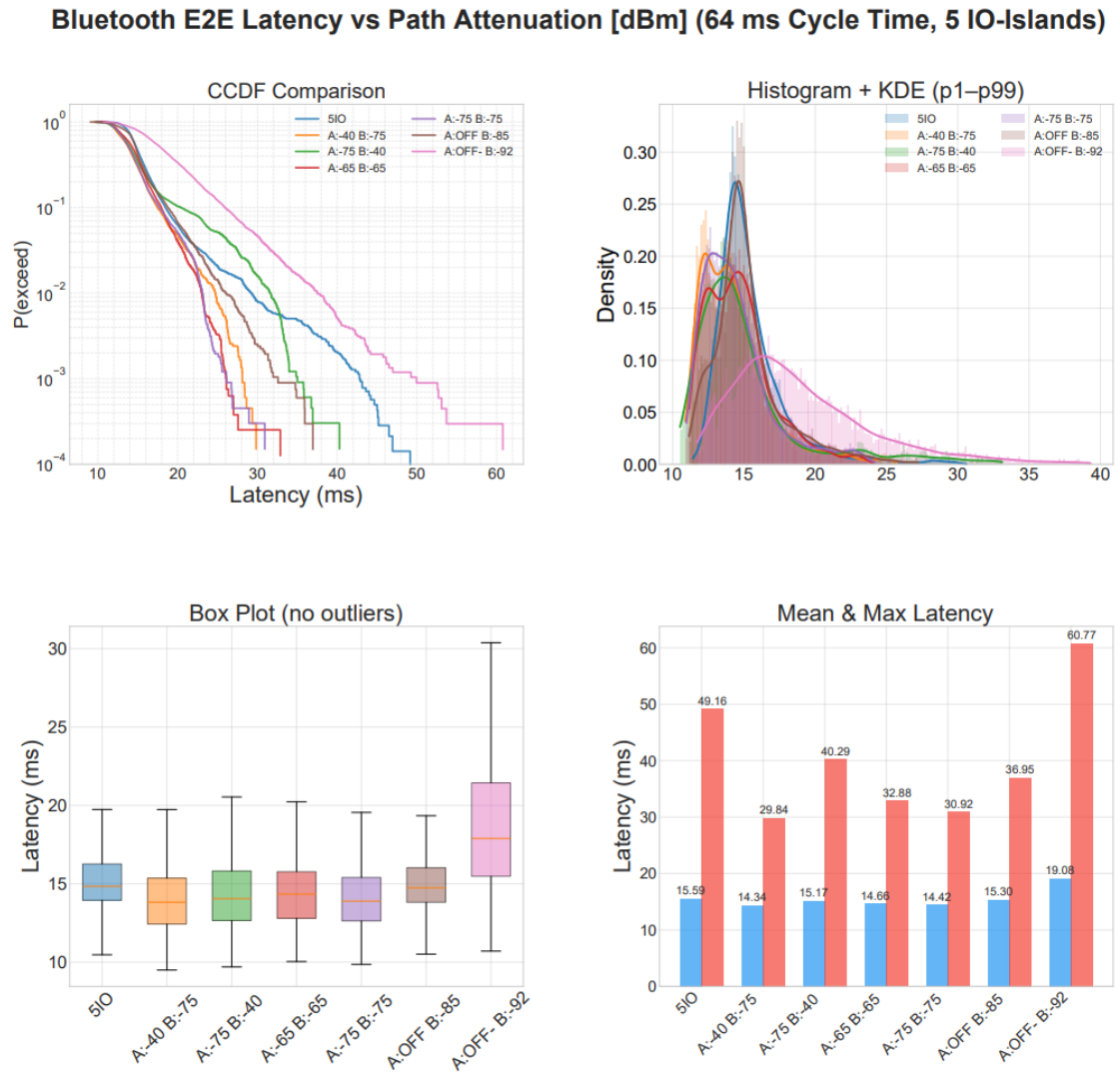


Figure 6.12: Bluetooth End-to-End latency distributions under progressive signal attenuation. **Top Left:** CCDF of latency thresholds. **Top Right:** Histogram and KDE of latency distributions. **Bottom Left:** Box plots of latency distributions. **Bottom Right:** Mean and maximum latency bar chart.

ms, and the IQRs do not expand.

Two mechanisms cause this stability. First, PRP seamlessly routes traffic over the healthy path if one line drops. Second, Bluetooth's AFH mechanism bypasses localized interference on the surviving link, preventing packet loss and maintaining the fixed TDMA timing floor down to an RSSI of -85 dBm [12].

6.5.1.2 The Noise Floor and Failure State

The link fails at the 'OFF / -92 dBm' configuration. The latency distribution shifts right, mean latency reaches 19.08 ms, and the system records 0.101% PROFINET packet loss.

At -92 dBm, the signal hits the hardware noise floor. The radio attempts MAC-layer retransmissions, but the low SNR forces the hardware to drop corrupted frames. Crucially, the 60.77 ms maximum latency remains well within the 1920 ms watchdog limit. This proves that unrecoverable physical signal loss, not protocol transmission delay, causes the packet drops.

6.5.2 Wi-Fi Performance Under Attenuation

The Wi-Fi architecture was subjected to the identical shielding methodology. These measurements also track MCS fallback, reflecting the negotiated bitrate at each attenuation level. Table 6.15 and Figure 6.13 detail the results.

Table 6.15 Wi-Fi's KPIs under various attenuation scenarios (64 ms cycle, 5 IO-Devices). "OFF" indicates that the line was disconnected to simulate connection loss.

Line A / Line B RSSI (dBm) (Bitrate (Mbps))	Net Load (% of 100 Mbps)	Total Rate (pps)	Mean (ms)	Max (ms)	P95 (ms)	P99 (ms)	Std Jitter (% cycle)	RFC 3550 (% cycle)	PN Loss (%)
5 IO Baseline	0.1034	156.2	0.6094	2.6781	0.7005	0.8187	0.098	0.039	0.000
A:-40 / B:-75 (A:54, B:48)	0.1255	156.2	0.6125	2.8489	0.7492	0.8348	0.110	0.034	0.000
A:-75 / B:-40 (A:36, B:54)	0.1034	156.2	0.5708	2.7638	0.6039	0.7730	0.126	0.035	0.000
A:-65 / B:-65 (A:54, B:54)	0.1104	156.2	0.6045	2.3672	0.6284	0.7606	0.079	0.036	0.000
A:-75 / B:-75 (A:48, B:48)	0.1255	156.2	0.5683	1.4080	0.5931	0.7362	0.063	0.032	0.000
A:OFF / B:-85 (B:24)	0.1254	156.2	1.4442	69.7715	0.8973	37.9292	9.029	2.531	0.007
A:OFF / B:-90 (B:18)	0.1253	156.2	1.5037	68.1577	0.8614	40.8102	9.419	2.664	0.029
A:OFF / B:-93 (B:6)	0.0744	156.2	2.1345	65.8033	7.7973	43.6292	9.841	3.663	2.423

Wi-Fi E2E Latency vs Path Attenuation [dBm] (64 ms Cycle Time, 5 IO-Islands)

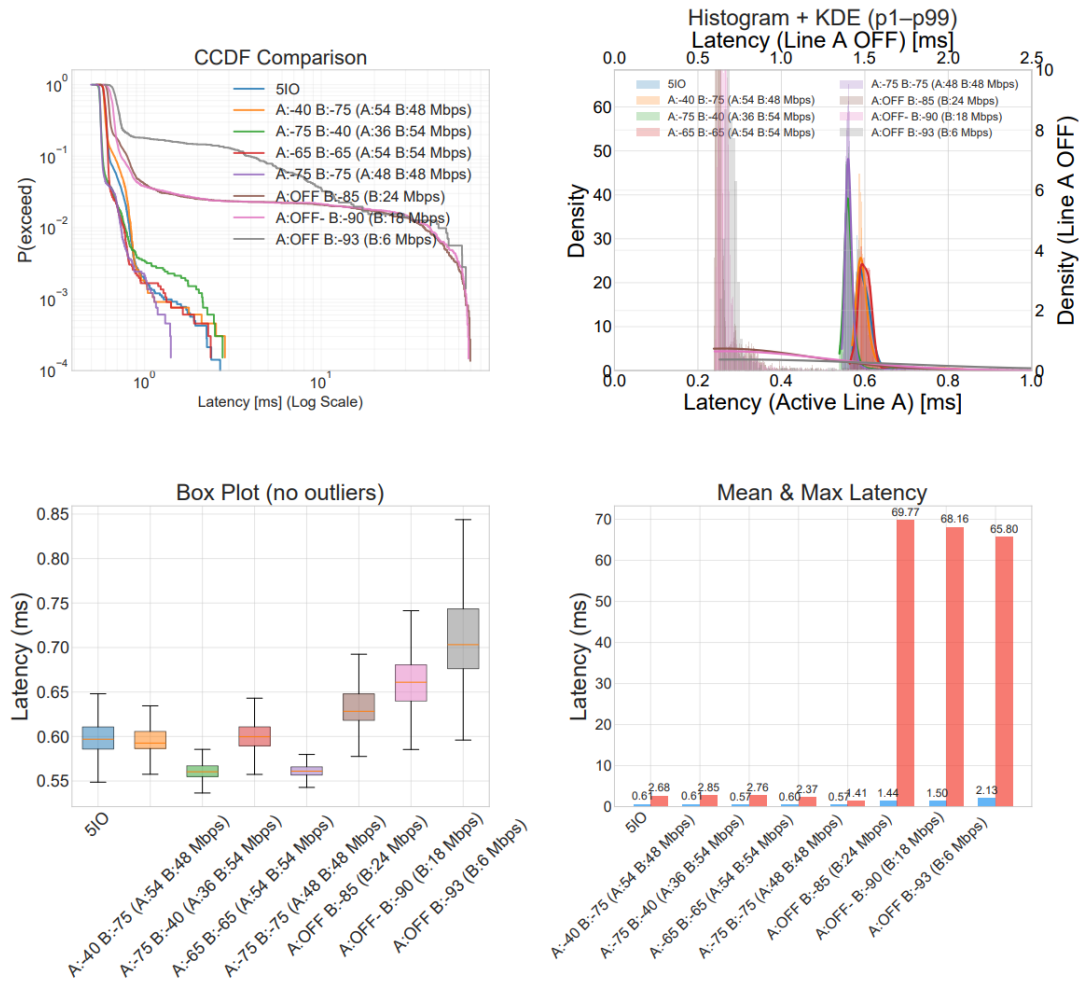


Figure 6.13: Wi-Fi End-to-End latency distributions under progressive signal attenuation. **Top Left:** CCDF showing the probability of exceeding latency thresholds. **Top Right:** Histogram and KDE showing the spread of the latency. **Bottom Left:** Box plot of latencies. **Bottom Right:** Mean and maximum latency bar chart.

6.5.2.1 Stability and PRP Deduplication

With both PRP lines active, the network masks physical signal degradation. Even at -75 dBm on both paths, the network performance is virtually indistinguishable from the Wi-Fi baseline. It maintains a mean latency around 0.60 ms and the maximum latency stays below 2.85 ms 6.13.

The hardware sustains a minimum 48 Mbps MCS bitrate at this attenuation. Because 48 Mbps easily accommodates the netload of 0.1255% (156.2 pps), PRP seamlessly delivers the fastest duplicate frame and preserves E2E timing determinism.

6.5.2.2 Single-Line Vulnerability and Packet Loss

The system behavior degrades starting at -85 dBm, the mean latency jumps to ~1.44 ms, and the standard deviation (jitter) expands from 0.06% to 9.0%. The CCDF plot illustrates this shift: the curves starting from -85 dBm and lower detach from the other curves and develop "fat tails", with P99 latencies above 37 ms.

The system begins dropping PROFINET packets (0.007%) at -85 dBm, escalating to 2.423% at -93 dBm. Because Wi-Fi does not hop frequencies, CSMA/CA retransmissions fail in the noisy environment, causing packet drops well before the absolute hardware noise floor.

As the signal further degrades, the performance of the system degrades accordingly.

6.5.2.3 MCS Fallback and Load Capacity Bottlenecks

To maintain a connection under attenuation, the IEEE 802.11 radio triggers dynamic MCS fallback. The bitrate drops from 54 Mbps at baseline to 6 Mbps at -93 dBm. While 6 Mbps sustains the 5 IO-Devices isolated PROFINET loop, it restricts total network capacity. In a shared IT/OT environment, a signal drop to -85 dBm reduces the bitrate to 24 Mbps. Standard background traffic would instantly saturate this degraded link.

Attenuation acts as a bottleneck on mixed-traffic throughput. To guarantee stability, the netload cannot exceed the negotiated bitrate under attenuation, as the possible throughput is limited by the negotiated bitrate.

This exposes the vulnerabilities related to shadowing and a "Sticky Client". Whether a Client "sticks" to a distant AP, or its LoS is obstructed, the resulting MCS fallback will reduce the available throughput. This highlights the necessity of the dual-Client PRP architecture: it ensures at least one link maintains a high-bitrate connection, preventing the entire system from bottlenecking due to dynamic shadowing, handovers, or cell-edge signal degradation.

6.5.3 Comparative Analysis: Bluetooth vs. Wi-Fi Under Attenuation

Table 6.16 summarizes how each technology handles physical signal degradation.

Table 6.16 Comparative summary of Bluetooth and Wi-Fi under progressive RF attenuation.

Characteristic	Bluetooth Architecture	Wi-Fi Architecture	Best Architecture
Predictability under Weak Signal	Stable Mean latency and jitter remain statically bounded even at -85 dBm. AFH smoothly routes around noise.	Volatile Degraded signal triggers retransmissions and MCS fallback, causing latency spikes and high jitter.	Bluetooth
Packet Loss	Robust Maintains 0.000% loss down to -85 dBm.	Vulnerable Begins dropping packets (0.007%) at -85 dBm, increasing with further attenuation.	Bluetooth
Load Capacity	Static Fixed Capacity Modulation profile is fixed. Baseline capacity is low but stable.	Dynamic (MCS Fallback) Bitrate drops dynamically (e.g., 54 → 6 Mbps), shrinking the available network capacity.	Topology Dependent

6.5.3.1 MAC-Layer Robustness Under Signal Degradation

Bluetooth demonstrates superior MAC-layer robustness under attenuation. It maintains a flawless 0.000% PROFINET packet loss down to -85 dBm. Because TDMA scheduling dictates its baseline latency and the AFH mechanism avoids the applied attenuation without dropping frames, its timing profile remains predictable regardless of signal strength, failing only at the -92 dBm noise floor. Conversely, Wi-Fi CSMA/CA mechanisms struggle with low SNR, leading to packet drops and jitter expansion at -85 dBm.

6.5.3.2 Engineering Assessment for Attenuation

If the deployment uses spatial diversity to ensure dynamic shadowing only degrades one path at a time, both architectures succeed. PRP successfully masks their respective vulnerabilities. However, PRP is a redundancy protocol, not a signal amplifier; it can only mask the impact of attenuation as long as one path remains healthy. If both paths degrade simultaneously, the technologies diverge:

- **Bluetooth** proves more suitable for dedicated PROFINET control links operating at the fringes of RF coverage. By leveraging TDMA slot scheduling and AFH, it maintains a predictable latency, low jitter, and flawless packet delivery much deeper into RF noise.

- **Wi-Fi** becomes structurally unstable at deeper RF attenuation (-85 dBm). To deploy Wi-Fi viably, engineers must stagger APs as described in Section 5.4, to guarantee that mobile clients remain within optimal signal boundaries (e.g., above -65 dBm) and prevent capacity loss associated with MCS fallback.

6.6 Seamless Handover Validation (P4)

Overcoming signal shadowing and “Sticky clients” within the ArcelorMittal infrastructure requires seamless handover. This validation scenario proves that the proposed redundant architectures completely mask handover delays.

6.6.1 Test Methodology

Manually disconnecting the Ethernet cables between the wireless bridges and the RedBoxes simulated a path disconnection handover, resulting in a loss of the Access Point connection. The test executed four distinct phases:

1. **Phase 1 (Pull Line A):** Simulates Client A disconnecting to roam, leaving only Line B active.
2. **Phase 2 (Pull Both):** A brief complete interruption to establish a zero-data baseline.
3. **Phase 3 (Pull Line B):** Simulates Client B disconnecting to roam, leaving only Line A active.
4. **Phase 4 (Both Restored):** Normal dual-path operation resumes.

6.6.2 Time-Series Analysis: The PRP Smoothing Effect

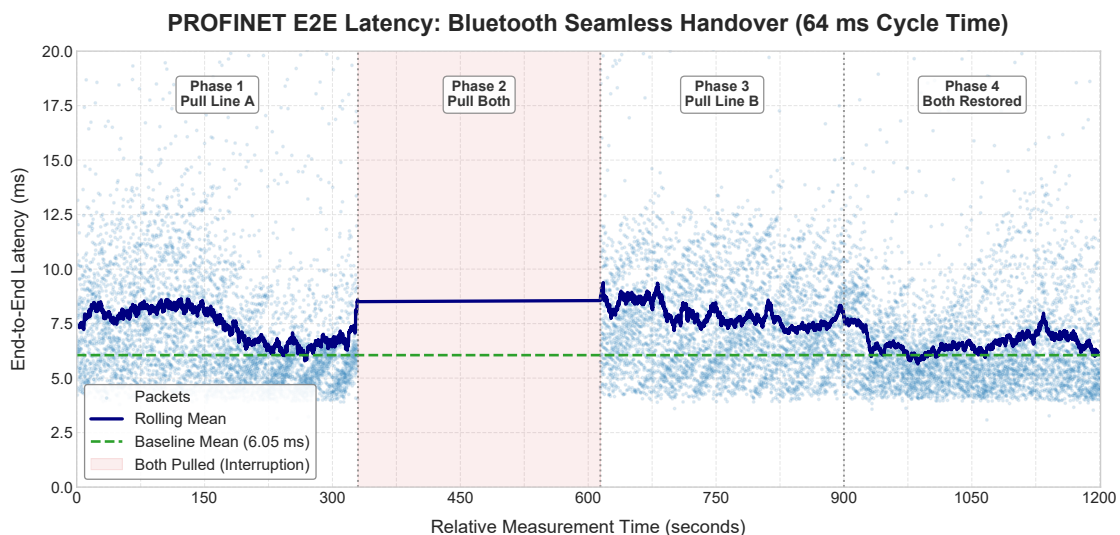


Figure 6.14: Bluetooth E2E latency time series during the seamless handover test.

The time-series plots (Figure 6.14 and Figure 6.15) proves the PRP architecture achieves seamless handover. During Phase 1 and Phase 3, the connection survives without triggering Fail-Safe timeouts.

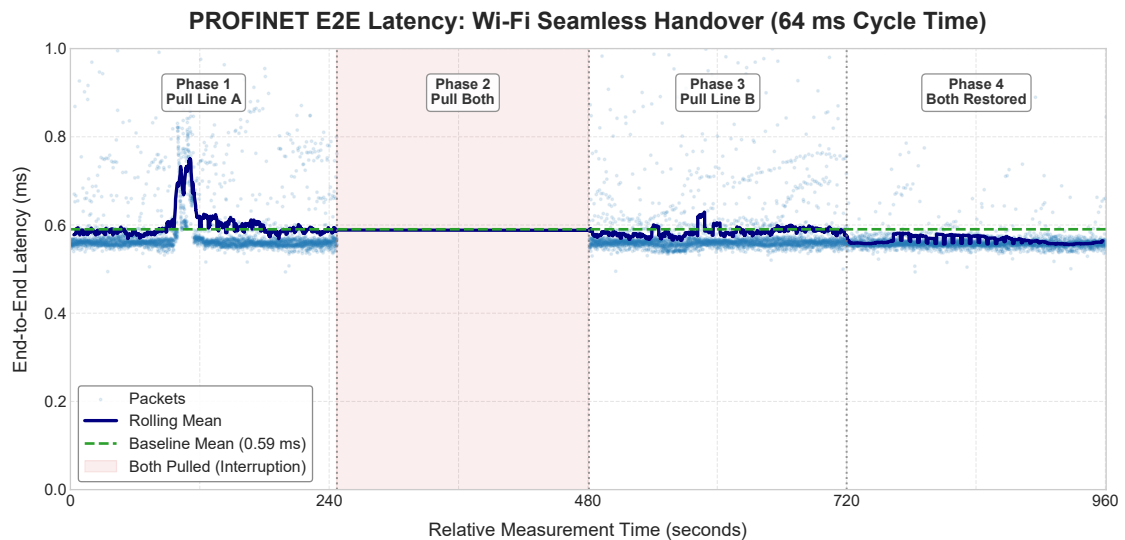


Figure 6.15: Wi-Fi E2E latency time series during the seamless handover test.

The data also highlights PRP's latency smoothing effect. When both lines are active (the pre-test baseline and Phase 4), PRP processes the fastest-arriving duplicate frame. Rather than merely masking latency spikes, this mechanism filters out all minor cycle-to-cycle timing differences. By evaluating both paths packet-by-packet and selecting the minimum latency, it achieves a tighter variance and a lower overall mean latency than either individual link.

Disconnecting a single cable removes this advantage, exposing the raw variance of the surviving link. During Phase 1 and Phase 3, the latency scatter widens and the rolling mean elevates. The Wi-Fi rolling mean spikes briefly to ~ 0.75 ms before settling near 0.60 ms. The Bluetooth rolling mean rises from a 6.05 ms baseline to ~ 8.50 ms. Restoring the second link immediately reinstates the smoothing.

6.6.3 Engineering Assessment for Seamless Handover

Deploying PRP over either Bluetooth or Wi-Fi completely neutralizes handover delays. One wireless Client maintains the critical control loop while the parallel Client scans and authenticates with a new Access Point. This architecture frees the system from standard single-Client roaming thresholds, guaranteeing uninterrupted network stability for mobile automation.

6.7 Key Takeaways

The measurement data establishes distinct operational boundaries for both wireless architectures. Table 6.17 maps the observed failure modes to actionable engineering guidelines.

6.7.1 Seamless Handover and Latency Smoothing

The primary objective of this research was to eliminate PROFINET handover failures. The data proves that deploying PRP across either Wi-Fi or Bluetooth successfully masks handover gaps. Furthermore,

Table 6.17 Matrix of key empirical findings and hardware limitations per test scenario.

Test Scenario	Wi-Fi Architecture	Bluetooth Architecture	Core Engineering Takeaway
P1: Baseline	Sub-millisecond mean latency (~ 0.6 ms), but vulnerable to rare statistical outliers.	Fixed ~ 6 ms processing floor with mathematically bounded latency spikes.	Wi-Fi offers lower mean latency; Bluetooth guarantees strict timing determinism.
P2: Network Load	Sustains mixed IT/OT loads but has airtime exhaustion under small-packet traffic.	Narrow 1 MHz physical channels, TDMA scheduling and protocol encapsulation restrict throughput.	Wi-Fi supports shared infrastructure; Bluetooth strictly requires a dedicated link.
P3: Attenuation	MCS fallback dynamically shrinks capacity, creating hidden bottlenecks for background IT traffic.	AFH and TDMA maintain a stable latency distribution down to the -92 dBm hardware noise floor.	Wi-Fi capacity fluctuates with signal strength; Bluetooth maintains determinism until absolute physical failure.
P4: Handover	Survives sequential disconnections seamlessly without PROFINET packet loss.	Survives sequential disconnections seamlessly without PROFINET packet loss.	PRP successfully eliminates the handover gap and smooths out latency variance.
Topology Constraints	Requires VLAN isolation to separate Line A and Line B traffic.	Requires dedicated point-to-point hardware for isolation.	Bridging redundant lines causes fatal MAC flapping; absolute path isolation is mandatory.

the PRP deduplication mechanism actively smooths the baseline latency by constantly selecting the fastest frame from the redundant paths, improving overall cycle-to-cycle stability.

6.7.2 Strict Path Isolation

The exploratory point-to-multipoint topology proves that logical network isolation is mandatory. Routing both PRP Line A and Line B through a single unmanaged switch merges the parallel paths into one collision domain. This triggers severe MAC flapping, hardware error states, and latency inflation. Industrial deployments must enforce strict path isolation using either dedicated point-to-point hardware connections or VLANs.

6.7.3 Throughput vs. Determinism Trade-off

The empirical data exposes a fundamental architectural choice between raw capacity and timing determinism. Wi-Fi operates as the only viable solution for shared IT/OT infrastructures. However, its total capacity is strictly bound to signal integrity; RF attenuation forces dynamic MCS fallback, which shrinks the available capacity and risks saturating the OT traffic with background IT traffic.

Conversely, Bluetooth's narrow 1 MHz physical channels and TDMA structure cannot support mixed IT/OT traffic. Yet, this scheduling, combined with AFH, grants Bluetooth exceptional RF determinism. It survives deep into the hardware noise floor without latency degradation, making it a robust, dedicated wireless cable replacement.

6.7.4 QoS Ineffectiveness in VAP Architectures

Finally, the load tests prove that IEEE 802.11e WMM (QoS) fails to prioritize PROFINET traffic in this specific deployment. Because the setup requires Virtual APs (VAPs) to maintain network isolation, the hardware segregates the IT and OT traffic streams into separate logical queues. This isolation bypasses the WMM protocol entirely, neutralizing cross-traffic prioritization.

Chapter 7

Industrial Deployment Recommendation and Cost Analysis for ArcelorMittal Gent

To eliminate the Fail-Safe standstills caused by dynamic shadowing and handover gaps within the ArcelorMittal Gent Steelshop, this research recommends the implementation of the **Redundant Wi-Fi Architecture**.

For a evaluation comparing standard wireless links against the redundant PRP architectures, refer to the full datasheet in Appendix A (Tables A.1 and A.2).

7.1 Why Wi-Fi over Bluetooth?

ArcelorMittal Gent's crane automation requires a network capable of simultaneously handling IT traffic and OT traffic (PROFINET). While the Bluetooth architecture maintains a consistent latency profile under signal attenuation until the signal drops below the hardware's minimum receiver sensitivity, its throughput limitations make it unsuitable for this mixed-traffic convergence.

Unlike Wi-Fi, which utilizes wide 20 MHz or 40 MHz channels, Bluetooth operates on narrow 1 MHz physical channels. Combined with TDMA time-slot allocations, this fundamental physical constraint restricts the maximum throughput, causing network saturation when attempting to route IT loads.

Therefore, this research recommends the redundant Wi-Fi architecture for the following reasons:

- **Multi-Device Scalability:** When measuring a typical setup (16 IO-Devices at a 64 ms cycle time, executed as a 4 IO-Devices test at 16 ms cycle time, (see Figures B.1 and B.2)), it reveals that the Bluetooth architecture cannot handle this IO-load, with a mean latency increasing to 30.3 ms and maximum latency reaching 46.4 ms. Conversely, the Wi-Fi architecture effortlessly handles this exact configuration, maintaining a stable sub-millisecond mean latency (~ 0.62 ms) and a maximum latency of just 2.6 ms.
- **Superior Overall Performance and Capacity:** The Wi-Fi setup delivers sub-millisecond baseline mean latency (~ 0.6 – 0.8 ms), minimized maximum latency, and sustains high IT network

loads without losing PROFINET frames.

- **IT/OT Convergence:** Wi-Fi supports the convergence of IT and OT traffic over the same infrastructure, provided the load remains below the negotiated bitrate at all times.
- **Future-Proof Scalability:** As automation complexity grows, data throughput demands will inevitably increase. The Wi-Fi architecture provides the necessary capacity to accommodate OT and IT expansions ensuring the infrastructure remains viable long-term.
- **Vulnerability Masking:** While a single Wi-Fi Client is vulnerable to handover gaps, fluctuating bitrates, and dynamic shadowing, applying PRP successfully neutralizes the impact of these.

Deploying PRP over Wi-Fi enables ArcelorMittal Gent to automate its cranes using a mixed OT and IT network and guaranteeing the deterministic reliability required by the PROFINET Watchdog.

7.2 Implementation and Scaling Strategy

7.2.0.1 Phase 1: Single Crane Trajectory Implementation

To support mixed traffic securely, the network must logically isolate the redundant PROFINET paths and physically segregate the IT network (VRON) to prevent bridging loops.

- **Static Infrastructure (Track Level):** The IO-Controller connects to a PRP module. This module duplicates the PROFINET frames and sends them via Line A and Line B to a Managed Switch. The managed switch ingests the IT traffic (Rest of Network (RON)) and applies strict VLAN tags: VLAN A (e.g. VLAN 10) for Line A, VLAN B (e.g. VLAN 20) for Line B, and a dedicated VLAN for the IT traffic (VRON, e.g. VLAN 30). Two APs broadcast these streams over non-overlapping 5 GHz channels, using VAPs to broadcast separate SSIDs for each VLAN.
- **Mobile Infrastructure (Crane Level):** The crane uses three independent Wi-Fi Clients to maintain strict hardware isolation. Client A connects to the Line A SSID and Client B connects to the Line B SSID. They strip the VLAN tags and forward the OT traffic to the crane's PRP module, which accepts and forwards the first frame and discards the duplicate. A third Client connects to the VRON SSID to exclusively handle IT traffic.

7.2.1 Phase 2: Scaling to the Full AMG Runway

When scaling this solution to cover the full length of the runway with multiple active cranes, spatial and frequency planning become critical to maintain isolation:

- item **Staggered AP Topology:** Engineers must install APs in a staggered layout, alternating between Line A and Line B. The RF coverage cells of APs on the same line must have zero physical overlap. This prevents bridging loops and ensures handovers on Line A and Line B never occur simultaneously.
- **Static Channel Planning:** Adjacent APs must operate on static, non-overlapping 5 GHz channels to eliminate Co-Channel Interference.

- **Aggressive Roaming Thresholds:** Configure the mobile clients to disconnect at a strict signal threshold (e.g., -70dBm) to proactively sever weak connections. The redundant active path maintains the connection, preventing Watchdog timeouts during the switch.

7.2.2 Advisable PROFINET Watchdog Configuration

Configuring the PROFINET Watchdog Timer requires balancing typical network performance with the absolute worst-case latency peaks. Translating the laboratory measurements directly to the ArcelorMittal Gent Steelshop requires caution. The environment introduces RF challenges, such as multipath fading, dynamic shadowing, and degraded SNR, which make exact latency predictions difficult. Furthermore, the probabilistic nature of Wi-Fi guarantees that transient latency spikes will occur, though they typically affect only isolated packets for very short durations.

Across all lab tests, the redundant Wi-Fi architecture in a stable configuration recorded an average latency around 1 ms and most of the time a maximum latency below 2 ms, with an absolute maximum latency of ~35.7 ms.

The typical maximum latency (12 ms) consumes only a negligible fraction of a 64 ms cycle time. Even a packet with the absolute maximum latency of 35.7 ms, still arrives well within a single 64 ms window. Theoretically, applying a industry standard PROFINET multiplier of 3 (192 ms Watchdog threshold) is sufficient. However, unless the industrial process dictates such a low threshold, it is advisable to configure a higher multiplier to maximize the safety margin. ArcelorMittal Gent's maximum desired Watchdog Timer of 2 seconds (2000 ms) is therefore supported by this architecture.

If operations demand a 16 ms cycle time, the average maximum latency (12 ms) easily fits within a single 16 ms cycle. However, the absolute maximum latency of 35.7 ms exceeds two full cycles. A multiplier of at least 3 (48 ms Watchdog Timer) becomes mandatory, but does not have a lot of safety margin. So a higher multiplier would be advisable and ArcelorMittal Gent's 2 seconds maximum desired Watchdog Timer is also possible.

7.3 Economic Feasibility and Cost Analysis

Table 7.1 outlines the estimated CAPEX utilizing current, undiscounted market pricing. The selected components directly reflect the hardware deployed during laboratory testing, substituting discontinued models with their modern equivalents. Crucially, the architecture designates the FL SWITCH 4300 strictly as the PRP redundancy box while retaining the Siemens SCALANCE XC208 for VLAN management. This hardware separation is mandatory; empirical testing has not yet verified whether the FL SWITCH 4300 can simultaneously execute PRP duplication and VLAN segregation without introducing processing latency. *Note: Cable costs are excluded from this CAPEX estimate as they vary depending on the specific application deployment.*

7.3.1 Segmented Infrastructure Costs

The total network hardware costs for a single operational zone and crane are divided as follows. Note that every FL SWITCH 4300 requires two FL SFP GT-H modules to convert the fiber ports to standard RJ45 copper for Line A and Line B.

Table 7.1 Estimated hardware component costs (undiscounted unit prices).

Component Category	Specific Hardware Model	Estimated Unit Price (EUR)
Wi-Fi Access Point/Client	Siemens SCALANCE W788-1 RJ45 [15]	1,282.65
Managed Switch	Siemens SCALANCE XC208 [16]	710.22
Bluetooth Module	Phoenix Contact FL EPA 2 RSMA [17]	757.35
PRP Module	Phoenix Contact FL SWITCH 4300 [17]	4,073.44
SFP to RJ45 Module	Phoenix Contact FL SFP GT-H [17]	282.98

- **Mobile Entity (Crane Level):**

- **Wi-Fi Architecture:** Requires 1 PRP Module (4,073.44 EUR), 2 SFP Modules (2 × 282.98 EUR), and 3 Wi-Fi Clients (3 × 1,282.65 EUR). **Total: 8,487.35 EUR.**
- **Bluetooth Architecture:** Requires 1 PRP Module (4,073.44 EUR), 2 SFP Modules (2 × 282.98 EUR), and 2 Bluetooth Modules (2 × 757.35 EUR). **Total: 6,154.10 EUR.**

- **Static Infrastructure Zone (Track Level):**

- **Wi-Fi Architecture:** Requires 1 PRP Module (4,073.44 EUR), 2 SFP Modules (2 × 282.98 EUR), 1 Managed Switch (710.22 EUR), and 2 Wi-Fi APs (2 × 1,282.65 EUR). **Total: 7,914.92 EUR.**
- **Bluetooth Architecture:** Requires 1 PRP Module (4,073.44 EUR), 2 SFP Modules (2 × 282.98 EUR), and 2 Bluetooth Modules (2 × 757.35 EUR). **Total: 6,154.10 EUR.**

7.3.2 Total System Capital Expenditure

To calculate the total investment for a full-scale deployment, let C represent the number of automated cranes and Z represent the number of static infrastructure zones required to ensure continuous RF coverage.

$$T_{\text{CAPEX}} = (C \times \text{Cost}_{\text{Mobile}}) + (Z \times \text{Cost}_{\text{Static}}) \quad (7.1)$$

For a minimum viable deployment consisting of a single crane operating within a single track zone ($C = 1, Z = 1$):

- **Total Wi-Fi CAPEX: 16,402.27 EUR**
- **Total Bluetooth CAPEX: 12,308.20 EUR**

While the Bluetooth setup represents a cheaper initial investment due to omitting the managed switch and requiring fewer hardware clients, it structurally fails to support the converging IT/OT requirements. Therefore, the Wi-Fi architecture remains the mandatory, future-proof investment for ArcelorMittal Gent.

Chapter 8

Conclusions and Future Work

The legacy single-path Wi-Fi network at ArcelorMittal Gent triggers frequent Fail-Safe states due to dynamic RF shadowing, fluctuating bitrates, and "Sticky Client" handovers. This thesis evaluated redundant wireless architectures to resolve this unreliability. Laboratory tests confirmed that applying PRP successfully eliminates handover gaps. By duplicating PROFINET frames over two independent wireless paths, the receiving RedBox accepts the fastest frame and discards the duplicate. This creates a resilient system with no single points of failure.

8.0.1 Successfully Masking the Handover Gap

The wireless PRP architecture successfully masked all handover gaps with zero packet loss. Simulating a handover event by disconnecting one path proved that the end-to-end PROFINET connection continues uninterrupted. The redundant path reliably delivered every frame, maintaining a consistent end-to-end latency and minimizing maximum latency. Furthermore, because the RedBox forwards the fastest arriving duplicate, the system automatically smooths out latency variance between the paths.

8.0.2 Evaluation of the Redundant Bluetooth Architecture

The Bluetooth architecture provides determinism and maintains link stability under signal attenuation.

- **Determinism under attenuation:** TDMA slots and AFH maintain a stable latency (~6.0–6.8 ms) and prevent packet loss until the signal drops below the absolute hardware noise threshold.
- **Throughput constraints:** Narrow 1 MHz physical channels, TDMA time-slot allocations, and protocol encapsulation collectively bottleneck the total throughput. The link saturates whether it is introduced to a high volume of small OT-like frames or a low frequency of large IT payloads

Consequently, the Bluetooth architecture only suits isolated, low-netload OT networks and cannot support mixed IT/OT environments.

8.0.3 Evaluation of the Redundant Wi-Fi Architecture

Logically isolating PRP paths via IEEE 802.1Q VLANs and Virtual Access Points (VAPs) offers superior performance and scalability for industrial environments.

- **High capacity and low latency:** The architecture delivers sub-millisecond baseline latency (e.g., $\sim 0.6\text{--}0.8$ ms) and successfully supports mixed IT/OT traffic (e.g., sustaining 65 Mbps of 1400-byte UDP packets). It will be stable provided the total load remains below the negotiated physical bitrate.
- **Small-packet vulnerability:** The MAC-level overhead of CSMA/CA creates a severe bottle-neck for small-packet transmissions. Many high-frequency small frames saturate the airtime, causing network saturation long before reaching its theoretical throughput limit.
- **MCS fallback risks:** Signal attenuation forces the hardware into lower modulation schemes (MCS fallback), drastically reduces the available throughput (e.g., dropping to a 6 Mbps link speed at -93 dBm). RF cell planning is therefore mandatory to prevent IT traffic from overwhelming OT frames during these signal dips.
- **QoS limitations:** Physically isolating the hardware queues via separate VAPs breaks standard WMM QoS prioritization between competing IT and OT streams.

8.0.4 Recommendation for ArcelorMittal Gent

This research recommends deploying the redundant Wi-Fi architecture with PRP for the overhead cranes at ArcelorMittal Gent. This solution successfully masks handover gaps while routing the required IT/OT traffic mix. It delivers superior performance, scalability and supports the necessary throughput, making it a future-proof investment compared to the isolated Bluetooth alternative.

8.1 Future Work

Future research must transition these laboratory findings into a production-ready system and investigate next-generation technologies to bypass current limitations.

8.1.1 Immediate Next Steps for ArcelorMittal Gent

- **Physical Rollout and Cell Isolation:** Engineers must deploy the staggered AP topology along the active crane runway. This requires optimizing AP placement, strictly enforcing cell boundaries, and fine-tuning Client roaming thresholds. AP coverage cells on the same redundancy line must never overlap to prevent bridging loops.
- **Real-World Movement Profiles:** Field tests must evaluate how the implemented architecture will behave on moving cranes. Tests need to quantify how things like crane speed, Doppler shifts, dynamic multipath reflections, SNR, ... impact handover latency, MCS fallback and the general performance itself.
- **Integrated Hardware Validation:** Future testing should evaluate if modern industrial switches (e.g., the FL SWITCH 4300) can simultaneously execute PRP duplication and complex VLAN

segregation without introducing processing latency. Proving this capability would eliminate the need for secondary managed switches, reducing overall CAPEX and architectural complexity.

8.1.2 Broader Technological Trajectories for Future Research

While this research validated an external PRP architecture over standard Wi-Fi, other research must evaluate next-generation wireless protocols:

- **Wi-Fi 6E and Wi-Fi 7:** Wi-Fi 6E provides an uncontested 6 GHz spectrum. The Orthogonal Frequency Division Multiple Access (OFDMA) mechanism in Wi-Fi 6 and 7 reduces MAC-level overhead, directly addressing the small-packet vulnerability identified in this research. Furthermore, Wi-Fi 7 introduces Multi-Link Operation (MLO) for simultaneous transmission over multiple frequency bands. Researchers must determine if MLO provides intrinsic handover resilience that makes external PRP redundancy boxes obsolete [18].
- **5G Private Networks:** 5G operates on licensed spectrum, guaranteeing QoS and sub-millisecond latency. Future studies should benchmark 5G architectures against this Wi-Fi PRP approach to directly compare industrial performance [18].
- **TSN and IEEE 802.1CB (FRER):** While PRP requires building two strictly disjoint physical networks and deploying external RedBox hardware, the industry is transitioning toward Time-Sensitive Networking (TSN). Future research should evaluate IEEE 802.1CB (Frame Replication and Elimination for Reliability (FRER)) [19]. Because FRER operates natively at the MAC layer within TSN-compatible switches, it can dynamically duplicate frames across flexible mesh topologies rather than strict parallel lines [20]. Researchers must investigate if wireless FRER can be used while maintaining seamless handover.

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Appendix A

Technology Datasheet: Wi-Fi vs. Bluetooth

Table A.1 + A.2 provides a comparative overview of the conventional single-path connections compared to the redundant (PRP) architectures from this study. The table is structured around the Key Performance Indicators (KPIs) for industrial automation. For characteristics that depend on external datasheets or were not directly measured, this is specifically indicated.

Table A.1 Comparative datasheet of the wireless technologies for PROFINET (Part 1 of 2).

Characteristic	Std. Wi-Fi	Std. Bluetooth	PRP over Wi-Fi	PRP over BT
Handover, Latency & Reliability				
Seamless handover?	No	No	Yes	Yes
Handover delay	Break-before-make (> 50 ms)*	Reconnect (seconds)*	Fully masked (0 ms)	Fully masked (0 ms)
Packet loss on handover?	Yes	Yes	No (0.000 %)	No (0.000 %)
Extra latency (app-to-app)	N/A	N/A	Very low (~10–50 μ s)	Very low (~10–50 μ s)
Avg. latency	Low	Moderate	Low (~0.6–0.8 ms)	Moderate (~6.0–6.8 ms)
Avg. Max. latency	Low	Moderate	Low (< 2 ms)	Moderate (>30 ms)
Jitter	Variable	Constant (TDMA)	Low (PRP smoothed, ~0.1 %)	Constant (PRP smoothed, ~4.0 %)
Network, Capacity & Spectrum				
Max. throughput	~54 Mbps (theoretical)	~3 Mbps (theoretical)	~54 Mbps (effective)	< 3 Mbps (effective)
Throughput degradation dependency	Variable (negotiated bitrate)	Constant (TDMA limits)	Variable (negotiated bitrate)	Constant (TDMA limits)
Max. nodes / application	Multiple (CSMA/CA)	Up to 7, 1 recommended (PANU)*	Multiple (strict planning)	1 (P2P)
Spectrum	5 GHz	2.4 GHz	5 GHz	2.4 GHz
Channels used at the same time / application	1 (e.g., Channels 36, 40, 44, 48)	1 (AFH, 79 channels)	2 (Each LAN)	2 (Each PANU + AFH)
Attenuation Prone?	Yes	No	Yes	No
VLAN Isolation?	Possible	No	Yes	No
QoS	WMM (802.11e)	N/A	WMM (fails across VAPs)	N/A

**Sourced from manufacturer datasheets (Siemens SCALANCE W700 series; Phoenix Contact FL BT EPA 2). Not formally tested in this study.*

Table A.2 Comparative datasheet of the wireless technologies for PROFINET (Part 2 of 2).

Characteristic	Std. Wi-Fi	Std. Bluetooth	PRP over Wi-Fi	PRP over BT
Suitability per Traffic Mix (++ / + / 0 / - / --)				
Strictly OT	0	+	++	++ (low IO count)
Strictly IT	+	--	++	--
Mixed OT/IT	+	--	++ (below bitrate limit)	--
Typical Config.: Controller + 16 IO-devices @ 64 ms**				
Mean latency	Not separately measured		0.62 ms	30.3 ms
Max. achieved latency	Not separately measured		2.61 ms	46.4 ms
Physical & Environmental Properties				
Expected range	~30–100 m (but higher possible, antenna dependent)*	~100 m*	Broad (staggered APs)	Limited (cascaded RedBoxes)
Resilience to multipath	Vulnerable (MCS fallback)	Highly resilient (AFH)	Mitigated (PRP path masking)	Highly resilient (AFH)
Max. RF output (EIRP)*	≤ 23 dBm (SCALANCE)	≤ 16 dBm (FL EPA)	≤ 23 dBm (SCALANCE)	≤ 16 dBm (FL EPA)
Boot-up duration	~10–15 s (Theoretical)*	~40–60 s (Theoretical)*	14.2 s (Measured)	54.6 s (Measured)
Typical Application/Use Case & Price				
Primary Use Case	Non-critical IT/OT (tolerant to handover gaps)	Isolated, static OT loops	Mission-critical, mobile IT/OT	Mission-critical, dedicated OT (low IO count)
Estimated CAPEX (per crane zone)	Low (~3,850 EUR)	Very Low (~2,300 EUR)	High (~16,400 EUR)	Moderate (~12,300 EUR)

*Sourced from manufacturer datasheets (Siemens SCALANCE W700 series; Phoenix Contact FL BT EPA 2). Not formally tested in this study.

** 16 IO-devices at 64 ms, validated as an equivalent 4-IO test at 16 ms.

Appendix B

Additional Measurements: 16 IO-Device Load at 64 ms Cycle Time

This appendix evaluates network performance under a typical industrial load, providing detailed end-to-end latency distributions for a simulated 16 IO-Device setup. To mathematically replicate the packet injection rate of 16 IO-Devices operating at a 64 ms cycle time, the laboratory setup utilized 4 physical IO-Devices configured at an aggressive 16 ms cycle time. Throughout these measurements, the PROFINET Watchdog Timer remained set to 1920 ms for both architectures.

Bluetooth Saturation

Figure B.1 illustrates the performance breakdown of the Bluetooth architecture under this scaled load. It proves that the Bluetooth's throughput limitations prevent it from processing this volume of PROFINET frames efficiently.

Wi-Fi Capacity Verification

Conversely, Figure B.2 provides the data proving the Wi-Fi architecture's scalability. During these specific measurements, the Wi-Fi infrastructure operated with a fixed physical bitrate of 54 Mbps. The graph confirms that this capacity is sufficient to process the PROFINET traffic, maintaining a sub-millisecond mean latency.

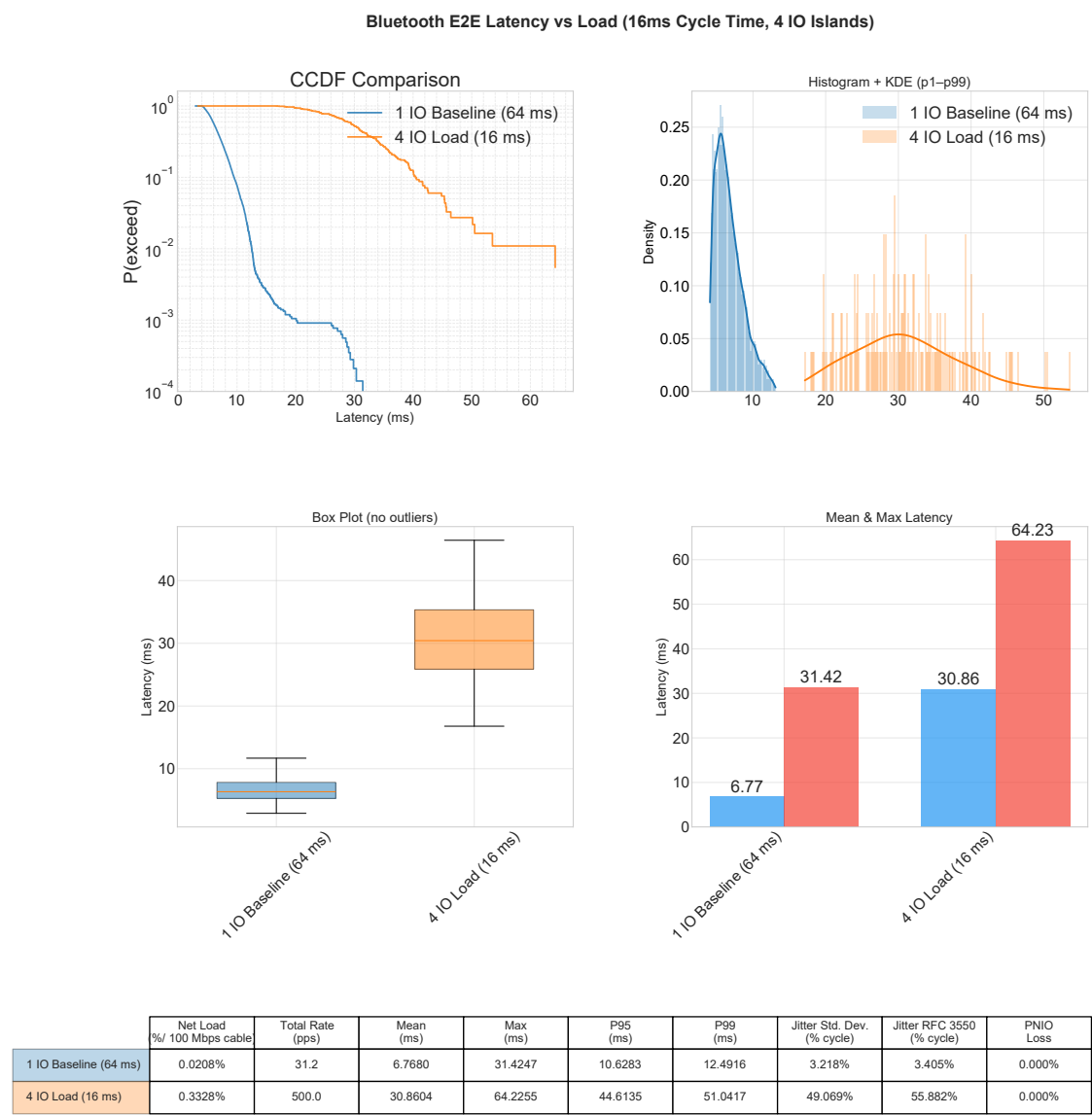


Figure B.1: Bluetooth end-to-end latency comparison under a simulated 16 IO-Device load (executed as 4 IO-Devices at a 16 ms cycle time).

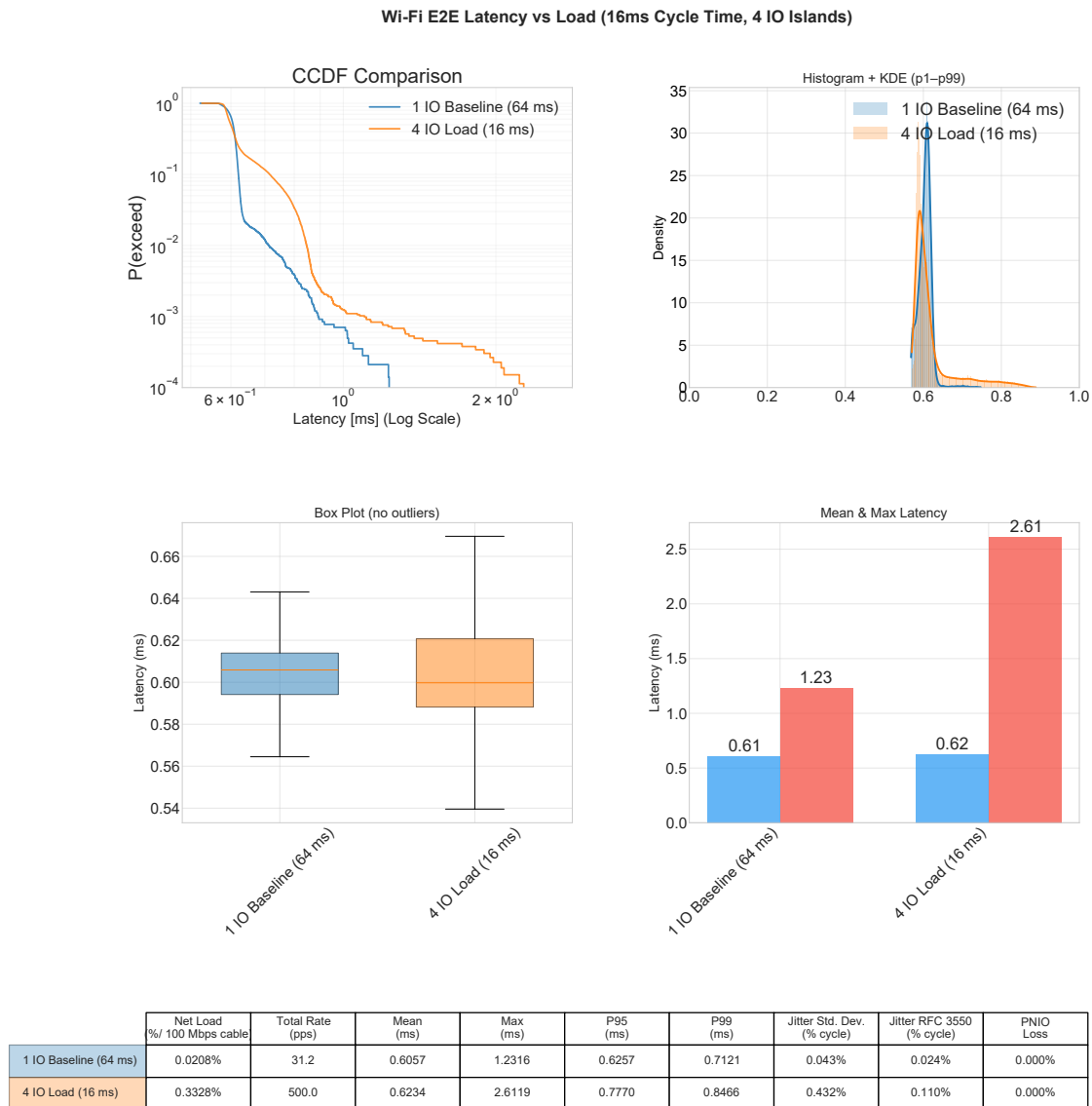


Figure B.2: Wi-Fi end-to-end latency comparison under a simulated 16 IO-Device load (executed as 4 IO-Devices at a 16 ms cycle time).

Appendix C

Beschrijving van deze masterproef in de vorm van een wetenschappelijk artikel

PROFINET via draadloze verbinding: De handover gap overbruggen voor seamless handover en redundantie

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Abstract—In industriële omgevingen zoals de staalfabriek van ArcelorMittal Gent zijn geautomatiseerde machines afhankelijk van draadloze netwerken voor hun werking en veiligheid. Het industriële protocol PROFINET vereist deterministische communicatie, bewaakt door watchdog-timers. Wanneer de communicatievertraging deze drempelwaarden overschrijdt, wordt een Fail-Safe toestand geactiveerd die de machine onmiddellijk stilzet. Dit onderzoek past het Parallel Redundancy Protocol (PRP), gestandaardiseerd onder IEC 62439-3, toe op draadloze verbindingen om de zogenaamde “handover gap” te elimineren. Twee proof-of-concept-architecturen werden gebouwd en geëvalueerd: een Bluetooth-opstelling met fysiek gescheiden point-to-point verbindingen en een Wi-Fi-opstelling met Virtual Local Area Network (VLAN)-gebaseerde logische isolatie. Experimentele evaluatie toont aan dat beide architecturen de handover gap succesvol maskeren met nul pakketverlies. De Wi-Fi-architectuur levert superieure prestaties met een baseline latency van $\sim 0,6$ ms en ondersteunt gemengd IT/OT-verkeer. De Bluetooth-architectuur biedt superieure robuustheid onder signaaldempingen dankzij Adaptive Frequency Hopping, maar is beperkt tot geïsoleerde OT-netwerken door een lage throughput. Dit onderzoek biedt een gevalideerd framework voor het implementeren van draadloze redundantie-architecturen voor betrouwbare PROFINET-communicatie in industriële omgevingen.

Index Terms—PROFINET, Parallel Redundancy Protocol (PRP), Seamless Handover, Handover Gap, Deterministische Communicatie, Wi-Fi, Bluetooth, Industriële Automatisering

I. INLEIDING

De transitie naar Industry 4.0 heeft de integratie van autonome en semi-autonome mobiele machines in industriële omgevingen fundamenteel versneld. In faciliteiten zoals de staalfabriek van ArcelorMittal Gent zijn overheadkranen essentieel voor het transport van vloeibaar staal. Om deze kritische assets te automatiseren is continue, deterministische communicatie tussen de centrale servers, de vaste Programmable Logic Controllers (PLC's) en de mobiele PLC's op de kraan een absolute noodzaak [1].

In industriële automatisering wordt communicatie doorgaans beheerd door real-time protocollen zoals PROFINET. In tegenstelling tot standaard Information Technology (IT) netwerken werkt PROFINET op deterministische cyclustijden, bewaakt door communicatie-watchdogs. Wanneer de communicatievertraging een gedefinieerde drempelwaarde overschrijdt,

gaat de PLC over in een **Fail-Safe** toestand om ongevallen te voorkomen [2]. De Watchdog Time wordt berekend als:

$$\text{Watchdog Time} = \text{Update Time} \times \text{Multiplier} \quad (1)$$

waarbij de industriestandaard een multiplier van 3 opeenvolgende gemiste frames is [2].

Hoewel dit Fail-Safe-mechanisme essentieel is voor de veiligheid, introduceert het paradoxaal genoeg nieuwe risico's. Een stilstaande kraan vormt een onverwacht obstakel met directe botsingsrisico's. Bovendien veroorzaken deze onderbrekingen kostbare productievertragingen en vereisen ze handmatige interventie [1].

Het bestaande enkelvoudige draadloze pad in de staalfabriek kampt met drie fundamentele problemen:

- **Dynamische Radio Frequency (RF) shadowing:** Grote metalen constructies, zoals aangrenzende kranen, blokkeren de Line-of-Sight (LoS) tussen de draadloze client en de vaste Access Points (AP's). Dit veroorzaakt ernstige multipath fading en plotselinge signaaluitval.
- **Bitrate-fluctuaties door MCS-degradatie:** Wisselende ruisniveaus, veroorzaakt door elektromagnetische interferentie (EMI), co-channel interference en shadowing, dwingen de radio tot een lagere Modulation and Coding Scheme (MCS). De resulterende bitrate-instortingen verhogen de airtime per pakket en introduceren jitter die de watchdog-drempelwaarden overschrijdt, zelfs wanneer het Received Signal Strength Indication (RSSI) niveau acceptabel lijkt.
- **Handover-latentie en het “Sticky Client”-fenomeen:** De draadloze client wacht te lang met het wisselen naar een beter AP, waardoor hij op een zwak signaal aan de celrand blijft werken. Dit maakt de verbinding extra kwetsbaar voor shadowing. Wanneer de handover uiteindelijk plaatsvindt, overschrijdt de “handover gap” vaak de watchdog-drempelwaarden.

Het doel van dit onderzoek is het ontwerpen, implementeren en evalueren van redundante draadloze architecturen die deze problemen elimineren. De voorgestelde oplossing is gebaseerd op het **Parallel Redundancy Protocol**, gestandaardiseerd onder IEC 62439-3 [3], dat elk bericht dupliceert en gelijktijdig verstuurt over twee onafhankelijke paden.

II. PRP EN DRAADLOZE TECHNOLOGIEËN

A. Parallel Redundancy Protocol

PRP bereikt naadloze redundantie door twee fysiek gescheiden, parallelle Local Area Networks (LAN's) te gebruiken: **LAN A** en **LAN B** [3]. Bij verzending wordt elk PROFINET-frame gedupliceerd en voorzien van een 6-byte **Redundancy Control Trailer (RCT)**. De twee kopieën worden gelijktijdig over de afzonderlijke paden verstuurd. Aan de ontvangtzijde accepteert het ontvangende apparaat (een Dual Attached Node of RedBox) het eerst aangekomen, foutloze frame en verwijdert het duplicaat op basis van het sequentienummer in de RCT. Dit resulteert in een hersteltijd van 0 ms bij het uitvallen van één pad.

B. Wi-Fi (IEEE 802.11)

Wi-Fi biedt een transparante Layer-2 bridge voor Ethernet-frames via het Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA) mechanisme [4]. De 5 GHz-band biedt voldoende niet-overlappende kanalen voor dichte AP-deployment. In een onbelast medium levert Wi-Fi sub-milliseconde latency. Het standaard "Break-before-Make"-handover-model introduceert echter een communicatie-onderbreking van 50 ms tot meer dan 300 ms [5], wat de PROFINET-watchdog kan triggeren.

C. Bluetooth (IEEE 802.15.1)

Bluetooth 2.1 + EDR biedt een geplande Time Division Multiple Access (TDMA) gebaseerde verbinding met Adaptive Frequency Hopping (AFH) [6]. Het PAN-profiel (Personal Area Networking) encapsuleert Ethernet-frames via BNEP (Bluetooth Network Encapsulation Protocol). Bluetooth mist echter native fast-roaming-mechanismen: een herverbinding met een nieuwe NAP (Network Access Point) duurt meerdere seconden, wat standaard roaming onmogelijk maakt voor PROFINET [7].

III. FIELD AUDIT: PROBLEEMANALYSE ARCELORMITTAL GENT

Een data-gedreven audit van de bestaande draadloze infrastructuur kwantificeerde de beperkingen. Over een periode van 12 maanden werden **1394 Watchdog Timeouts** geregistreerd, gemiddeld 4 keer per dag. Dit maakt geautomatiseerde kraanbediening onmogelijk met de huidige infrastructuur.

Fig. 1 toont de ruimtelijke verdeling van Fail-Safe toestanden. Hoewel sommige clusters zich bevinden rond permanente fysieke obstakels, treden veel storingen op over het volledige traject. Dit wijst op **dynamische shadowing** door aangrenzende kranen als dominante oorzaak, naast statische dode zones.

Analyse van een specifieke operationele dag bevestigde drie technische bottlenecks: (1) dynamische shadowing door aangrenzende kranen die de LoS naar AP's blokkeren; (2) bitrate-instoringen ondanks een acceptabel RSSI-niveau (-65 tot -72 dBm), veroorzaakt door variërende ruisniveaus (EMI, multipath scattering, co-channel interference) die de hardware

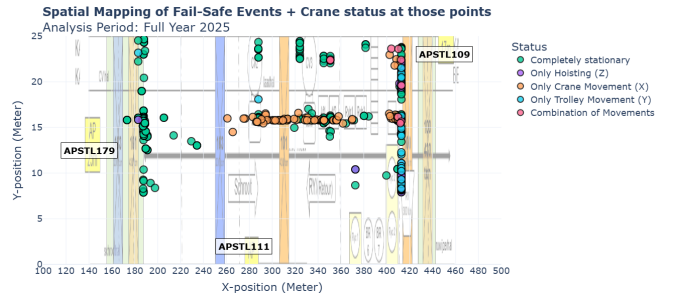


Fig. 1. Jaarlijkse ruimtelijke verdeling van Fail-Safe toestanden over het kraantraject.

dwingen een lagere MCS te gebruiken. Hierdoor stijgt de airtime per pakket en ontstaat jitter die de watchdog triggert; en (3) het "Sticky Client"-fenomeen, waarbij de client te lang verbonden blijft met een veraf gelegen AP en daardoor aan de celrand werkt met minimale linkmarge. Deze drie bottlenecks bevestigen de noodzaak van **ruimtelijke padredundantie**, **handover-maskering** en **bitrate-stabilisatie**. Dit zijn vereisten waaraan enkel een redundante architectuur met twee onafhankelijke draadloze paden kan voldoen.

IV. ARCHITECTUUR EN IMPLEMENTATIE

Twee redundante architecturen werden geïmplementeerd en geëvalueerd als proof of concept.

A. Bluetooth Point-to-Point Architectuur

Deze opstelling bereikt fysieke padscheiding door twee onafhankelijke point-to-point Bluetooth-verbindingen. De LAN A-poort van de statische RedBox verbindt met een eerste NAP, de LAN B-poort met een tweede NAP (Fig. 2). Aan de mobiele zijde verbinden twee PAN User (PANU) clients elk exclusief met hun respectieve NAP.

Om het bereik uit te breiden wordt een **Cascaded PRP**-topologie gebruikt, waarbij secundaire RedBoxen meerdere Bluetooth-zones combineren met overlappende dekkinggebieden. Wanneer de mobiele eenheid zich verplaatst, garandeert PRP dat ten minste één verbinding actief blijft.

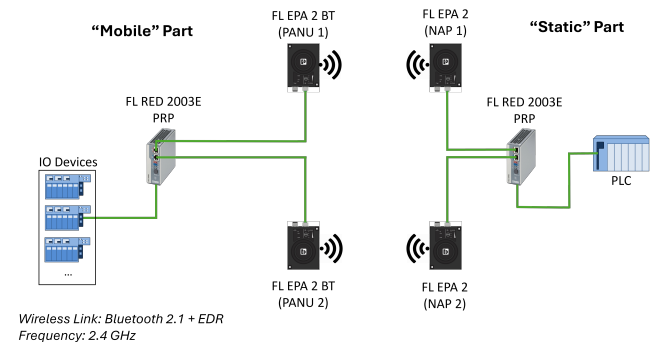


Fig. 2. Topologie van de Point-to-Point Bluetooth-architectuur met fysieke padscheiding.

B. Wi-Fi Architectuur met VLAN-isolatie

De Wi-Fi-architectuur gebruikt IEEE 802.1Q Virtual Local Area Networks (VLAN's) voor logische padscheiding (Fig. 3). Een managed switch scheidt het verkeer in VLAN A (Line A), VLAN B (Line B) en een VRON (Rest of Network) voor IT-verkeer. Elke fysieke AP configureert meerdere Virtual Access Points (VAPs) die afzonderlijke Service Set Identifiers (SSID's) uitzenden, elk gekoppeld aan een specifiek VLAN.

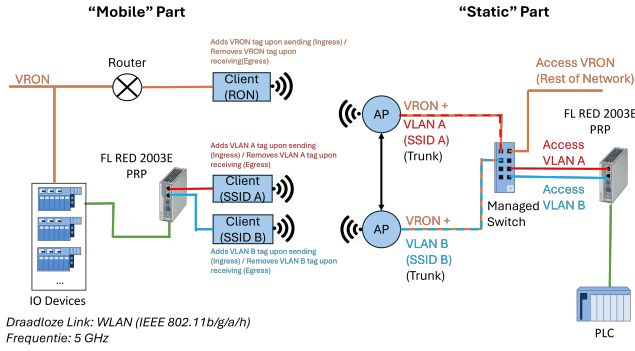


Fig. 3. Logische en fysieke topologie van de redundante Wi-Fi-architectuur met VLAN-isolatie.

Aan de mobiele zijde verbinden onafhankelijke clients met hun respectieve SSID's. Doordat de AP's fysiek versprongen zijn, vindt een handover op Line A en Line B nooit gelijktijdig plaats. PRP maskeert de tijdelijke disconnectie op één lijn terwijl de andere actief blijft.

V. MEETRESULTATEN EN ANALYSE

De architecturen werden geëvalueerd in vier scenario's: baseline (P1), netwerkbelasting (P2), signaaldemping (P3) en seamless handover (P4). De End-to-End (E2E) latency werd gemeten met een Hilscher netAnalyzer (10 ns resolutie) en verwerkt via een custom Python-analysepipeline.

A. Baseline-prestaties (P1)

Tabel I vat de baseline-resultaten samen bij een 64 ms cyclustijd.

TABLE I
BASELINE E2E LATENCY VERGELIJKING BIJ 64 MS CYCLUSTIJD.

Architectuur	Mean (ms)	Max (ms)	Std Jitter (% cycle)	PN Loss (%)
Bluetooth	6,77	31,42	3,218	0,000
Wi-Fi	0,61	1,23	0,043	0,000

Wi-Fi behaalt een gemiddelde latency van $\sim 0,6$ ms dankzij het onbelaste CSMA/CA-medium, terwijl Bluetooth een vaste ~ 6 ms verwerkingsbodem vertoont door de TDMA-polling en BNEP-encapsulatie. Beide architecturen bereiken 0,000% PROFINET-pakketverlies.

B. Netwerkbelasting (P2)

De totale netwerkcapaciteit van Wi-Fi is strikt gebonden aan de onderhandelde fysieke bitrate (54 Mbps tijdens de metingen). Wi-Fi blijft zeer stabiel bij 40 Mbps achtergrondverkeer (1400-byte User Datagram Protocol (UDP) pakketten), met een gemiddelde latency van $\sim 1,0$ ms. Wanneer de belasting deze fysieke grens overschrijdt (bijv. bij 60 Mbps), begint de stabiliteit af te nemen. Boven 67 Mbps treedt volledige airtime-saturatie op met 3,42% pakketverlies. Bij kleine pakketten (64 bytes) daalt dit breukpunt echter drastisch naar ~ 5 Mbps, aangezien de hoge Medium Access Control (MAC) layer overhead per pakket de beschikbare airtime veel sneller uitput.

Bluetooth verzadigt reeds bij 0,3 Mbps (1400-byte) door smalle 1 MHz kanalen en TDMA-fragmentatie, en bij 0,1 Mbps (64-byte) door uitputting van polling-slots. Dit bevestigt dat Bluetooth **uitsluitend geschikt is voor dedicated Operational Technology (OT) verkeer**.

C. Signaaldemping (P3)

Tabel II vergelijkt de prestaties onder progressieve signaaldemping.

TABLE II
PRESTATIES ONDER SIGNAALDEMPING MET ÉÉN LIJN UITGESCHAKELD.

Scenario	Mean (ms)	Max (ms)	Jitter (% cyc)	Loss (%)
<i>Bluetooth</i>				
-75 dBm	14,42	30,92	4,000	0,000
-85 dBm	15,30	36,95	4,455	0,000
-92 dBm	19,08	60,77	8,331	0,101
<i>Wi-Fi</i>				
-75 dBm (48 Mbps)	0,5683	1,41	0,063	0,000
-85 dBm (24 Mbps)	1,44	69,77	9,029	0,007
-93 dBm (6 Mbps)	2,13	65,80	9,841	2,423

Bluetooth handhaaft een deterministisch latency-profiel tot nabij de hardware noise floor (-92 dBm) dankzij de TDMA-slottoewijzing en het FHSS-mechanisme, dat geïnterfereerde frequenties dynamisch vermijdt via AFH. Wi-Fi is kwetsbaarder: MCS-fallback verlaagt de bitrate van 54 Mbps naar 6 Mbps, waardoor de beschikbare capaciteit drastisch krimpt en pakketverlies eerder optreedt. Wi-Fi blijft redelijk stabiel tot -85 dBm.

D. Seamless Handover (P4)

De handover-validatie simuleerde handover door sequentieel Ethernet-kabels los te koppelen. Fig. 5 en Fig. 4 toont de tijdreeks voor Wi-Fi en Bluetooth.

Beide architecturen overleefden de sequentiële disconnecties **zonder pakketverlies** en zonder het triggeren van een Fail-Safe toestand. Bovendien toonde de data een **latency-smoothing effect**: wanneer beide lijnen actief zijn, selecteert PRP consistent het snelst aangekomen frame, wat resulteert in een lagere gemiddelde latency en strakkere verdeling dan elk individueel pad.

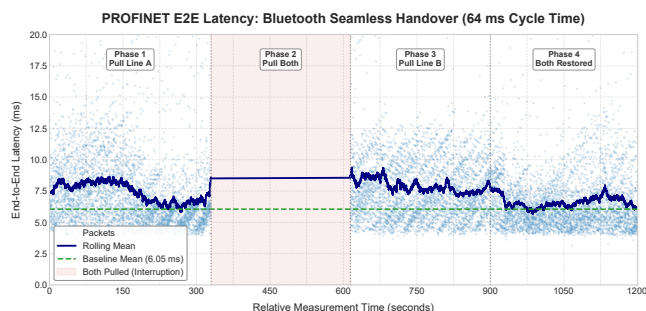


Fig. 4. Bluetooth E2E latency time series during the seamless handover test.

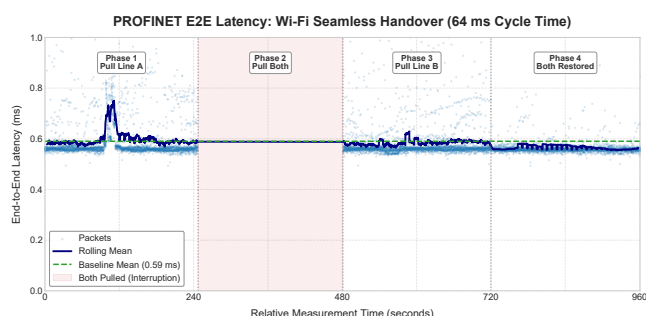


Fig. 5. Wi-Fi E2E latency tijdreeks tijdens de seamless handover test: Line A pull, beide pull, Line B pull, beide hersteld.

VI. VERGELIJKING EN AANBEVELING

Tabel III vat de kernbevindingen samen.

TABLE III
SAMENVATTING VAN DE EMPIRISCHE BEVINDINGEN PER TESTSCENARIO.

Scenario	Wi-Fi	Bluetooth
Baseline	Sub-ms latency (~0,6 ms), sporadische outliers	Vaste ~6 ms floor, voorspelbare spreiding
Load	Gemengd IT/OT verkeer, stabiele werking enkel als throughput onder de negotiated bitrate blijft	Zeer lage throughput, enkel geschikt voor OT-links
Demping	MCS-fallback verkleint capaciteit dynamisch	Stabiel tot noise floor dankzij TDMA/FHSS
Handover	Naadloze handover met 0% loss	Naadloze handover met 0% loss

Op basis van deze resultaten wordt de **redundante Wi-Fi-architectuur met PRP** aanbevolen voor ArcelorMittal Gent. De voornaamste redenen zijn:

- **IT/OT-convergentie:** Wi-Fi is de enige architectuur die gemengd IT- en OT-verkeer ondersteunt over dezelfde infrastructuur.
- **Superieure prestaties:** Sub-milliseconde baseline latency, geminimaliseerde maximum latency en hoge throughput bieden ruime marges voor de watchdog-timer.

- **Schaalbaarheid:** Door AP's versprongen te plaatsen met niet-overlappende kanalen wordt het bereik uitgebreid zonder bridging loops.
- **Kwetsbaarheidsmaskering:** PRP neutraliseert effectief de handover gap, de fluctuerende bitrate en de shadowing-kwetsbaarheid van Wi-Fi.

De geschatte Capital Expenditure (CAPEX) voor een minimale deployment (1 kraan, 1 zone) bedraagt **16.402 EUR** voor Wi-Fi, tegenover 12.308 EUR voor Bluetooth. Ondanks de hogere initiële investering biedt de Wi-Fi-architectuur de noodzakelijke capaciteit en toekomstbestendigheid.

VII. CONCLUSIE EN TOEKOMSTIG WERK

Dit onderzoek heeft aangetoond dat het toepassen van PRP over draadloze verbindingen de handover gap succesvol elimineert voor industriële PROFINET-communicatie. Door elk frame te dupliceren over twee onafhankelijke draadloze paden, accepteert de ontvangende RedBox het snelst aangekomen frame en verwijdert het duplicaat, wat resulteert in naadloze handover met nul pakketverlies.

De Wi-Fi-architectuur met VLAN-isolatie biedt de beste combinatie van lage baseline latency, geminimaliseerde maximum latency, hoge throughput en IT/OT-convergentie. De Bluetooth-architectuur biedt superieure RF-robustheid onder signaaldemping dankzij AFH en TDMA, maar is door haar throughputbeperkingen beperkt tot geïsoleerde OT-netwerken.

Een belangrijke bijkomende bevinding is dat Wi-Fi Multimedia (WMM) Quality of Service (QoS) (IEEE 802.11e) geen effectieve prioritering biedt in deze architectuur, aangezien de VAP-scheiding de hardware-wachtrijen isoleert en daardoor WMM-cross-traffic-prioritering neutraliseert.

Toekomstig werk omvat:

- **Veld demonstratie:** Deployment op een actieve kraanbaan met optimalisatie van AP-plaatsing, cellgrenzen en roaming-drempels.
- **Wi-Fi 6E/7:** Evaluatie van Orthogonal Frequency Division Multiple Access (OFDMA) voor reductie van MAC-overhead bij kleine pakketten en Multi-Link Operation (MLO) als potentieel intrinsiek redundantiemechanisme.
- **5G Private Networks:** Benchmarking van 5G met gegarandeerde QoS tegenover de Wi-Fi PRP-aanpak.
- **Time-Sensitive Networking (TSN) en IEEE 802.1CB (FRER):** Onderzoek naar Frame Replication and Elimination for Reliability als toekomstig alternatief voor PRP, opererend op MAC-layer niveau binnen TSN-compatibele switches.

DANKWOORD

De auteur bedankt Prof. François Rottenberg en Prof. Philippe Saey voor hun academische begeleiding. Daarnaast gaat de dank uit naar ArcelorMittal Gent (Nathan Baes en Tycho Van Hemelryck) voor de begeleiding, de industriële context en de operationele data. Tot slot bedankt de auteur Pieter-Jan Deltour van Phoenix Contact voor de industriële Bluetooth hardware en de DRAMCO-onderzoeksgroep voor de ondersteuning bij de laboratoriumvalidaties.

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Appendix D

Poster

PROFINET over wireless: Bridging the handover gap for seamless handover and redundancy

Implementation and Evaluation of Parallel Redundancy Protocol over a Wi-Fi or Bluetooth Network for Industrial Automation

Problem:

- Problem:** Latency could exceed watchdog thresholds, triggering a fail-safe state causing the crane to halt.
- Cause:** Radio Frequency (RF) shadowing forces mobile cranes to switch Access Points (APs).
- Consequence:** Roaming introduces handover latency.

Goal:

- Design:** A wireless architecture that guarantees continuous connectivity.
- Achieve:** Seamless handover and redundancy.
- Maintain:** Safety control loops without relaxing the watchdog timers.

Investigated Wireless Local Area Networks (WLAN) Architectures:

Architecture 1: Redundant WLAN with Wi-Fi

- Logical Isolation:** Virtual local area networks (VLANs), trunking and Virtual Access Points to separate redundancy lines.
- Staggered AP placement:** Handover for Line A and Line B occurs at different locations.
- Seamless Handover:** Parallel Redundancy Protocol (PRP) masks the handover latency even if a path is shadowed, as another VLAN with another AP is already active.

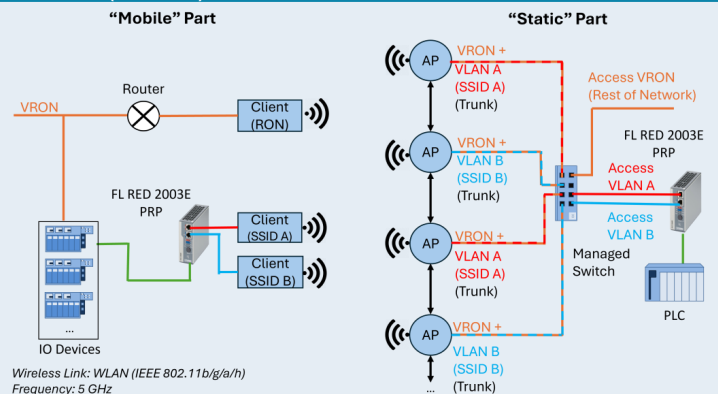


Figure 1: Wi-Fi architecture with logical VLAN isolation.

Architecture 2: Redundant WLAN with Bluetooth

- Physical Isolation:** Redundant line separation through point-to-point links. No VLANs needed.
- Seamless Redundancy:** PRP duplicated frames ensure communication even if a path is shadowed.
- Cascaded Extension:** Bluetooth clients do not roam, range is extended by nesting Redundancy Boxes, seamlessly combining multiple overlapping coverage zones.

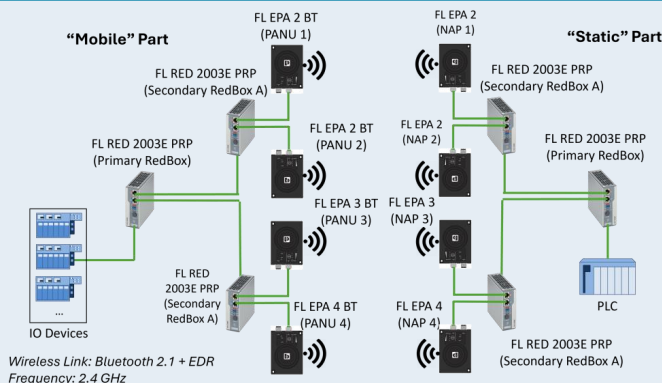


Figure 2: Cascaded Bluetooth architecture with physical point-to-point isolation.

Measurements and Conclusion:

- Seamless Handover:** The handover gap is masked successfully.
- Wi-Fi:**
 - + Lower latency and jitter and higher throughput
 - + Mix of short (Operational Technology (OT)) and longer (Information Technology (IT)) frames
 - More sensitive to attenuation.
- For higher throughput and lower max. latency applications
- Bluetooth:**
 - + Attenuation resistant
 - Only short (OT) frames
 - Lower performance, but within update time limits
- For lower throughput applications

	Wi-Fi	Bluetooth
Average Latency	✓ Very Low (≈ 0,7 ms)	~ Moderate (≈ 6,5 ms)
Average Jitter	✓ Very Low (≈ 0,1%)	~ Moderate (≈ 4%)
Max. Latency	✓ Very Low (≈ 1,2 ms)	~ Moderate (> 30 ms)
Max Throughput	✓ High (Bit rate)	✗ Low (< 3 Mbps)
Stability under Load	✓ High	~ Moderate
Attenuation Prone?	✓ Yes	✗ No
OT / OT & IT frames?	Typically OT & IT	Typically OT
VLAN needed	Yes	No
Price	More Expensive	Cheaper

Table 1: Comparison of implemented wireless architectures.

Appendix E

GenAI code of conduct for students

Generative AI (GenAI) assistance tools can be used to generate text, image, code, video, music, or combinations of these. It includes typical tools like (but this list is not limited to): ChatGPT, Google Gemini, MS Copilot, Midjourney, Claude.ai, Perplexity.ai, Dall-E, etc.

Student Information

Student name: Elias Neels

This form is related to my Master thesis.

Title Master thesis: PROFINET over wireless:

Bridging the handover gap for

seamless handover and redundancy

Promoter: François Rottenberg

Use of GenAI Assistance

Please indicate with "X" (possibly multiple times) in which way you were using GenAI:

☐ I did not use any GenAI assistance tool.

☒ I did use GenAI assistance. In this case, specify which ones (e.g., ChatGPT, ...): Gemini, Claude

GenAI Assistance Used As/For

☐ MS Copilot

☒ Language assistance

- ☐ Search engine
- ☐ Literature search
- ☒ Short-form input assistance
- ☒ Generating programming code
- ☐ Generating new research ideas
- ☐ Generating blocks of text
- ☒ Other (specify): Rephrasing parts of my own written work

Code of Conduct for Different Uses

Language Assistant for Reviewing or Improving Texts

This use is similar to using spelling and grammar check tools, and you do not have to refer to the use of GenAI in the text. Be careful:

- Using GenAI tools on texts you did not write yourself to cover up plagiarism (by paraphrasing original texts) is not allowed.

Search Engine for Information or Existing Research

This use is similar to a Google search or checking Wikipedia. If you then write your own text based on this information, you do not have to refer to the use of GenAI in the text. Be careful:

- Be aware that the output of the GenAI tool cannot be guaranteed as a 100% reliable source of information.
- The output may not be entirely correct and is limited due to the databases it uses. Knowledge evolves and may change over time, and it may be that the database of the GenAI tool is not up to date.

Literature Search

This use is comparable to a Google Scholar search. Be careful:

- Be aware that the output is restricted to the database it is built on. After this initial search, look for scientific sources and conduct your own analysis.
- GenAI tools (like ChatGPT) may output no or wrong references. As a student, you are responsible for further checking and verifying the accuracy of references.

Short-form Input Assistance

This use is similar to Google Docs powered by generative language models.

Generating Programming Code

The use of GenAI for coding should be explicitly allowed by the teacher. If used for coding, correctly mention the use of GenAI assistance and cite it.

Generating New Research Ideas

Further verify whether the idea is novel or not. It is likely that it is related to existing work, which should be referenced.

Generating Blocks of Text

Inserting blocks of text without quotes and a reference to GenAI assistance in your report or thesis is not allowed. Be careful:

- When you literally copy elements from a conversation with a GenAI tool, quote them between quotation marks and refer to them according to the specified reference style.
- Describe the use of the GenAI tool (tool name, version, date, etc.) in the method section and optionally add the full conversation as an attachment.

Other

Contact the professor of the course or the promoter of the thesis. Inform also the program director. Motivate how you comply with Article 84 of the exam regulations. Explain the use and added value of ChatGPT or another AI tool.

Further Important Guidelines and Remarks

- GenAI assistance cannot be used related to data or subjects under Non-Disclosure Agreement.
- GenAI assistance cannot be used related to sensitive or personal data due to privacy issues.
- Take a scientific and critical attitude when interacting with GenAI assistance and interpreting its output.
- As a student, you are responsible for complying with Article 84 of the exam regulations: your report or thesis should reflect your own knowledge, understanding, and skills.

Exam Regulations Article 84

“Every conduct individual students display with which they (partially) inhibit or attempt to inhibit a correct judgement of their own knowledge, understanding and/or skills or those of other students,

is considered an irregularity which may result in a suitable penalty. A special type of irregularity is plagiarism, i.e., copying the work (ideas, texts, structures, designs, images, plans, codes, ...) of others or prior personal work in an exact or slightly modified way without adequately acknowledging the sources."

Additional Reading

More information about being transparent on the use of GenAI assistance and about citing and referencing GenAI can be found on the student website.

A Few Final Words

If you are uncertain whether or not you should declare your use of AI tools, discuss the matter with your teacher or promoter. It is safer to declare AI use when it is not needed than to withhold that declaration when it is required.

Finally, remember that advanced AI tools are new and that they can do things they could not do until recently. It is important to follow up on the most recent developments in AI technologies and communicate openly with your teachers, assistants, supervisors, and peers.